BT501 Final Term PPTS by Tabih Fatima Topics 51 to 120 Fall 2022 Admin : All About Biotech

DNA Sequencing Technologies

Lecture no 51

- Determination of the nucleotide composition and order in a gene or genome is a foundational technology in modern medical biotechnology.

- DNA sequencing enables determination of the exact nucleotide sequence of a DNA fragment.

Examples

- DNA sequences can often reveal something about the function of the protein or RNA molecule encoded in a gene
- From predicted cofactor binding sites, transmembrane domains, receptor recognition sites, or DNA-binding regions.
- Provide information about the regulation of a gene.
- Reveal mutations that contribute to genetic diseases or the relatedness among the individuals
- Development of hypotheses about the evolutionary relationships among organisms.

Methods

- Maxam and Gilbert invented DNA sequencing by chemical cleavage method
- Sanger developed dideoxy chain termination method for sequencing.
- Many different sequencing technologies have been developed, including **pyrosequencing**, sequencing using **reversible chain terminators**, and sequencing by ligation.

- In general, all of these methods involve

I. Enzymatic addition of nucleotides to a primer based on complementarity to a template DNA fragment

II. Detection and identification of the nucleotide(s) added

- The techniques differ in the method by which the nucleotides are extended

I. employing either DNA polymerase to catalyze the addition of single nucleotides (sequencing by synthesis) or

II. ligase to add a short, complementary oligonucleotide (sequencing by ligation),

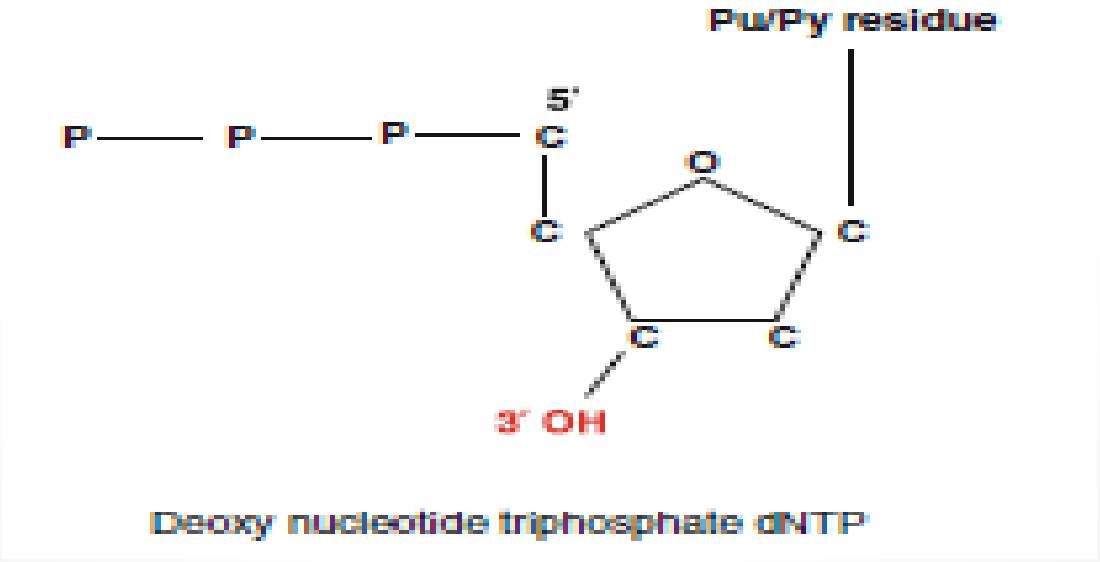
III. and in the method by which the addition is detected.

Dideoxynucleotide Procedure

Lecture no 52

Principle

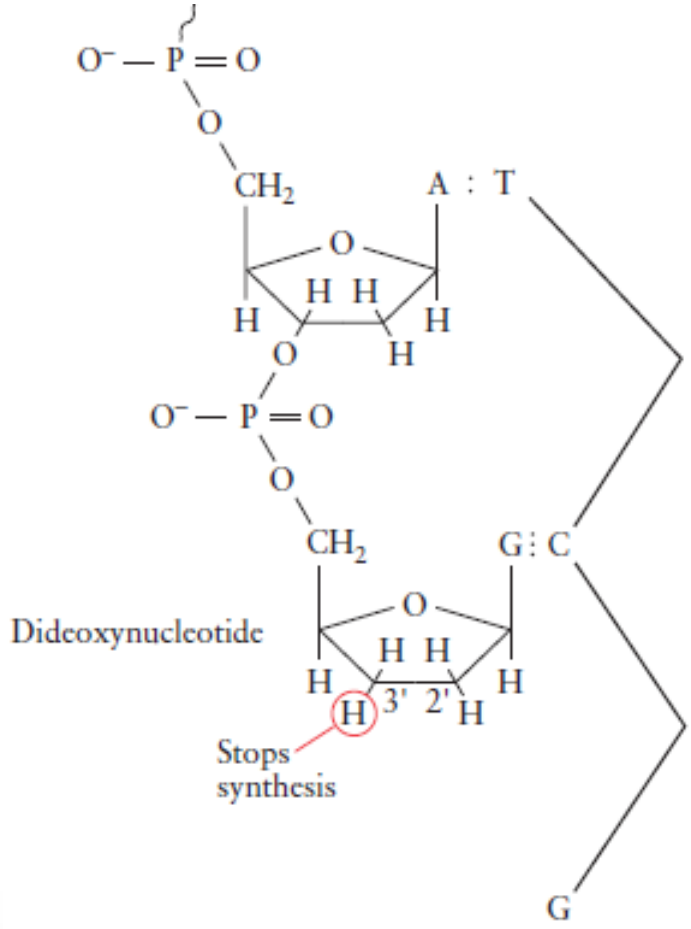
- DNA synthesis, addition of a nucleotide triphosphate requires a free hydroxyl group on the 3′ carbon of the sugar of the last nucleotide of the growing DNA strand.



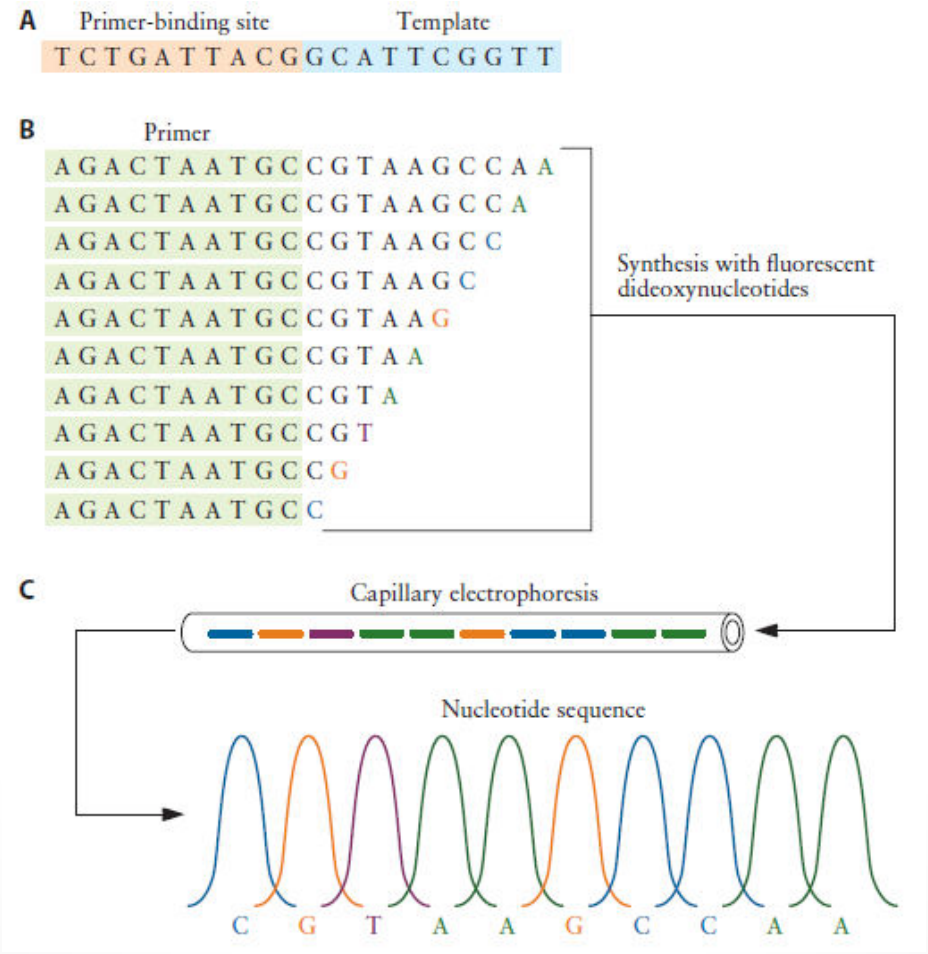
Deoxy nucleotide triphosphate dNTP

- A synthetic dideoxynucleotide that lacks a hydroxyl group at the 3′ carbon of the sugar moiety is incorporated at the end of the growing chain, DNA synthesis stops because a phosphodiester bond cannot be formed with the next incoming nucleotide (Fig. B).

- The termination of DNA synthesis is the defining feature of the dideoxynucleotide DNA sequencing method.


O⁻ — P = O
O
CH2
A : T
O
H H
H
H
O
H
O⁻ — P = O
O
CH2
G : C
Dideoxynucleotide
O
H H
H
H
H
3' 2'
H
Stops
synthesis
G

- A synthetic oligonucleotide primer (~17 to 24 nucleotides) anneals to a predetermined site on the strand of the DNA to be sequenced
- The reaction tube contains a mixture of the four deoxyribonucleotides and four dideoxynucleotides
- Each dideoxynucleotide is labeled with a different fluorescent dye
- These are applied to a polymer in a long capillary tube that enables separation of DNA fragments that differ in size by a single nucleotide.

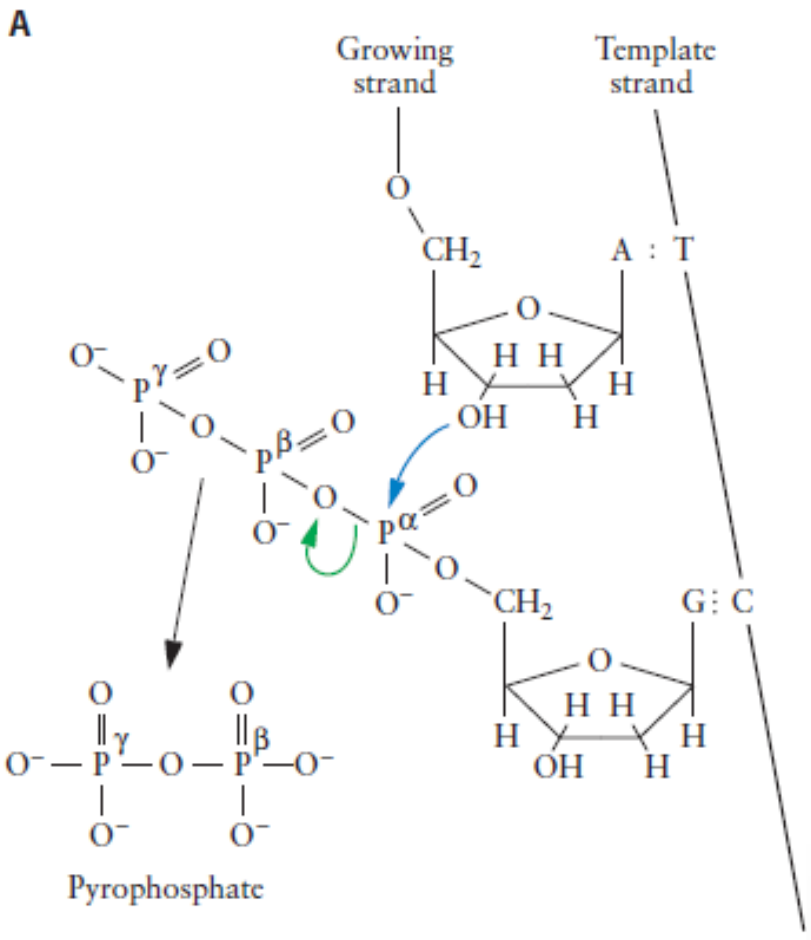

A
Primer-binding site
Template
T C T G A T T A C G G C A T T C G G T T
B
Primer
A G A C T A A T G C C G T A A G C C A A
A G A C T A A T G C C G T A A G C C A
A G A C T A A T G C C G T A A G C C
A G A C T A A T G C C G T A A G C
A G A C T A A T G C C G T A A G
A G A C T A A T G C C G T A A
A G A C T A A T G C C G T A
A G A C T A A T G C C G T
A G A C T A A T G C C G
A G A C T A A T G C C
Synthesis with fluorescent dideoxynucleotides
C
Capillary electrophoresis
Nucleotide sequence
C G T A A G C C A A

Pyrosequencing

Lecture no 53

- Pyrosequencing was the first of the next-generation sequencing technologies

- The basis of the technique is the detection of **pyrophosphate** that is released during DNA synthesis.

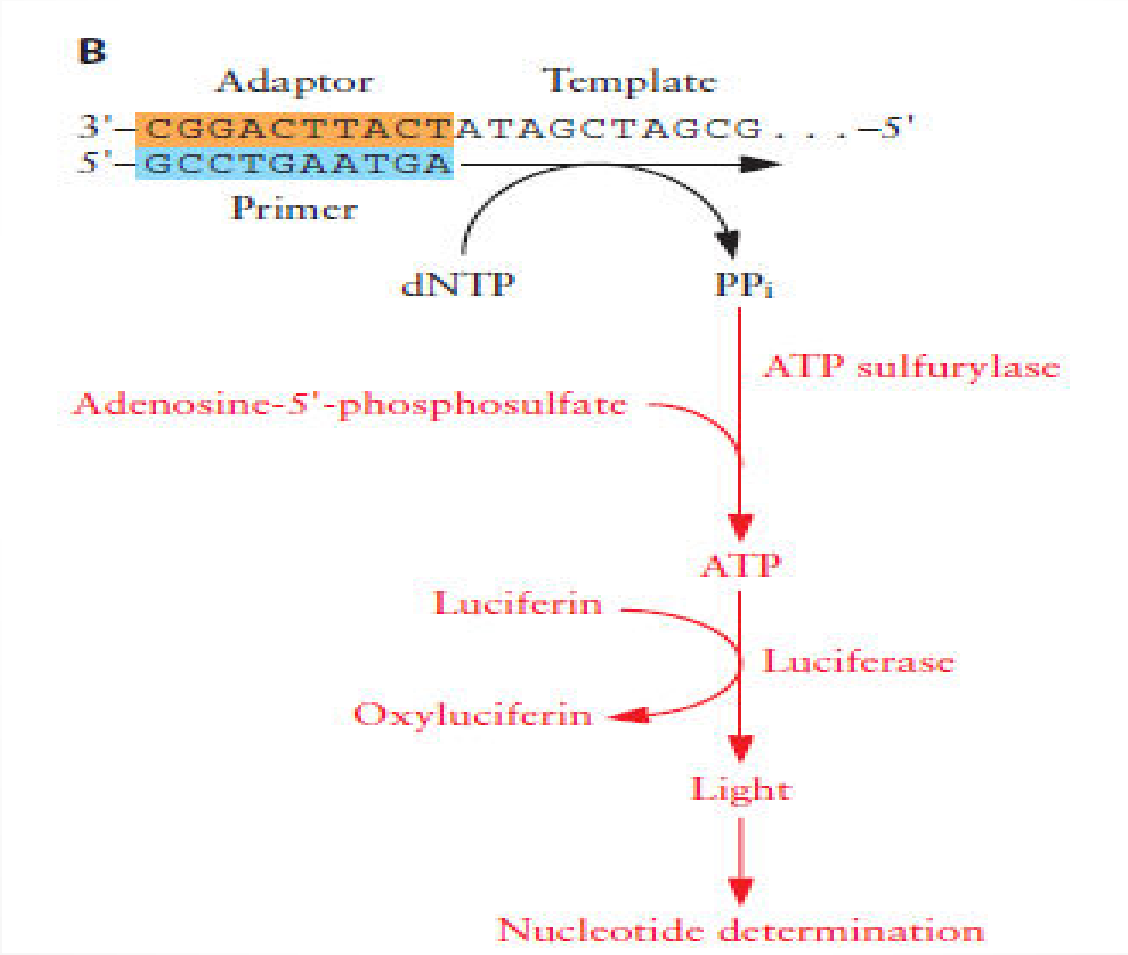

A
Growing strand
Template strand
A : T
G : C
Pyrophosphate

- A short DNA adaptor that serves as a binding site for a sequencing primer is first added to the end of the DNA template
- The pyrophosphate released following incorporation of a nucleotide is detected indirectly after enzymatic synthesis of ATP
- Pyrophosphate combines with adenosine-5′-phosphosulfate in the presence of the enzyme ATP sulfurylase to form ATP

- ATP drives the conversion of luciferin to oxyluciferin by the enzyme luciferase, a reaction that generates light

- Detection of light after each cycle of nucleotide addition and enzymatic reactions indicates the incorporation of a complementary nucleotide.

- sequences containing tracts of up to eight identical nucleotides in a row can be determined.


B
Adaptor
Template
3'–CGGACTTACTATAGCTAGCG . . .–5'
5'–GCCTGAATGA
Primer
dNTP
PPi
ATP sulfurylase
Adenosine-5'-phosphosulfate
ATP
Luciferin
Luciferase
Oxyluciferin
Light
Nucleotide determination

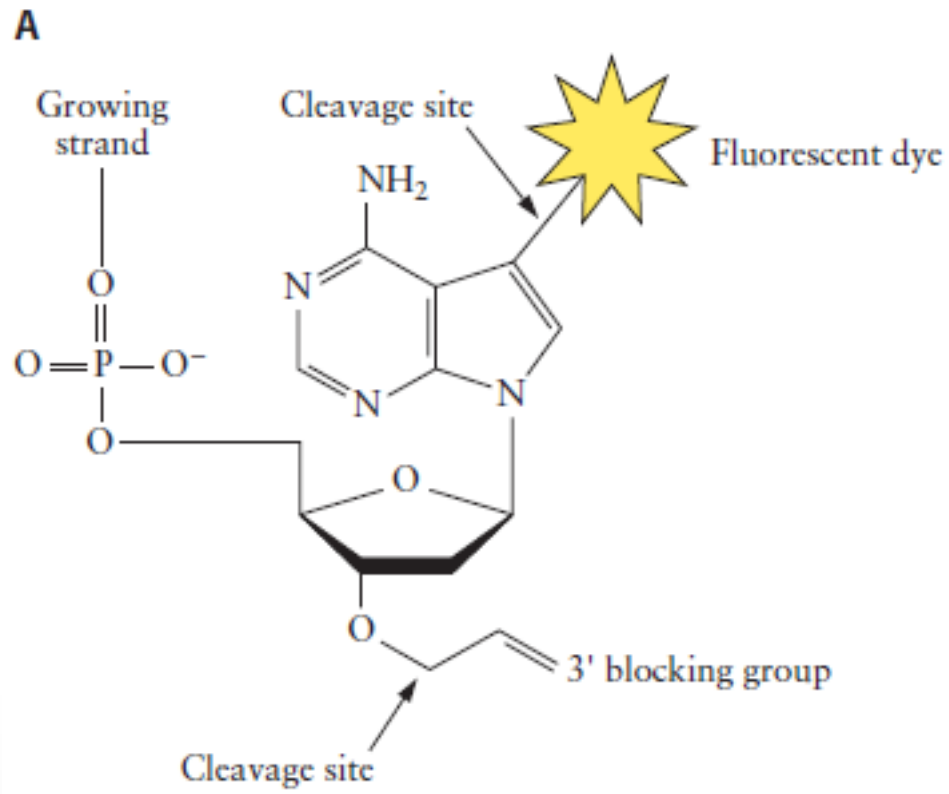
Sequencing Using Reversible Chain Terminators

Lecture no 54

- The sequence of a DNA fragment could be determined more rapidly if all the nucleotides were added together in each cycle

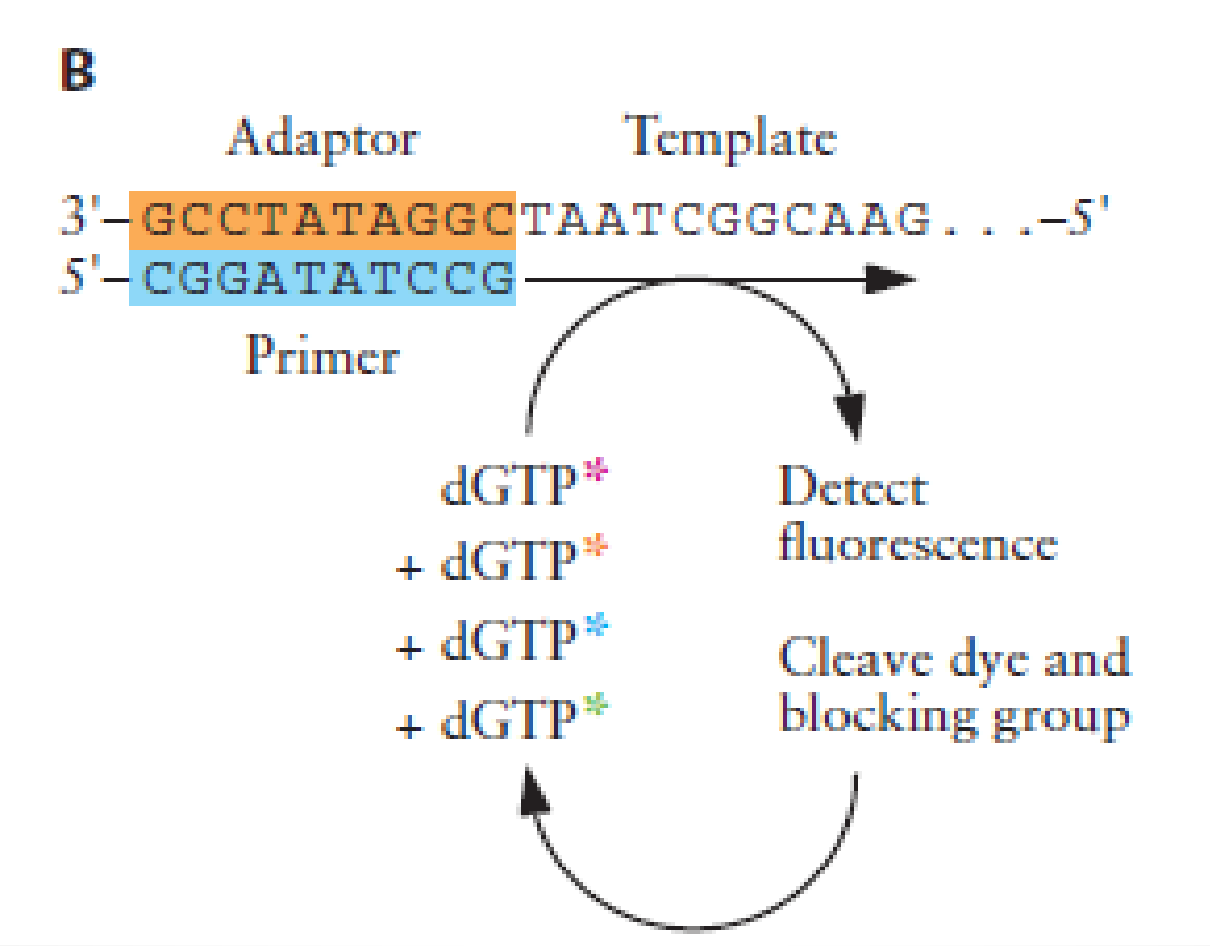
- Synthetic nucleotides known as reversible chain terminators have been designed to meet these criteria and form the basis of some of the next-generation sequencing-by-synthesis technologies.

- Reversible chain terminators are deoxynucleoside triphosphates with two important modifications:
- (i) a chemical blocking group is added to the 3′ carbon of the sugar moiety to prevent addition of more than one nucleotide during each round of sequencing, and
- (ii) a different fluorescent dye is added to each of the four nucleotides to enable identification of the incorporated nucleotide


A
Growing strand
Cleavage site
Fluorescent dye
NH2
N
N
N
O
O
O
O
O
O
O−
P
3' blocking group
Cleavage site

- The **fluorophore** is added at a position that does not interfere with either base-pairing or phosphodiester bond formation

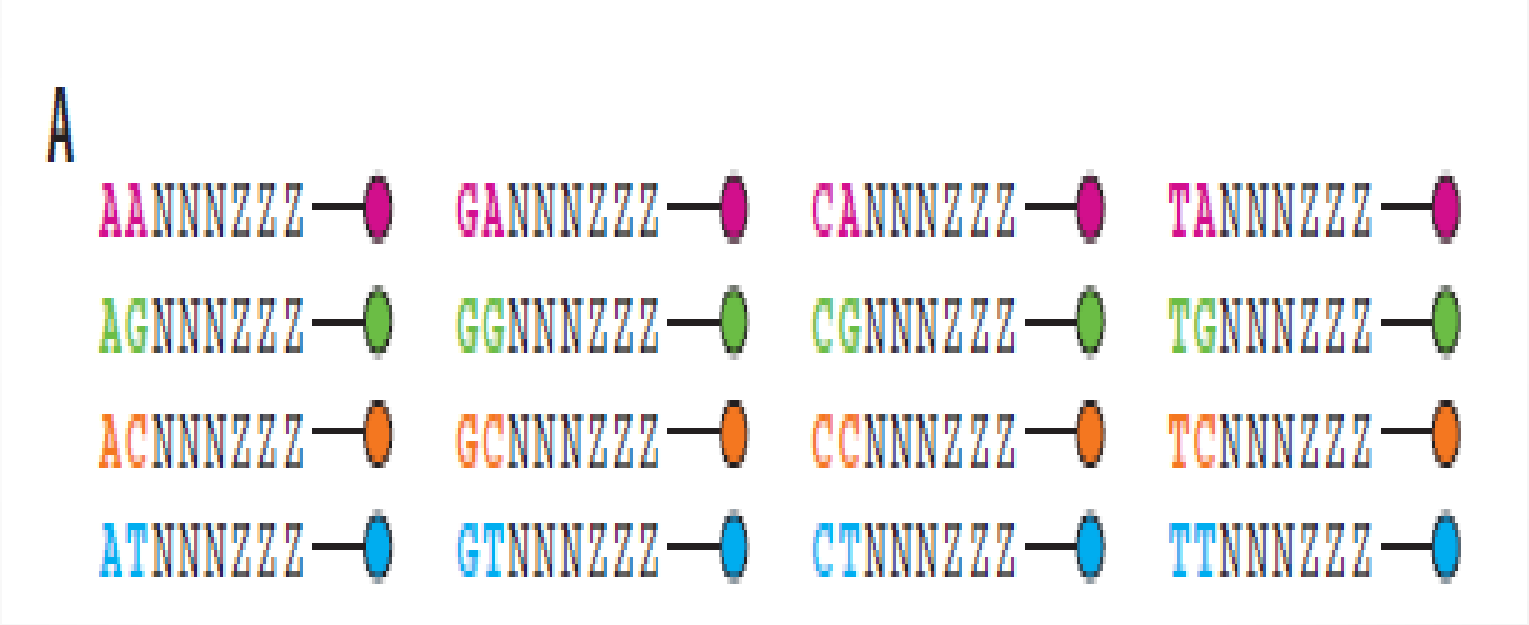
- After recording fluorescent emissions, the fluorescent dye and the 3′ blocking group are removed


B
Adaptor
Template
3'–GCCTATAGGCTAATCGGCAAG . . .–5'
5'–CGGATATCCG
Primer
dGTP*
+ dGTP*
+ dGTP*
+ dGTP*
Detect fluorescence
Cleave dye and blocking group

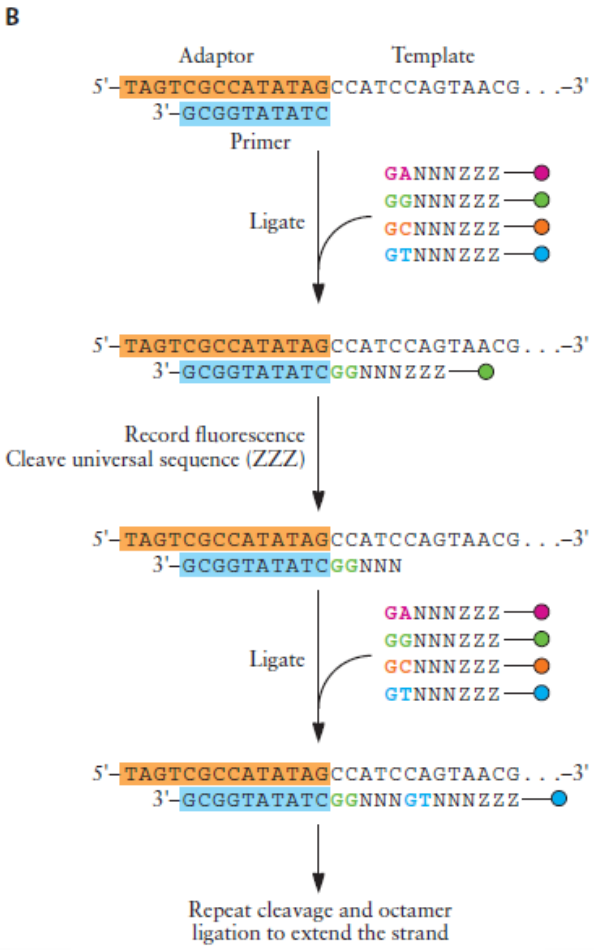
Sequencing by Ligation

Lecture no 55

- Sequencing by ligation extends the DNA strand by ligation of short oligonucleotides, in a template-dependent fashion, and utilizes the enzyme ligase rather than DNA polymerase.
- The oligonucleotides are eight nucleotides in length (octamers) with two known nucleotides at the 3′ (query) end, any nucleotide in the next three (degenerate) positions, and a sequence that is common (universal) to all of the oligonucleotides at the 5′ end


A
AANNNZZZ
GANNNZZZ
CANNNZZZ
TANNNZZZ
AGNNNZZZ
GGNNNZZZ
CGNNNZZZ
TGNNNZZZ
ACNNNZZZ
GCNNNZZZ
CCNNNZZZ
TCNNNZZZ
ATNNNZZZ
GTNNNZZZ
CTNNNZZZ
TTNNNZZZ

- A short nucleotide adaptor is joined to the 5′ ends of the DNA templates that are to be sequenced
- After denaturation of the template DNA, a primer binds to the adaptor sequence and provides a 5′ phosphate end for ligation to the 3′ hydroxyl end of an adjacent hybridized octamer.
- Sequential cycles of octamer ligation extend the complementary strand, and each cycle enables determination of the query dinucleotide sequence


B
Adaptor
Template
5'–TAGTCGCCATATAGCCATCCAGTAACG. . .–3'
3'–GCGGTATATC
Primer
Ligate
GANNNZZZ
GGNNNZZZ
GCNNNZZZ
GTNNNZZZ
5'–TAGTCGCCATATAGCCATCCAGTAACG. . .–3'
3'–GCGGTATATCGGNNNZZZ
Record fluorescence
Cleave universal sequence (ZZZ)
5'–TAGTCGCCATATAGCCATCCAGTAACG. . .–3'
3'–GCGGTATATCGGNNN
Ligate
GANNNZZZ
GGNNNZZZ
GCNNNZZZ
GTNNNZZZ
5'–TAGTCGCCATATAGCCATCCAGTAACG. . .–3'
3'–GCGGTATATCGGNNNGTNNNZZZ
Repeat cleavage and octamer
ligation to extend the strand

One ligation cycle consists of the following steps

- (i) Pools of different octamers, each tagged with a different fluorescent dye, are added along with T4 DNA ligase. Reaction conditions are set so that ligation occurs only if the bases at the 3′ end of an octamer (i.e., the query and degenerate nucleotides) are complementary to the template sequence.
- (ii) After washing away of the nonligated octamers and other components, the fluorescence signal is recorded. The identity of the dinucleotides in the query position is determined by the distinctive fluorescence emitted.
- (iii) The fluorescent dye is removed after each cycle by cleaving the terminal universal nucleotides. Cleavage provides a free end for ligation of another octamer in the next ligation cycle.

- To determine the nucleotide sequence of the intervening regions (i.e., between the identified dinucleotides), the primer and octamers are removed from the template strand, and the process is repeated using a new primer that is set back on the template DNA by one nucleotide from the first primer
- In total, the entire process of primer extension by serial octamer ligation is repeated five times, each time using a different primer that is offset from the previous primer by one nucleotide


C
Adaptor
Template
5'–TAGTCGCCATATAGCCATCCAGTAACGAC . . .–3'
3'–GCGGTATATCGGNNNGTNNNTGNNN . . .
Primer 1
5'–TAGTCGCCATATAGCCATCCAGTAACGAC . . .–3'
3'–AGCGGTATATCGNNNGGNNNTTNNN . . .
Primer 2
5'–TAGTCGCCATATAGCCATCCAGTAACGAC . . .–3'
3'–CAGCGGTATATCNNNAGNNNATNNN . . .
Primer 3
5'–TAGTCGCCATATAGCCATCCAGTAACGAC . . .–3'
3'–TCAGCGGTATATNNNTANNNCANNN . . .
Primer 4
5'–TAGTCGCCATATAGCCATCCAGTAACGAC . . .–3'
3'–ATCTGCGGTATANNNGTNNNTCNNN . . .
Primer 5

The Immune Response

Lecture no 56

Overview of Infection and Immunity

- Immunology is the study of the immune system
- The interaction of immune system cells, tissues, and molecules to infectious microbes elicits an **immune response**

Role of the immune system in health and disease

Defensive role	Consequence
Infection	Reduced immunity increases susceptibility to infection (e.g., HIV).
	Vaccination increases immunity and protects against infections.
Immune responses to tissue transplants and novel proteins	Barriers to transplantation and gene therapy
Tumors	Possible immunotherapy for cancer

Benefits of vaccination for some infectious diseases

Disease	Highest no. of cases (yr)	No. of cases in 2004	Decrease (%)
Diphtheria	206,939 (1921)	0	100.0
Haemophilus influenzae type b infection	20,000 (1984)	16	99.9
Hepatitis B	26,611 (1985)	6,632	75.1
Measles	894,134 (1941)	37	99.9
Mumps	152,209 (1968)	236	99.9
Pertussis	265,269 (1934)	18,957	96.8
Polio (paralytic)	21,269 (1952)	0	100.0
Rubella	57,686 (1969)	12	99.9
Tetanus	1,560 (1923)	26	98.3
TB	17,500 (1993)	14,511	96.7

Issues

- The immune response is the major barrier to successful organ transplantation, which is being used more frequently as a therapy for organ failure

- Abnormal immune responses are the causes of many inflammatory diseases with serious morbidity and mortality.

Antibodies

- Antibodies, protein products synthesized by B cells during an immune response.
- Antibodies have emerged as valuable reagents for the development of laboratory diagnostic technologies for both clinical research and clinical trials.
- Antibodies designed to block or eliminate potentially harmful molecules and cells are in extensive use for the treatment of immunological diseases, cancers, and other types of disorders.

Functions of the Immune System

Lecture no 57

Functions of the immune system

Function	Cell type
Immune recognition	Leukocytes (neutrophils, macrophages, and natural killer cells) and lymphocytes (T cells and B cells)
Contain and eliminate infection	Effector cells (T cells and B cells) and molecules (complement proteins, enzymes, and antibodies)
Immune regulation	Regulatory lymphocytes
Immunological memory	Memory lymphocytes

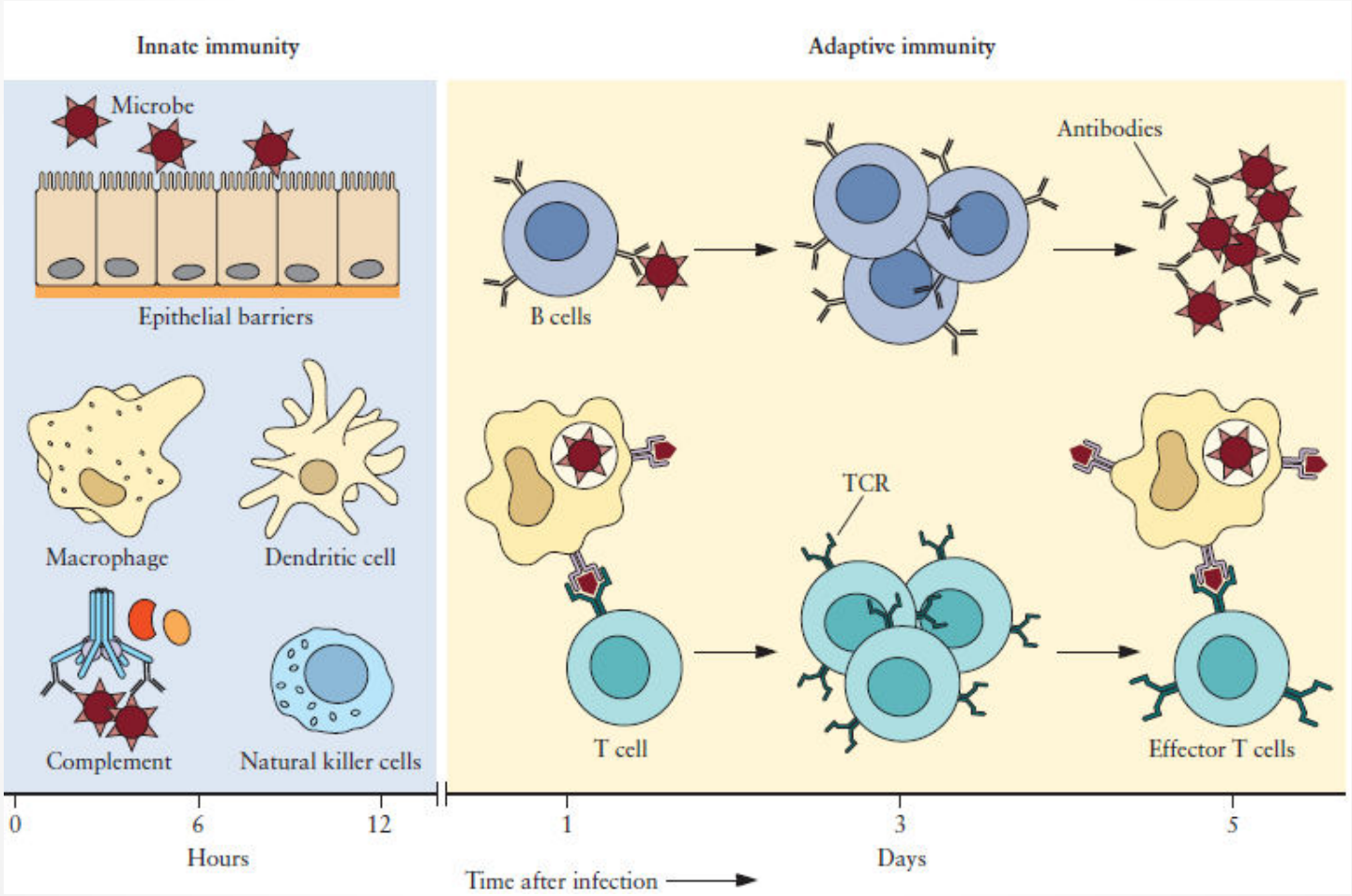
- A very significant challenge facing immunologists today is the identification of relevant cellular pathways and molecule that generate long-lasting immunity (i.e., memory) to pathogens, with the aim of identifying novel **immunotherapies** using cells and/or antibodies that eradicate such pathogens.

Efforts By: AARISH ALI

Types of immunity

- Two major types of immunity
- **innate immunity** provides a mechanism of host defense that mediates the initial protection against microbial infections.

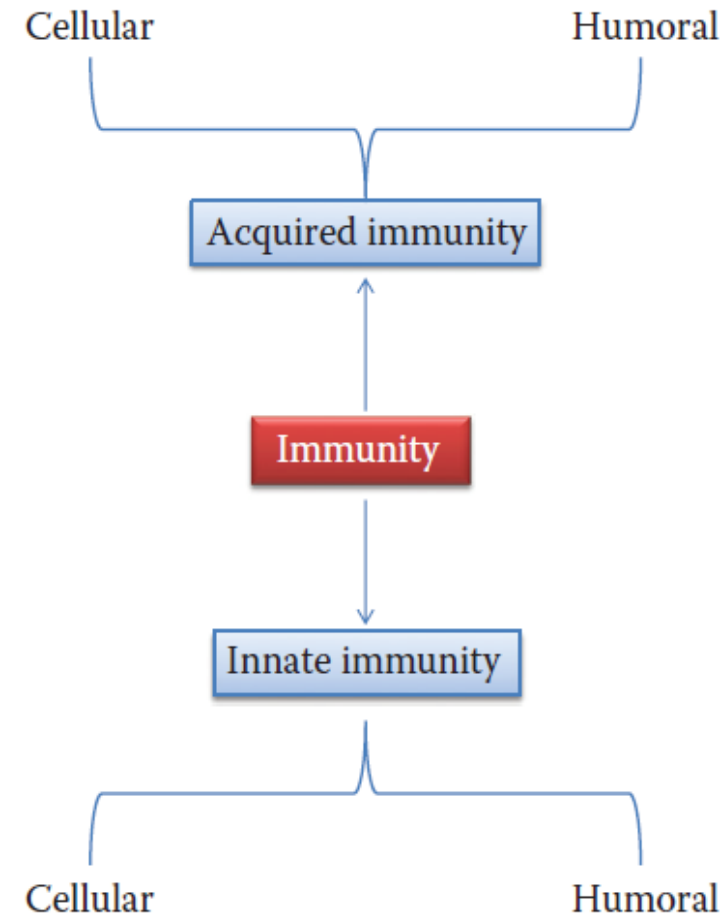
- **adaptive immunity**, which develops more slowly and confers specificity against a foreign antigen, mediates the later and more vigorous defense against infections.


Innate immunity
Adaptive immunity
Microbe
Epithelial barriers
Macrophage
Dendritic cell
Complement
Natural killer cells
B cells
Antibodies
TCR
T cell
Effector T cells
0
6
12
Hours
1
3
5
Days
Time after infection

Innate immune system

Lecture no 58

- The innate immune system contains cells and mechanisms that defend or fight against infection by other microorganisms in a generalized manner, which suggests that the cells of the innate system identify and respond to pathogens in a common way, and it does not convene long-lasting or protective immunity to the host


Cellular
Humoral
Acquired immunity
Immunity
Innate immunity
Cellular
Humoral

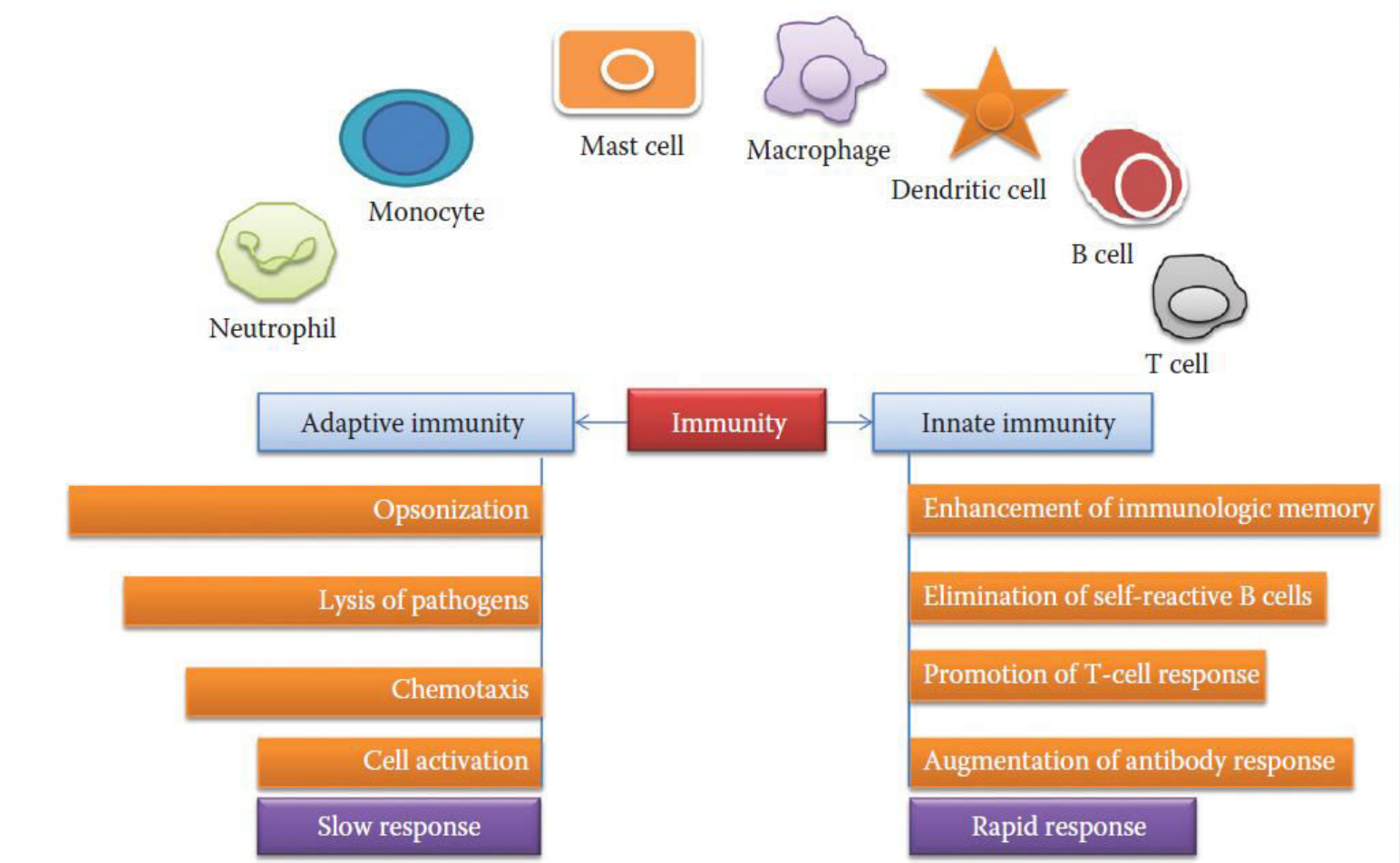
- It has been reported that innate immune systems offer immediate defense against infection and are found in all plants and animals.

- It has been found that the innate immune system generally consists of three components—membrane barrier, inflammation, and complement system—to protect the cells from infections

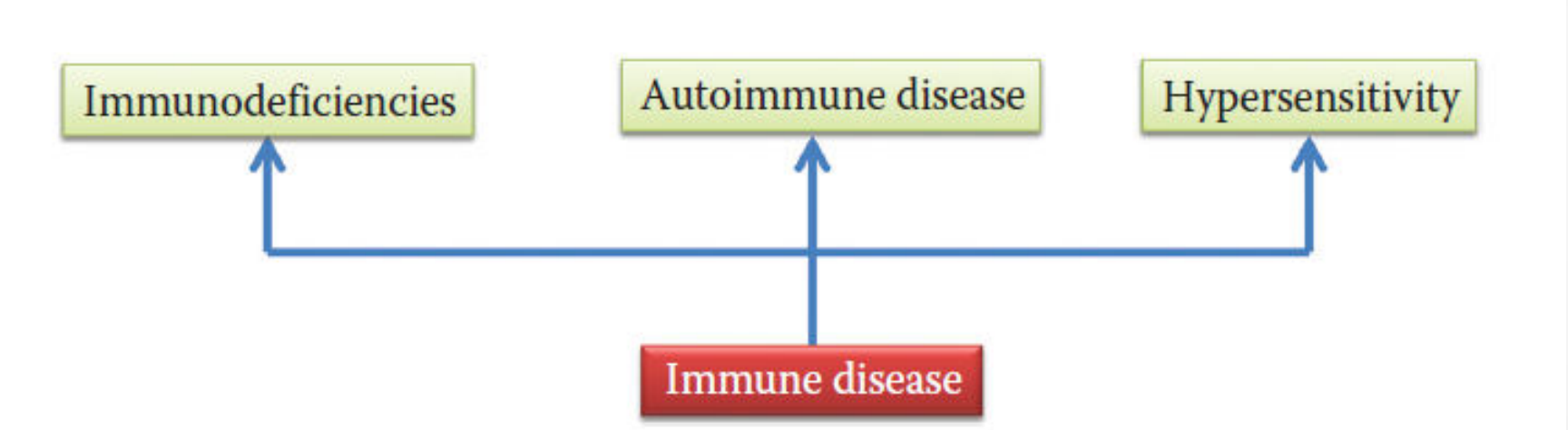
- In case pathogens can successfully evade the innate response, a host body possesses another layer of protection called the adaptive immune system, which can be activated by the innate response.

- At this point, the immune system familiarizes its response during an infection to increase its recognition of the pathogen.

Difference between innate and adaptive (acquired) immunity



Types of immune diseases



Cellular network of innate immune system

Lecture no 59

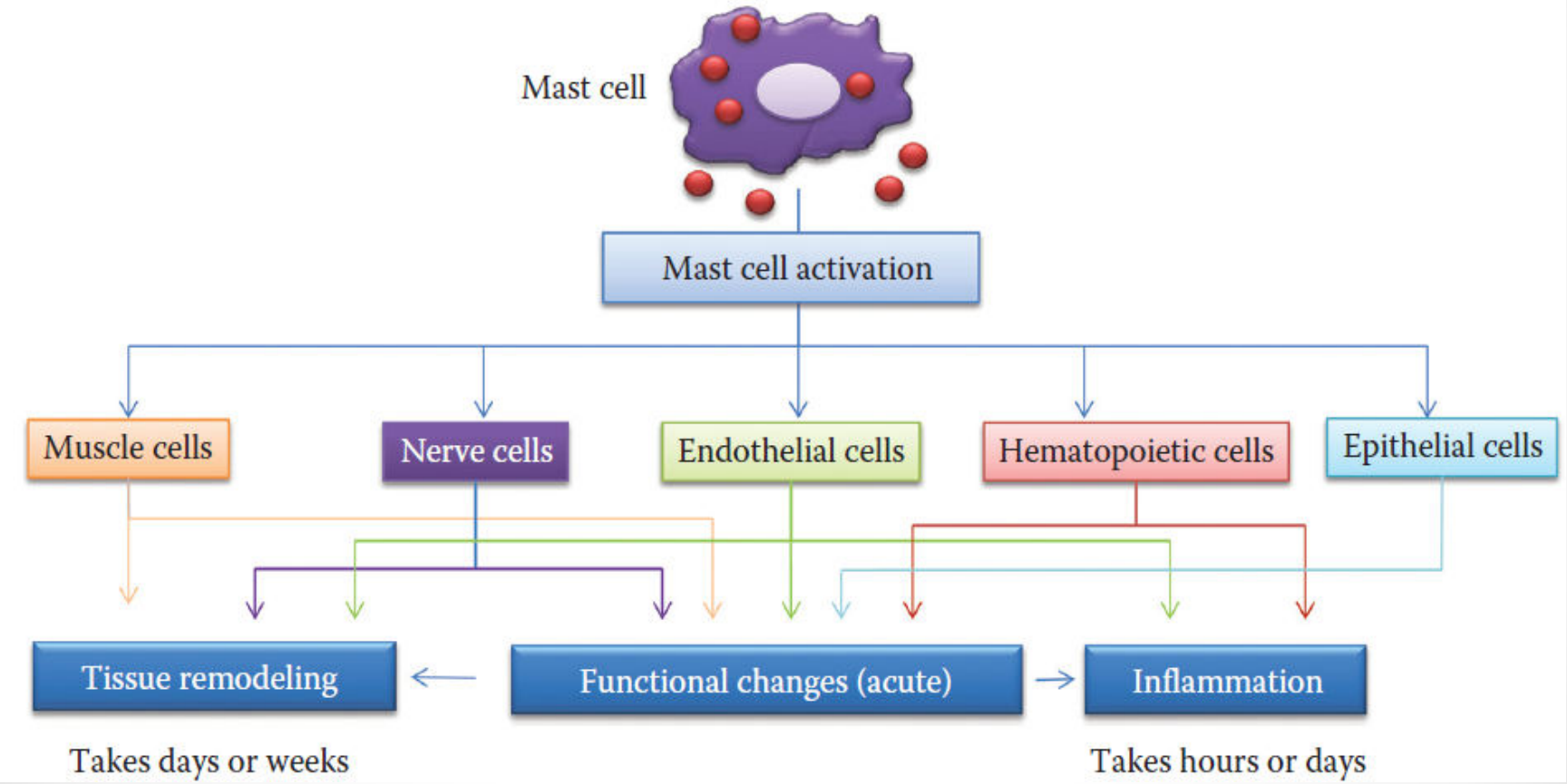
- The white blood cells (WBCs), identified as leukocytes, are different from other cells of the body because they are normally associated with a particular organ

- Leukocytes are the products of hematopoietic stem cells present in the bone marrow

Mast cells

- Mast cells are closely associated with defense against pathogens and wound healing
- Release some granules that are equipped with histamine and heparin and chemotactic cytokines into the host cell environment.
- It has been reported that histamines open blood vessels, causing irritation or inflammation, and recruit neutrophils and macrophages to defend the host cells

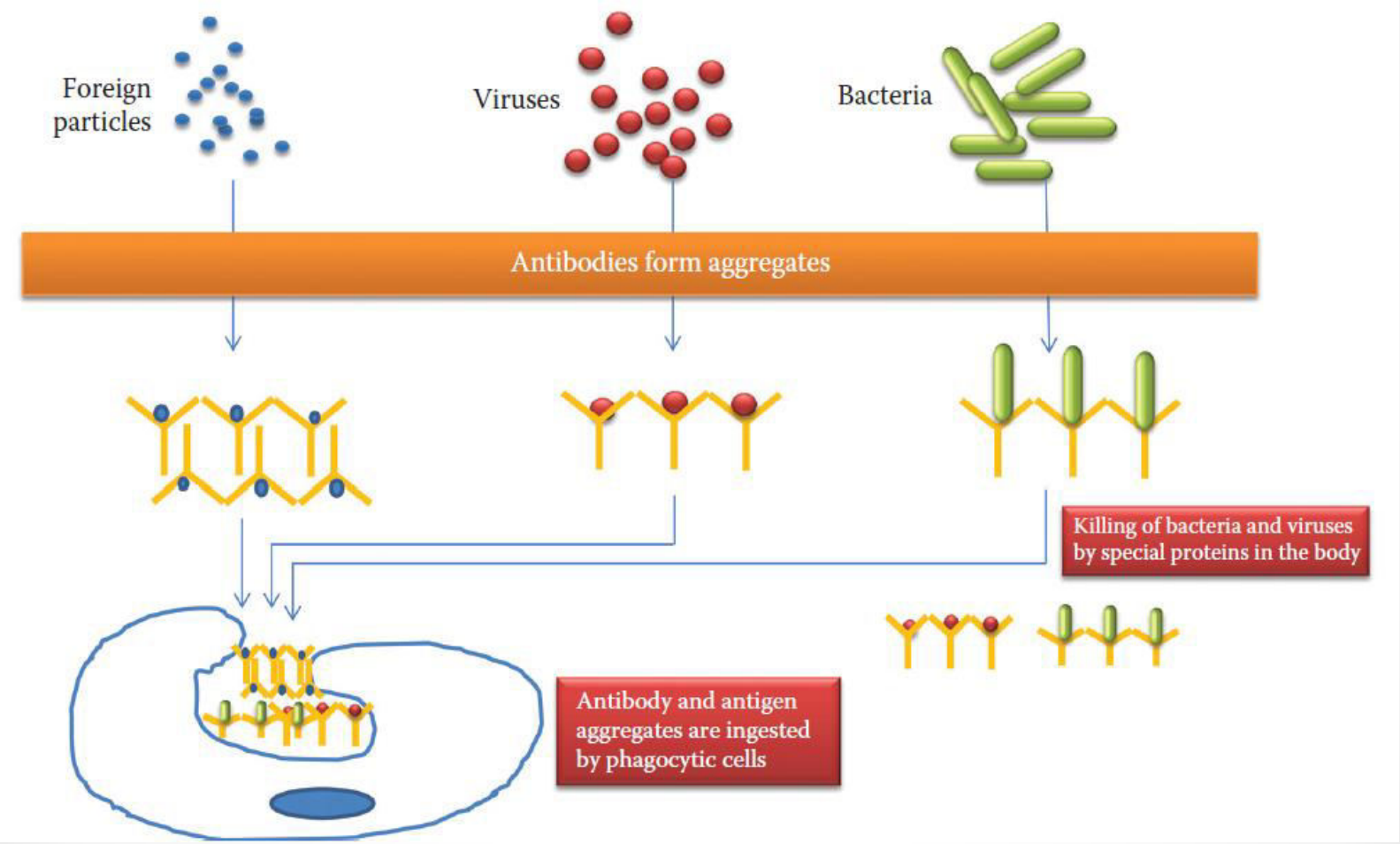
Mast cells and their role in innate and acquired immunity



Phagocytes

- Phagocytes are basically immune cells that engulf pathogens or particles to protect the body from any infection
- The lysosome, which contains enzymes and acids, starts killing and digesting the foreign particle or organism.
- It has been described that phagocytes generally patrol the human body searching for pathogens.
- There are various types of phagocytic cells reported in the human immune system that include neutrophils, macrophages, and dendritic cells (Figure 2).

Role of phagocytic cells in immune protection



Cellular network of innate immune system

Lecture no 60

Macrophages

- Macrophages are large leukocytes that are able to move outside the vascular system.
- It has been reported that these macrophages are the most efficient phagocytes, and can kill large numbers of bacteria or other cells or microorganisms.
- The attachment of anti-bacterial molecules to receptors of a macrophage terminates the bacterial population through respiratory spurt, causing the release of reactive oxygen species.
- It has been reported that pathogens can also stimulate the macrophage to produce chemokines, which command other cells to the site of infection.

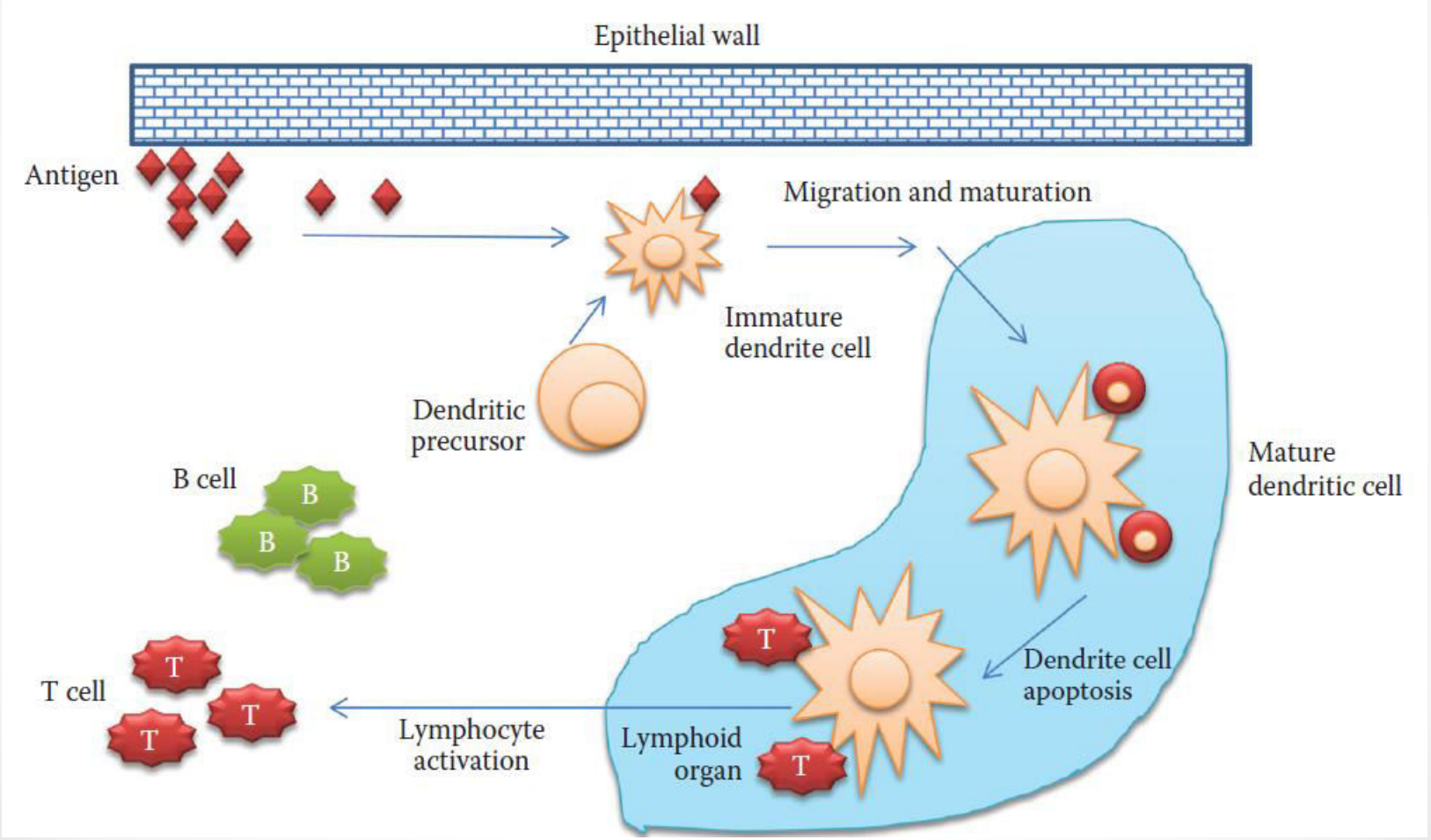
Neutrophils

- Because of the presence of granules in their cytoplasm, neutrophils, eosinophils, and basophils are known as granulocytes.
- Neutrophils are also known as polymorphonuclear cells (PMNs)
- It has been reported that neutrophil granules contain a variety of toxic elements that destroy or inhibit the growth of bacteria and fungi.
- Like macrophages, neutrophils attack pathogens by activating a respiratory spurt that contains oxidizing agents, including free oxygen radicals, hydrogen peroxide, and hypochlorite.

Dendritic cells

- Dendritic cells are phagocytic cells present in human tissues that are in contact with the external atmosphere, primarily the skin, where they are referred to as Langerhans cells, and the inner mucosal lining of the nose, stomach, intestines, and lungs.

Dendritic cells and their biological actions



Basophils and eosinophils

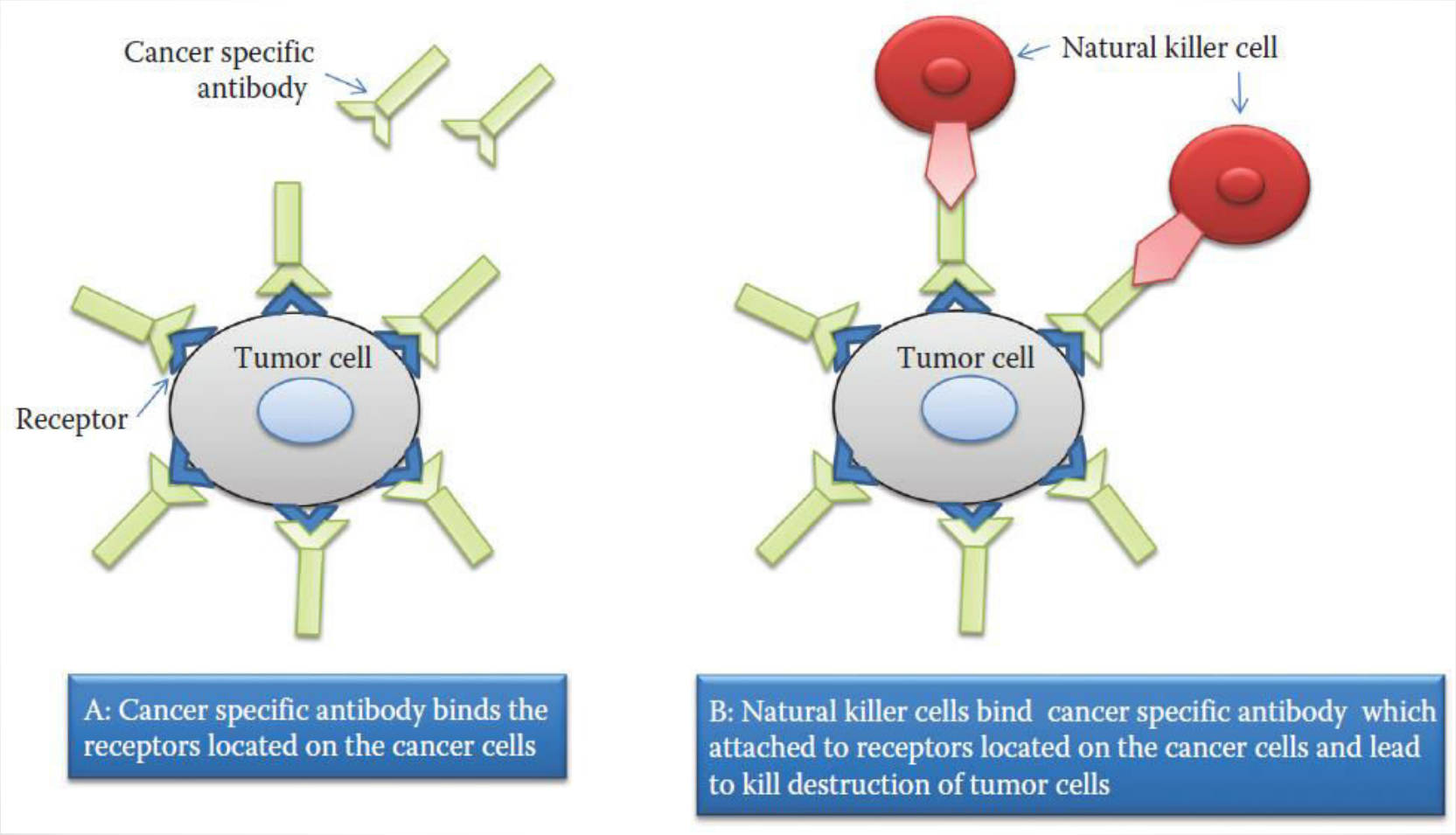
- Basophils and eosinophils are known to relate to the neutrophils.
- The basophils release histamine upon parasite entry, which are critical in the defense against parasites, and are also involved in allergic reactions.
- These basophils during activation secrete a highly toxic protein and free radicals that instantly kill bacteria and parasites; however, they are also known to cause cell and tissue damage during allergic-related reactions.

Natural killer cells

- Natural killer (NK) cells are an important member of the innate immune defense system that does not attack invading pathogens or microbes.

- In contrast to phagocytes, NK cells can terminate compromised host cells, such as cancerous cells or virus-infected cells. They are called NK cells because of an initial idea that these NK cells do not require activation to kill cells

NK cell destroying cancerous cells



Adaptive immune system

Lecture no 61

- The adaptive immune system is basically a defense system in the human body that remembers the entry and action of each and every pathogen

- Adaptive immune response is antigen specific and needs the recognition of specific antigens during antigen presentation

- The special adaptive capability of immune cells is maintained in the body by memory cells.

Lymphocytes

- The lymphocytes belong to the adaptive immune system.
- There are two different types of lymphocytes, B cells and T cells.
- These cells are generally derived from hematopoietic stem cells in the bone marrow.

- B cells take part in the humoral immune response, while T cells are mainly involved in cell-mediated immune response.
- Both B and T cells contain receptor molecules that distinguish specific targets.
- For example, T cells recognize a pathogen only after antigens are processed and presented in combination with major histocompatibility complex (MHC) molecule.

Types of T-cells

- 3 types
- Helper T-cells
  - Stimulate Cytotoxin T-cells
- Cytotoxin T-cells
  - Kill virally Infected Cell
- Suppressor T-cells
  - Suppress further killing of cells

T-Cells

Killer T cells

&

Helper T cells

Lecture no 62

Killer T cells

- Killer T cells are a subtype of T cells that can either kill cells that are infected with viruses and pathogens, or make them dysfunctional.
- It has been shown that killer T cells are activated when T cell receptor binds to antigens of another MHC cell
- The recognition of this MHC–antigen complex is aided by a coreceptor localized on the T cell, which is also called CD-8.

- It has been suggested that T cells travel throughout the body in hunt of cells where the MHC-I receptors contain this antigen.
- As soon as an activated T cell makes contact with such cells, it releases cytotoxins that form holes in the target cell's plasma membrane, and subsequently allow ions, water, and toxins to enter inside the cell.
- *granulysin* (a protease) persuades the target cell to undergo apoptosis.
- More significantly, destroying host cells by T cells is particularly important in preventing the replication of viruses.

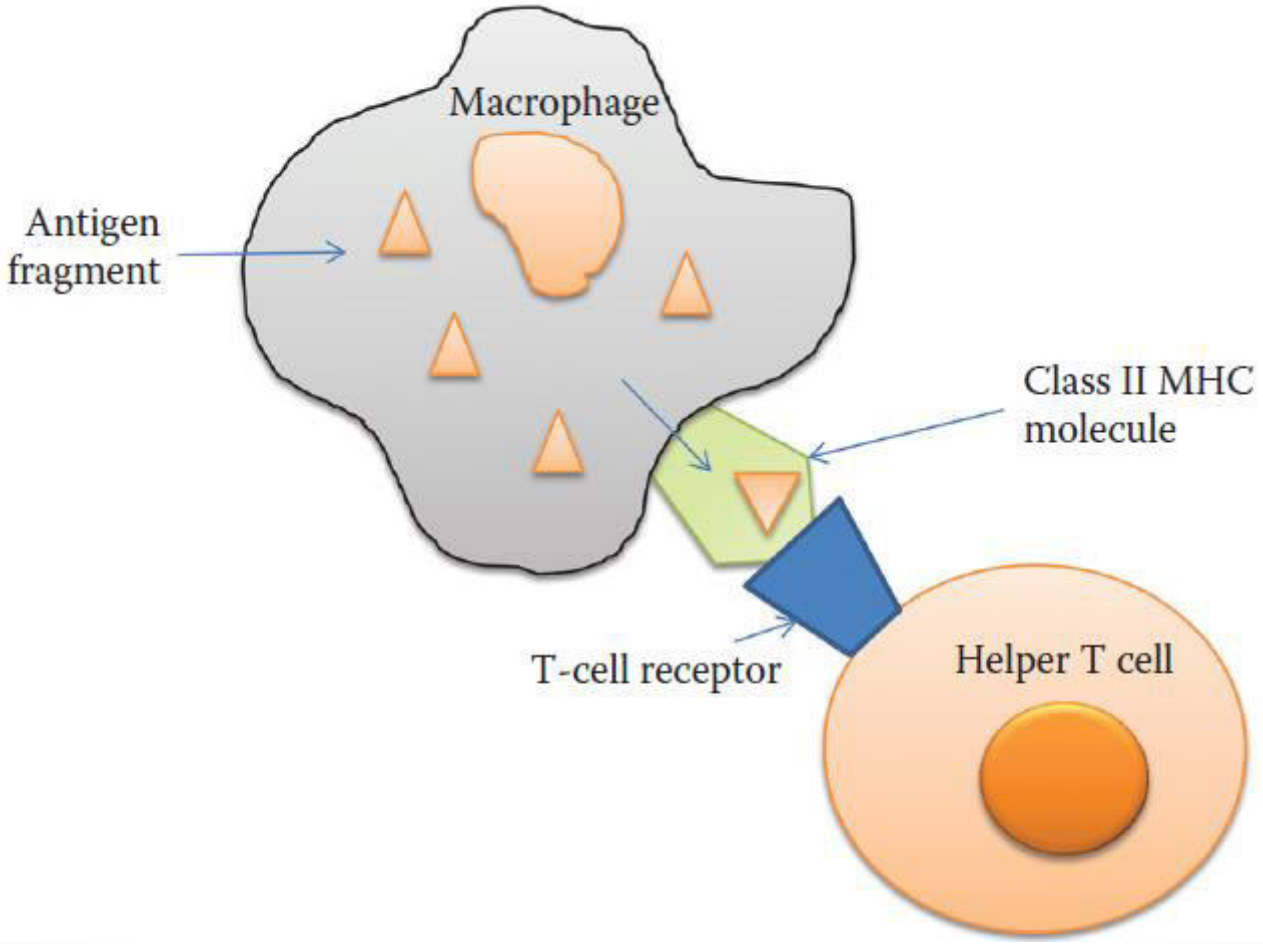
Helper T cells

- The activation of helper T cells can result in the release of cytokines as well as stimulation of macrophages such as killer T cells and B cells
- It has been reported that these helper T cells have no cytotoxic activity and do not kill infected cells
- Helper T cells express T cell receptors that identify antigens attached to Class-II MHC molecules.

- It has been reported that helper T cells have a weaker association with the MHC–antigen complex than killer T cells.
-
- Moreover, helper T cell activation also requires longer duration of assurance with an antigen-presenting cell.

- In addition, helper T cell activation results in upregulation of molecules expressed on the T cell surface, which provide extra stimulatory signals normally required to activate antibody-producing B cells.

Function of T helper cells



B lymphocytes and antibodies

Lecture no 63

- The B lymphocytes and antibody cells detect pathogens when the antibodies on their surface attach to a specific foreign antigen.

- It has been described that this antigen–antibody complex is taken up by the B cell and processed by proteolysis into peptides

- Later on, the B cells display these antigenic peptides on their surface of MHC Class-II molecules

- As soon as B cells are activated, they begin to multiply, and their offspring discharges millions of copies of the antibody that recognizes this antigen.

- Furthermore, these antibodies can circulate in blood and stick to pathogens expressing the antigen, destroying these pathogens with phagocytes.

Immunological memory

- When B cells and T cells are activated and begin to reproduce, some of their offspring can become long-lived memory cells. This process is called immunological memory.

- Memory cells can remember each and every specific pathogen encountered and can enhance a strong response if the pathogen is detected or enters again.

- This kind of immunological response is basically adaptive in nature, because it occurs during the lifetime of an individual.

- It has been reported that immunological memory can be classified as passive short-term memory or active long-term memory

Passive short-term memory & Active long-term memory

Lecture no 64

Passive memory

- It has been reported that newborn infants have no prior exposure to pathogens and are particularly vulnerable to infections.
- During this critical period of child development, the mother provides immunity against any kind of infection by producing a particular type of antibody known as immunoglobulin-G, which is transported into the baby directly across the placenta.
- These antibodies help the infant to fight against any infection.

- Development of immunity between the mother and child is called passive immunity because the fetus does not make any antibodies on its own, it only borrows them from the mother.

- It has been reported that passive immunity is only for a short duration of time, up to several months.

- Interestingly, now, it is possible to induce protective passive immunity in the infant by artificial ways.

Active memory and immunization

- It has been suggested that long-term active memory is acquired following infection by activating B cells and T cells.
- Active immunity can be produced by vaccination.
- The main principle behind vaccination is called immunization where a patient is injected with an antigen developed from a nonactive or killed form of pathogens to inspire the body to develop specific immunity against that particular pathogen.

Immune diseases

Lecture no 65

- The human immune system is remarkable in protecting the body from illness

- Under disease conditions, the immune system may fail to defend the body against various infections. This is called an immune-compromised immune system.

- This condition makes the body vulnerable to various viral, bacterial, or fungal infections..

- The immune-related diseases can be classified into three broad categories

- ❑ Immunodeficiency
- ❑ Autoimmunity
- ❑ Hypersensitivities

Immunodeficiency

- Immunodeficiency is a physiological condition where the immune system's capacity to fight infectious disease has become dysfunctional or entirely absent.
- In most cases, immunodeficiencies are acquired diseases, but some people are born with defects in the immune system, which is known as a primary immunodeficiency.
- It has been reported that a person who has an immunodeficiency of any kind is considered to be immune-compromised

Primary immunodeficiency

- A number of rare diseases have been reported susceptible to infections from childhood onward and many of these diseases are hereditary and autosomal recessive or X-linked in nature.

- There are more than 80 primary immunodeficiency syndromes reported in humans and they are generally grouped by their malfunctioning of lymphocytes or granulocyte cells.

- The treatment of primary immunodeficiencies depends on the nature of the defect, and may involve antibody infusions, long-term antibiotics, and stem cell transplantation.

Acquired immunodeficiency

- An immune deficiency that is caused due to external processes is called secondary or acquired immunodeficiency.
- The common causes for secondary immunodeficiency are malnutrition, aging, and particular medications
- It has been reported that there are many specific diseases that directly or indirectly impair the immune system
- Immunodeficiency is also the hallmark of AIDS, caused by HIV.

Autoimmunity

Lecture no 66

- In a healthy person, immunity generally protects the person from all infectious diseases, but in some individuals, it works against itself and destroys the person's immune system.

- This condition is generally known as an autoimmune disease. It has been suggested that any disease fallouts from such an abnormal immune response are called an autoimmune disease.
-
- Autoimmunity is frequently caused by a deficiency of germ cell development that results in the development of a condition where body cells work against their own cells and tissues.

- There are several examples of autoimmunity disease such as celiac disease, Churg–Strauss syndrome, Hashimoto's thyroiditis, Graves' disease, idiopathic thrombocytopenic purpura, diabetes mellitus type 1, systemic lupus erythematosus, Sjögren's syndrome, and rheumatoid arthritis (RA).

- Autoimmune diseases can be classified into systemic autoimmune disease and localized autoimmune disorders, depending on the clinical and pathological features of each disease.

- The examples of systemic autoimmune diseases are Sjögren's syndrome, scleroderma, RA, systemic lupus erythematosus, and dermatomyositis. It has been proposed that these disease conditions incline to be associated with autoantibodies to antigens, which are not tissue specific.

Name of Syndromes That Affect Various Body Organs

Name of syndrome	Affected organ/tissue
Diabetes mellitus type 1	Pancreas
Celiac disease, pernicious anemia	Intestine
Pemphigus vulgaris, vitiligo	Skin
Autoimmune hemolytic anemia, idiopathic thrombocytopenic purpura	Blood
Myasthenia gravis	Brain

Classifications of cell transplantation

Lecture no 67

- In cell transplantation, the cells or tissues can be classified based on their utilization in different parts of the body.

- We have given a few examples where cell-based transplantation is effectively used in clinical research.

Autologous cells

- The autologous cell transplantation is basically a transplantation in which cells obtained from patients can be transplanted into the same patient.

- One significance of the autologous transplantation is that there is a high rate of success as the transplanted cells easily can be accepted by the patient's body and there are few problems associated with ejection and pathogen transmission

- A few shortcomings in the autologous cell transplantation as the availability of required cells to be transplanted are found in very low in numbers and also genetic disease-suitable autologous cells are not available.

- Another disadvantage of autologous cell transplantation is that older patients do not have sufficient numbers of autologous cells to be used for cell transplantation.

- Autologous cells also must be cultured from samples before they can be used and these are lengthy procedures, so autologous cell transplantation might take a long time.

- In recent times there has been a trend toward the use of human mesenchymal stem cells (MSCs) isolated from bone marrow and fat cells.

- These MSCs can be differentiated into a variety of tissue types including bone, cartilage, fat, and nerve cells. A large number of cells can be easily and quickly isolated from a fat source, consequently opening the potential for large numbers of cells to be quickly and easily obtained.

Classifications of cell transplantation

Lecture no 68

Allogeneic cells

- Allogeneic cells come from the body of a donor of the same species; for example, the cells from one person (donor) can be used by another person (patient).
- There are some ethical restraints on the use of human cells for *in vitro* studies, however, the use of dermal fibroblasts from human foreskin has been demonstrated to be clinically safe and consequently a viable choice for tissue engineering of skin.

Xenogeneic cells

- The cells which are isolated from two different species are called as xenogeneic cells; for example, the cells from one species (sheep) can be used in another species (human).

- In particular, animal cells have been widely used in experiments aimed at the construction of cardiovascular implants which later can be used in humans.

Syngeneic or isogenic cells

- This type of cell isolates from genetically identical individuals, such as twins, clones, or extremely inbred research animal models.

ORGAN TRANSPLANTATION

Lecture no 69

ORGAN TRANSPLANT SUCCESSES AND FAILURES

- It has been over 50 years since the first successful human organ transplant was done—a kidney transplanted from one identical twin to another—and the procedure has become almost routine. The list of organs transplanted includes not just kidneys, but hearts, livers, lungs, and the pancreas.
- The number of transplants performed is impressive, considering both how complicated the surgery is and how strong a fight the immune system launches against the new organ.

- Because human beings have two kidneys and can function just fine with only one, kidney transplants can come from living donors.

- In recent years, methods to transplant just a part of a liver have succeeded because the liver can repair and regenerate itself. A few hundred people each year, often children, receive a lobe of a liver from a family member.

- Human-to-human organ transplantation has risks. The human immune system is finely tuned to recognize and destroy invaders
- A major risk exists if antibodies to the donor cells are present in the recipient's blood at the time of transplant.
- This rapid antibody reaction to the cells lining the organ blood vessels is called **hyperacute rejection**.
- The ABO blood group system, critical in blood transfusion, is also important for some, but not all, transplanted organs. An organ donor must be a suitable ABO type for the patient or ABO antibodies in the recipient's blood will cause hyperacute rejection.

- Scientists have discovered and developed a number of **immunosuppressive drugs**
- **Corticosteroids** and the cancer chemotherapy drug azathioprine were the first drugs used to suppress the immune system for organ transplants.
- In the 1970s, the introduction of **cyclosporine**, the product of a soil fungus, dramatically improved one-year graft and transplant patient survival.

- Other widely used chemical immunosuppressive drugs include tacrolimus, mycophenolate mofetil, and sirolimus.

- Several monoclonal antibodies that remove or shut down T lymphocytes (such as muromonab, daclizumab, and basiliximab) are used to prevent or treat rejection.
- A preparation of animal antibodies raised against human lymphocytes is still used for the same purpose.
- The success of organ transplantation is measured by organ and patient survival. Over 80% of kidney patients and 65% of their grafts survive for five years.

TAKING ORGANS FROM OTHER ANIMALS

- Because the transplant waiting list continues to grow, scientists have explored nonhuman animals as a possible source of organs.

- Despite the shock and repulsion it sometimes causes, the idea of **xenotransplantation**, or transplantation of an organ from one species to another, deserves some serious consideration

Transgenic organisms

Lecture no 70

- Transgenic organisms are genetically modified organisms (GMOs) which have inserted DNA of different species origin.
- The inserted DNA/gene is known as transgene.
- Transgene is introduced into the organism, using recombinant DNA technology.
- The gene must be transmitted through the germline so that every cell, including germ cells, of the body contains the same modified genetic material.

Basic Requirements

The process requires some basic steps for successful transfer:

- The extraction of nucleic acid (DNA/RNA). It is the first step in the genetic engineering process.

- After DNA extraction and selection of suitable gene, the next step is gene cloning . Gene cloning has four stages: (1) generation of DNA fragments, (2) ligation to a vector, (3) propagation in a host cell, and (4) selection of the transformed host.

- Gene is designed in such a way that it is under appropriate promoter and correct regulatory elements. Selectable marker gene may be incorporated for selection of the transformants from non-transformants.

- Incorporation of reporter gene allows detection of gene expression in transformed cells and enhancers are essential for efficient transcriptional control.

- The gene of interest (GOI) is ligated to selectable marker gene and reporter gene. It is helpful for screening, detection, and identification of the transgene containing host cell. Reported genes may be β-glucuronidase gene (GUS) or green fluorescent protein or luciferase gene. GUS produces blue product when it acts on its substrate, and thus cells appear blue, whereas GFP and luciferase transformed cells emit light when excited with appropriate wavelength of light

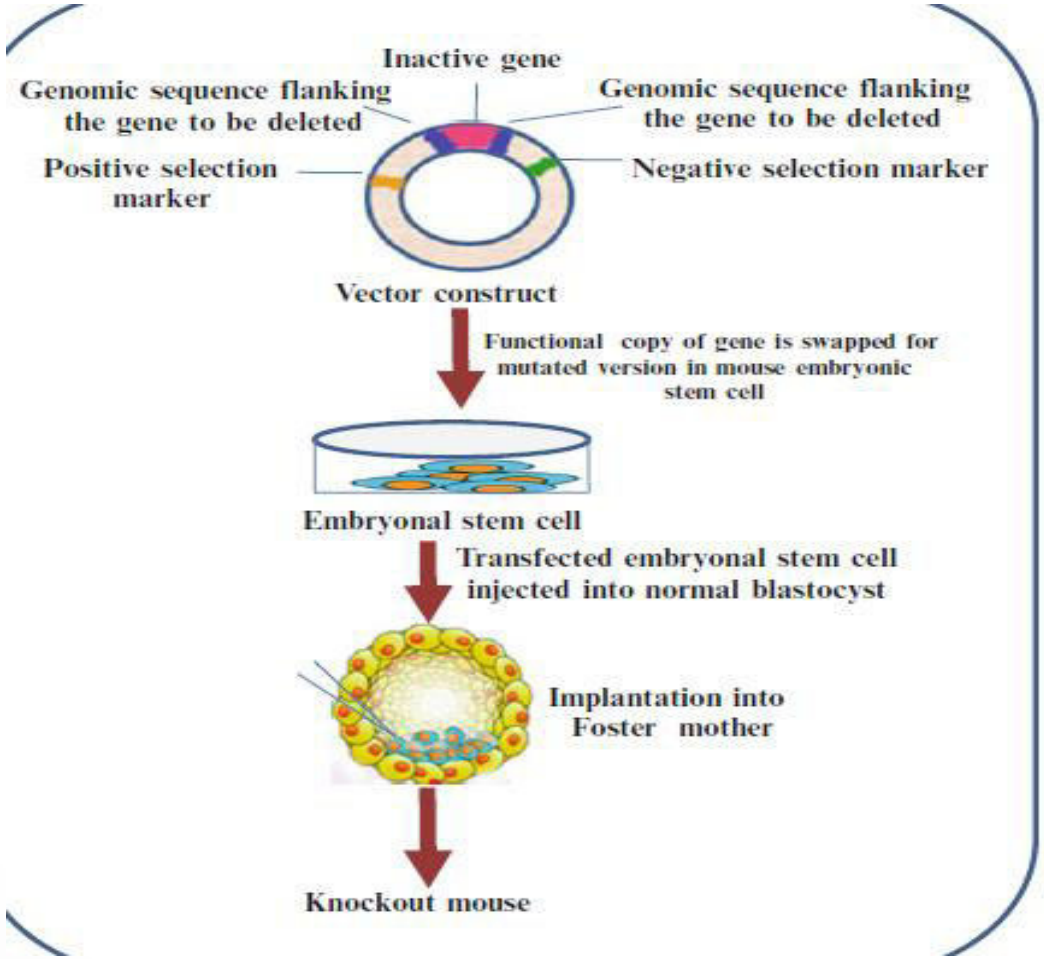
Production of Transgenic Animals

Lecture no 71

- There are two basic strategies for producing transgenic animals.
- These are the so-called “gain of function” or “loss of function” transgenics.
- The “gain of function” relies on addition of a cloned fragment of DNA to an animal’s genome to obtain new expression of a gene product that did not previously exist in that cell or tissue type.

- “Loss of function” is a more direct way to determine the function of a lost/inactive gene after its alteration

- For example, the mouse is obese or diabetic. These types of mice are often called knockouts, and from several years now, it has been possible to generate “ knockout” mice , which carry specifically defined mutations in the gene of interest using gene targeting in mouse embryonic stem cells .


Inactive gene
Genomic sequence flanking the gene to be deleted
Genomic sequence flanking the gene to be deleted
Positive selection marker
Negative selection marker
Vector construct
Functional copy of gene is swapped for mutated version in mouse embryonic stem cell
Embryonal stem cell
Transfected embryonal stem cell injected into normal blastocyst
Implantation into Foster mother
Knockout mouse

Transfer of Transgene in the Animal

Lecture no 72

- The essence of this technology resides in successful gene transfer in the intended animal and its insertion in the genome.

- The insertion of gene in the genome can be random or targeted.

- There are several methods of producing transgenic animals:

❑ DNA microinjection,

❑ retrovirusmediated gene transfer,

❑ embryonic stem cell -mediated transfer, and

❑ liposomemediated DNA transfer into cells and embryos.

- A few modifications are also there for creating transgenics:

- ❑ Electroporation of DNA into the sperm, ova, or embryos and

- ❑ Nuclear transfer with somatic or embryonic cells.

Pronuclear Microinjection

Lecture no 73

- Microinjection was the first technique used to generate transgenic mice by successful gene transfer using SV40 DNA directly into blastocoel cavity of preimplantation embryos and then their implantation in the foster mother.
- Eventually, the gene of interest may randomly integrate in the embryonic genome of some embryonic cells and ultimately contributes to germline. The technique was then extended to other animals (livestock).

- The method has poor efficiency and usually results in a high percentage of mosaics in which not all cells of the animal contained the transgene.
- The time and cost of screening for germline transmission from the major mosaic animals prohibits the generation of more transgenic animals.
- The method can also lead to high variability in transgene expression between animals due to mosaicism, variable efficiency of transgene integration, and chromosomal position effects.

- The major advantage of this method is its applicability to a wide variety of species.
- This method involves the direct injection of the gene construct into the cells to ensure guaranteed successful transfer of transgene to target cells.
- Presently the established technique involves the injection of DNA into the pronuclei of the fertilized egg. The host cell is held in position with a suction pipette and DNA is transferred into the nucleus through a fine needle.

- Thereafter the embryos with the transgene are cultured in vitro up to the morula stage and then transferred to pseudopregnant foster mothers.
- The resulting animal may be transgenic or chimeric for transgene insertion.
- Five to 40 % of mice developing from manipulated eggs contain the transgene. Once the transgene is transmitted through the germline, it tends to be stably inherited over many generations.
- The transgene insertion is accompanied by deletions and rearrangements occurring in genomic DNA.

- The limitation of microinjection is that it only allows for the random addition of exogenous DNA rather than targeting to specific sites.
- DNA targeting is necessary in generating gene knockouts, for instance, to model human diseases.
- The success of pronuclear microinjection is evident in the generation of transgenic pigs, goats, sheep, and cattle; but its limitations have hindered the progress of transgenics in livestock.

DNA Transfer Through Virus

Lecture no 74

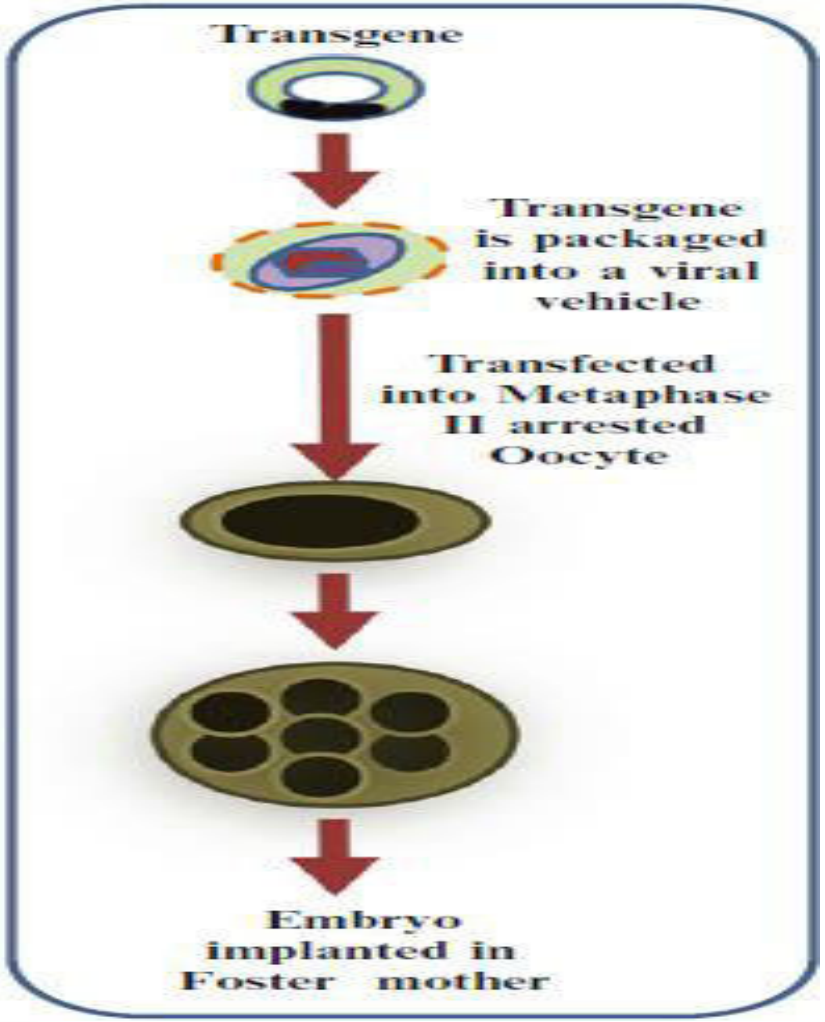
- Gene transfer mediated by means of a carrier or vector, generally, a virus or a plasmid, provides increased probability of expression.
- Virus has natural ability to stably introduce DNA to the host upon infection; therefore recombinant retroviruses and replication defective adenoviruses can be efficiently used as vectors.
- Retrovirus is efficiently capable of infecting dividing cells.

- Initial attempts to produce transgenics by retroviral infection of early embryos invariably resulted in genetic mosaics caused by multiple insertion sites of the transgene.
- The short window of opportunity for the viral preintegration complexes to reach the embryonic chromatin during M-phase explains the delayed viral insertion, resulting in cell lineages with different insertion sites or no insertion at all.

- A significant advancement of this technique has been achieved recently by exposing metaphase II oocytes to transgene containing retrovirus.

- Arrested oocytes are particularly appropriate because they have undergone nuclear envelope breakdown and remain at metaphase II for a longer period compared with the M-phase of somatic cells.

Virus-mediated transgenic production



- This maximizes the probability of preintegration complexes gaining access to the oocyte chromatin.

- The infected embryos are implanted in the uterus of foster mother to get germline transfer of transgene.

- Adenovirus represents an alternative to retrovirus as vectors to insert recombinant DNA into the mammalian germline.

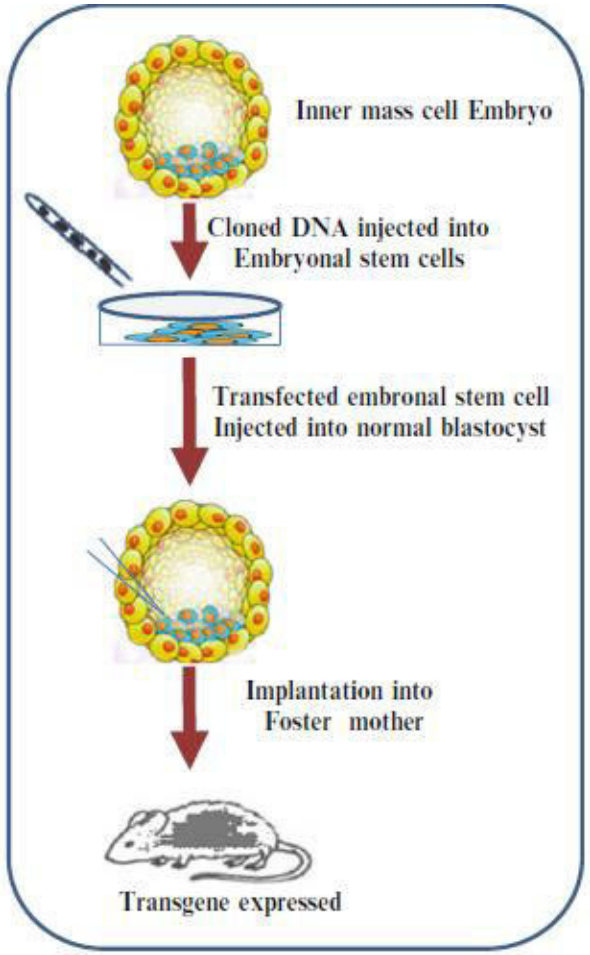
- The disadvantages of this technique are the possible interference of viral regulatory elements with the expression of genes and the susceptibility of virus to de novo methylation resulting in silencing of the genes.
- Though viruses are highly adapted to the process of gene transfer however, using virus as the vector may also produce health hazards as they may potentially arouse cancer/leukemia .

Embryonic Stem Cell Mediated Gene Transfer

Lecture no 75

- Embryonic stem cells are obtained from inner cell mass of blastocyst.
- This method involves prior insertion of the desired DNA sequence by homologous recombination into an in vitro culture of embryonic stem (ES) cells, and the transformed cells can be selected using standard markers.
- Cells containing the desired DNA are incorporated into the blastocoel of host's embryo at the blastocyst stage, where they mix with inner cell mass (Fig.1).

Embryonic stem cell-mediated gene transfer



- Unlike the other two methods, which require live transgenic offspring to test for the presence of the desired transgene
- This method allows testing for transgene at the cell stage.
- This is advantageous as there is no convenient way to select for eggs or embryos that have taken up foreign DNA.

- ES cell-mediated gene transfer is the method of choice for gene inactivation, the so-called knockout method.
- This technique is of particular importance for the study of the genetic control of developmental processes.
- This technique works particularly well in mice. It has the advantage of allowing precise targeting of defined mutations in the gene via homologous recombination.

Sperm-Mediated Transgenesis

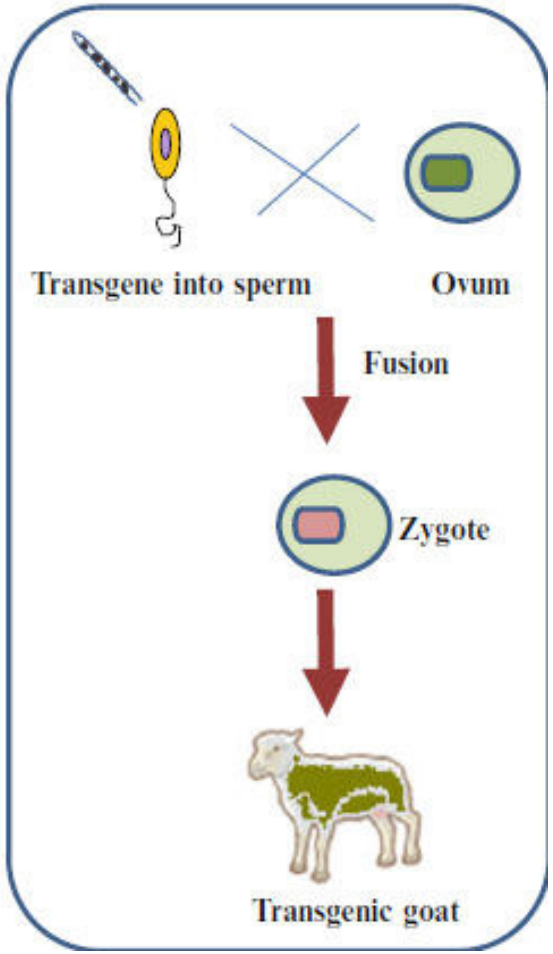
Lecture no 76

- The ability of sperm cells to carry exogenous DNA spontaneously into the oocyte during fertilization was first reported by Brackett and coworkers in 1971.
- However, the concept of sperm-mediated transgenesis rested for 18 years until scientists reported the use of spermatozoa as DNA carriers to produce transgenic mice.
- The scientific community was challenged by an unsuccessful attempt to replicate the experiment met this work with skepticism.

- Transgene delivery by sperm cells was used to produce transgenic animals in a wide variety of species, including cattle, pigs, rabbits, frogs, and zebra fish.
- Sperm-mediated transgenesis is the most straightforward technique to produce transgenic animals.
- The spermatozoa are incubated with the DNA containing the gene of interest followed by in vivo or in vitro insemination.

- DNA binds to the sperm's plasma membrane through specific DNA-binding proteins. After internalization, they are carried into the oocyte upon fertilization (Fig.1).
- The most appealing characteristics of using sperm as vectors to produce transgenic animals are its simplicity (no embryo manipulation is required) and the possibility of performing mass production of genetically modified animals through in vivo or in vitro insemination of many oocytes. On the other hand, this technique also has limitations.

Sperm-mediated gene transfer

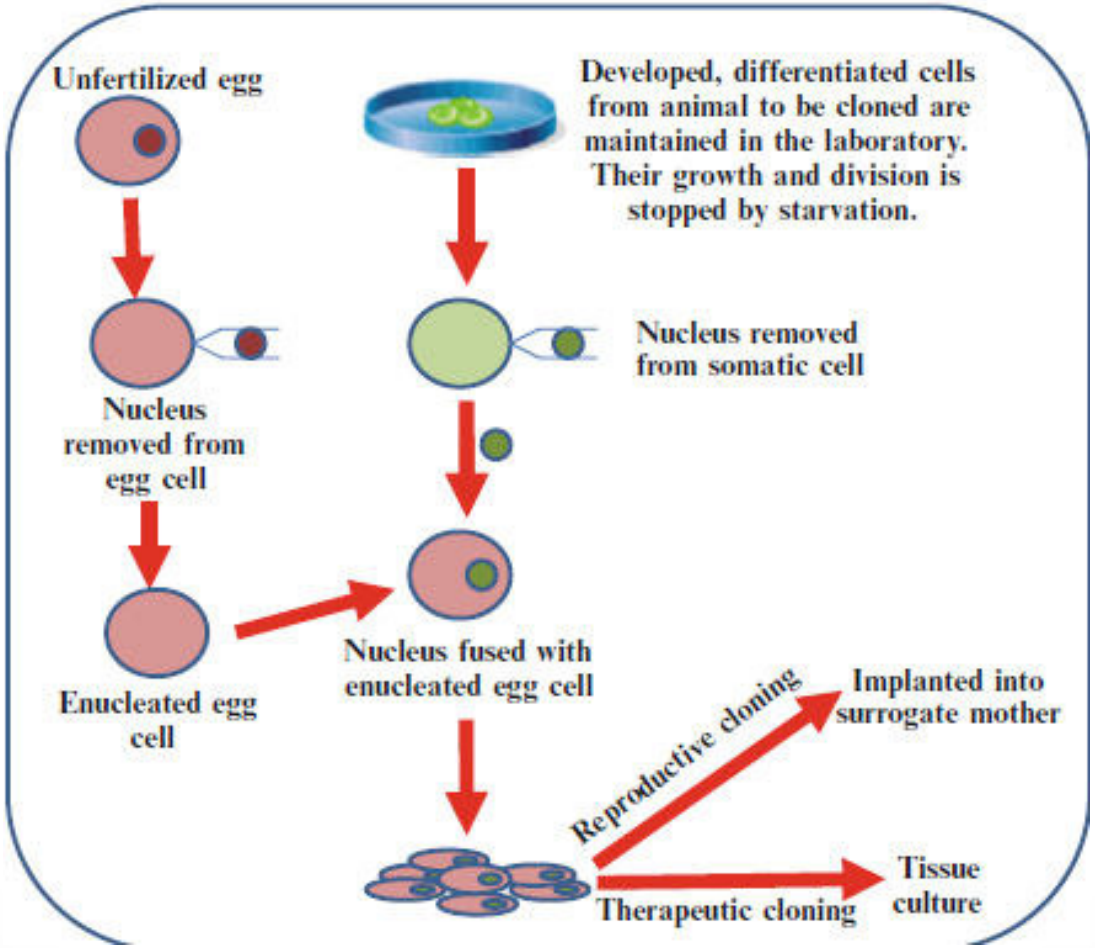


- Recently scientists reported successful generation of transgenic pigs carrying a human gene (human decay accelerating factor, hDAF).
- In this study, approximately 80 % of pigs had integrated the construct containing an hDAF minigene; from this 80 %, approximately 53 % expressed the foreign protein at different levels.
- There is no doubt that this success represents an important step toward production of humanized pig organs and tissues for human transplantation. If these findings are readily reproducible, sperm- mediated gene transfer may displace in short time the popular pronuclear microinjection method.

Somatic Cell Nuclear Transfer

Lecture no 77

- Due to the absence of proven ES cells and the recent advances in nuclear transfer (NT), current emphasis for creating transgenic animals is on somatic cell nuclear transfer (SCNT).
- Nuclear transfer is a technique that is used to create a genetically identical copy, or a clone, of an animal.
- Nuclear transfer commonly involves the transfer or placement of a donor nucleus into the cytoplasm of an enucleated MII oocyte (Fig.1).


Unfertilized egg
Developed, differentiated cells from animal to be cloned are maintained in the laboratory. Their growth and division is stopped by starvation.
Nucleus removed from somatic cell
Nucleus removed from egg cell
Nucleus fused with enucleated egg cell
Enucleated egg cell
Reproductive cloning
Implanted into surrogate mother
Tissue culture
Therapeutic cloning

- Donor cells can originate from a wide variety of cell types ranging from embryonic blastomeres all the way up to adult cells .
- Initial work on nuclear transfer was focused on using embryonic blastomeres as a donor source; the limited number of cells available in an early embryo hampered the process.

- Fetal or adult cells have been used successfully to clone all major livestock species, including sheep, cattle, goats, and swine.

- Somatic cell nuclear transfer has facilitated the ability to make transgenic animals by circumventing most of the shortcomings of other transgenic techniques.
- First, the sex of the animal can be predetermined by choosing the donor material (i.e., male or female tissue).
- Second, the use of cell culture to propagate donor cells can lead to large numbers of transgenic cells that can be frozen and stored for long periods of time.

- Transgene structure and expression can be tested by molecular techniques, such as PCR , Southern blot analysis , fluorescence in situ hybridization, and Western blot analysis , before initiating nuclear transfer and transferring the embryo to a recipient cow with a lengthy gestation time of 9 months
- This technique can be used for the addition of DNA at random sites and targeted insertion of DNA by homologous recombination, which is vital in modulating specific gene expression as well as creating gene knockouts

Medical Applications of Transgenic Animals

Lecture no 78

- In medicine the transgenic animals may serve as:
- Disease models

- Knockouts for studies related to gene function

- Quality trait improvement, and

- Pharmaceutical production.

Transgenic Animals as Disease Model

- Certain transgenic animals are produced for specific traits and serve as disease model.
- Animals are genetically manipulated to exhibit disease symptoms; thus they can be used for studies related to causes of the disease and effective treatment.
- For example, Oncomouse® developed by Harvard scientists is genetically modified and carries a gene that promotes the development of various human cancers.

- Transgenic animals have enabled scientists to understand the role of genes in specific diseases.
- By either introducing or inactivating particular genes, researchers can discover the root causes of diseases associated with gene defects.
- Transgenic animals allow more effective treatments to be developed. Having found the genes implicated in a disease, scientists can then target these or design other therapies which act by influencing their expression.

- Transgenic animals help test the safety of new medicines and vaccines.
- Because transgenic models can highlight specific characteristics such as certain mechanisms involved in the formation of tumors, they can demonstrate more clearly the possible side effects of new therapies.
- Their use in early toxicity trials may also serve to prevent the subsequent use of a larger number of animals in the development phase.

Xenotransplantation

Lecture no 79

- The gap between the demand for and the availability of human organs for transplantation is growing, a tendency that is not likely to reverse in the near future.

- The number of patients on waiting lists for organs has been growing every year, reflecting both an increased demand for organs for transplantation and a pronounced shortage of donors.

- Transgenic animals are being developed by some companies to provide new organs for transplantation such as kidneys, livers, and hearts.
- Pigs have long been considered as an alternative source of organs for xenotransplantation.
- Pigs seem to meet most of the requirements for an ideal animal donor: they are anatomically and physiologically similar to humans with organs of appropriate size, they are prolific, and they can be maintained under specific pathogen-free conditions.

- There are barriers to pig-to-human xenotransplantation; one of the main challenges is to avoid the hyperacute immunological rejection of the grafted tissue.
- When organs or tissues are transplanted between discordant species like pig to human, the host's immune system initiates a fast reaction known as hyperacute rejection (HAR).

Therapeutic proteins

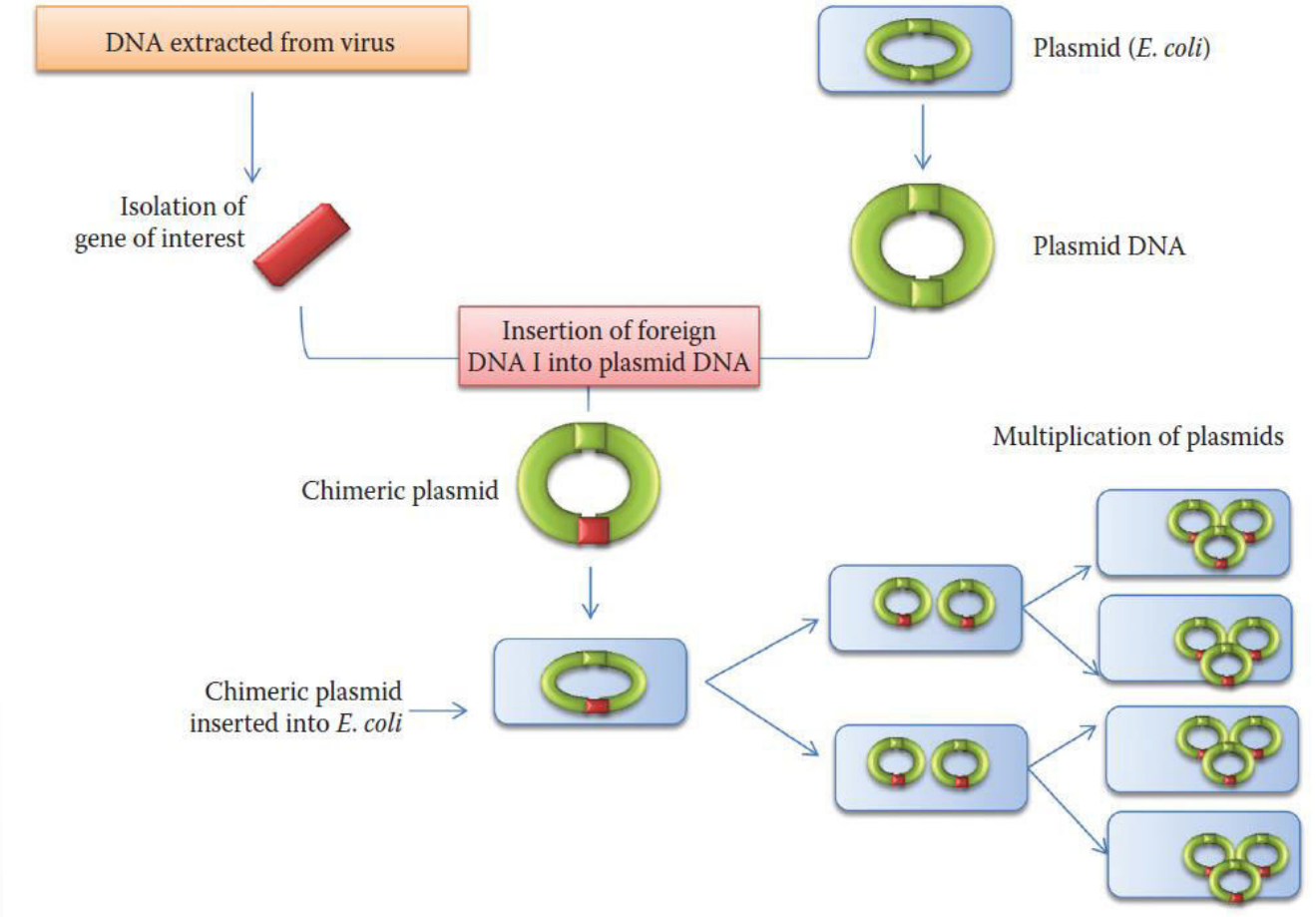
Lecture no 80

- One of the main discoveries in r-DNA technology was the production of biosynthetic human insulin, which was the first biomolecule made through r-DNA technology.
- Later on, this bioengineered insulin became the first biotechnology product approved by the U.S. Food and Drug Administration (FDA).
- It has been reported that insulin is basically an ideal candidate because it is a relatively simple protein and is therefore relatively easy to manipulate for genetic engineering.

- The first step in making recombinant insulin is the introduction of a specific gene sequence (oligonucleotide) that codes for insulin production in humans into the bacteria *E. coli*; it has been reported that only 0.94% of bacteria can pick up the sequence.
- Nevertheless, because the life cycle of *E. coli* is only about 30 min, there is no problem in generating a million copies of *E. coli* in a short period of time to be used for inducing insulin production.

- Among all applications of r-DNA technology, therapeutic proteins have been extensively used.
- Therapeutic proteins are those proteins that are either removed from human cells or engineered in the laboratory for pharmaceutical use.
- In addition to human insulin, other human proteins such as follicle-stimulating hormone, plasminogen, erythropoietin, and growth hormones have been created by using r-DNA technology (Figure 1).

Making of chimeric proteins



- There are many proteins essential to for the normal function of the human body and unfortunately some people fail to produce sufficient amounts of these proteins in their body
- These essential proteins include various blood-clotting factors causing hemophilia, insulin, growth hormone, and other proteins.
- These proteins can be synthesized outside of the human body by using recombinant technology.

Recombinant human insulin

Lecture no 81

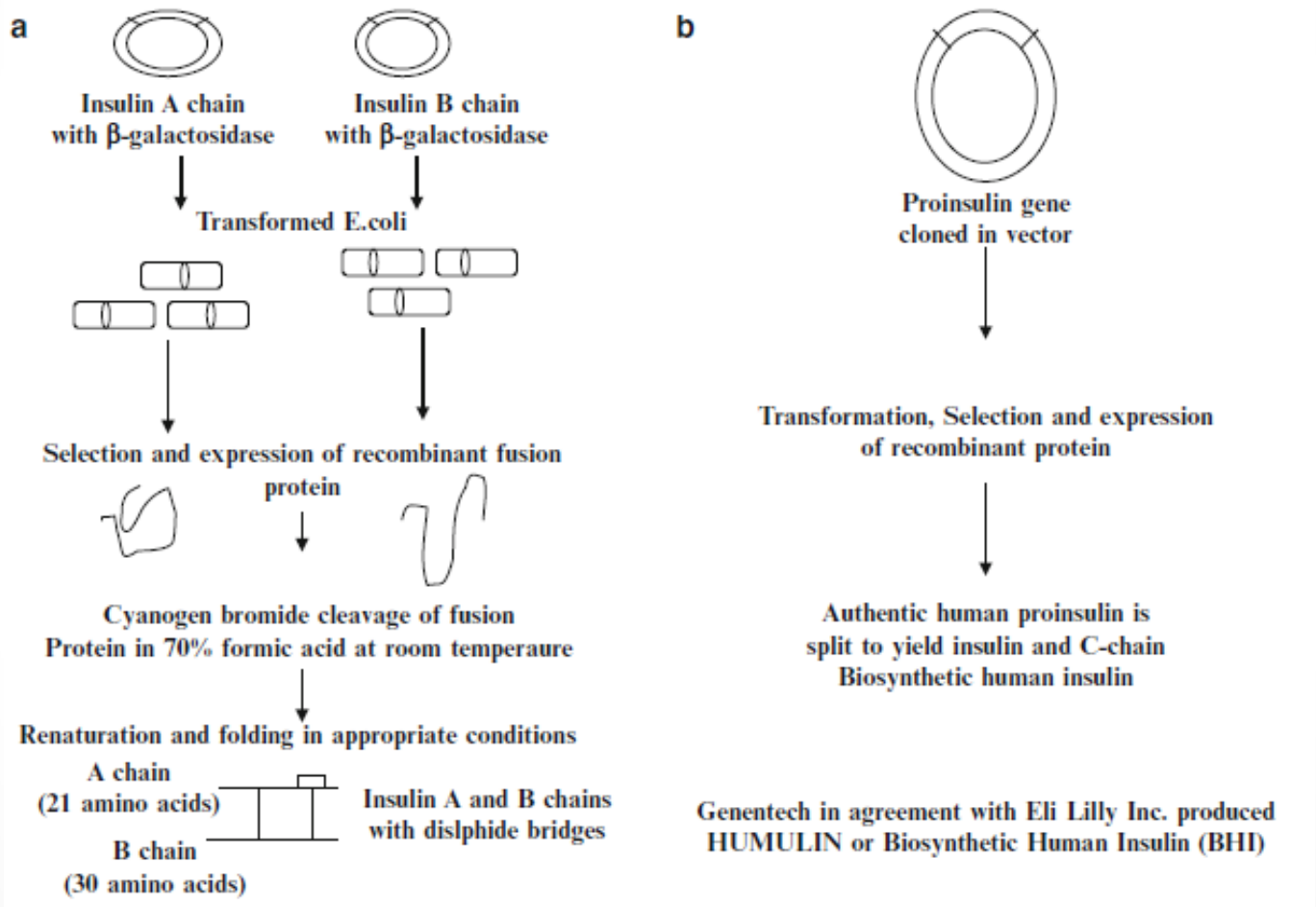
- Insulin is a peptide hormone consisting of 51 amino acids. It is secreted by β cells of islets of Langerhans of pancreas.
- The hormone is responsible for maintaining normal blood glucose level in blood.
- Insulin is stored in the form of proinsulin which contains two polypeptide chains, A and B, and is connected with a third peptide C chain, which before secretion is cleaved with production of insulin and C-chain.
- The cleavage results in the removal of C-chain, and the A (21 amino acids) and B chain (30 amino acids) are linked by disulfide linkage to form mature insulin.

- In the beginning, efforts were made to isolate mRNA for pre- and proinsulin from rat islets of Langerhans of pancreas and to synthesize cDNA.
- Thereafter, it was inserted into a plasmid. The recombinant plasmids were transferred into the *E. coli* cells, which secreted proinsulin.
- Scientists have chemically synthesized DNA sequences for two chains, A and B, of insulin and separately inserted into two pBR322 plasmids by the side of β-galactosidase gene.

- The recombinant plasmids were separately transferred into *E. coli* cells which secreted fused β galactosidase-A chain and β-galactosidase-B chain separately.

- These chains were isolated in pure form by detaching from β-galactosidase with yields of about 10 mg/24 g of healthy and transformed cells.

Production of recombinant insulin



Recombinant human growth hormone

Lecture no 82

- Growth hormone is produced by the anterior lobe of the pituitary gland and is released in multiple pulses.
- The hGH is encoded by GHN gene cluster (an array of five closely related genes), which is localized on chromosome 17.
- It belongs to diverse gene family that has evolved by gene duplication events and has lots of structural similarity and some common functions.
- Of the various forms, the predominant form of hGH is 22 kDa protein of 191 amino acids with two disulfide bonds.

- GH does not control the functions directly, but acts on certain hormones or somatomedins for its activity, for example, insulin-like growth factor 1 (IGF-1).

- In clinical conditions, the therapy of growth hormone is given in the treatment of dwarfism, bone fractures, skin burns, bleeding ulcers, and AIDS.

- Recombinant human growth hormone (rhGH) is 22 kDa consisting of long chain of amino acids.
- It is used in deficiency disorders of growth hormone. In children, it is used for growth abnormality as short stature and is used in chronic renal insufficiency.
- GH is one of the most widely used hormones with the estimated market of more than 1.7 billion USD.
- Long experience in its administration has proven the therapy as safe and effective in various conditions of growth abnormality.

- Earlier pituitary-derived hGH was used but later on it was prohibited when found associated with Creutzfeldt–Jakob disease.
- Because of recombinant DNA technology , safe and abundant recombinant hGH was produced in various heterologous systems.
- As the non-glycosylated human growth hormone was biologically active, thus, the preferred system for its production is *E. coli* , which allows its rapid and economical production in large amounts.

Recombinant hGH (rhGH) is now used to treat:

- GH-deficient (GHD) short-stature children.
- Acceleration of wound healing.
- Increase in insulin-like growth factor (IGF)-1 levels.
- rhGH increases IGF-1, osteocalcin, type I procollagen pro-peptide (PICP), and bone density, when administered to children with GHD

Recombinant blood clotting factor VIII

Lecture no 83

- Factor VIII is one of the important factors of all blood clotting factors.

- Deficiency of factor VIII causes bleeding disorder called hemophilia A. The hemophilia may be mild to severe depending upon factor VIII concentration in the body.

- In moderate and severe factor VIII deficiency, there can be spontaneous bleeding episodes in the joints.
-
- Hemophilia A affects 1 in 5,000–10,000 males.
- Replacement therapy is the treatment option for hemophiliacs either with human plasma-derived factor VIII (pdFVIII) or recombinant FVIII (rFVIII).

- Transfection of HEK 293 cell cultures in serum-free suspension is being tried for optimal yield.

- Recombinant factor VIII (rFVIII) is produced by culturing mammalian cells as baby hamster kidney (BHK) or Chinese hamster ovary cells (CHO), using large-scale bioreactors . Standardizations are done to maximize yields

Recombinant Tissue Plasminogen Activator

Lecture no 84

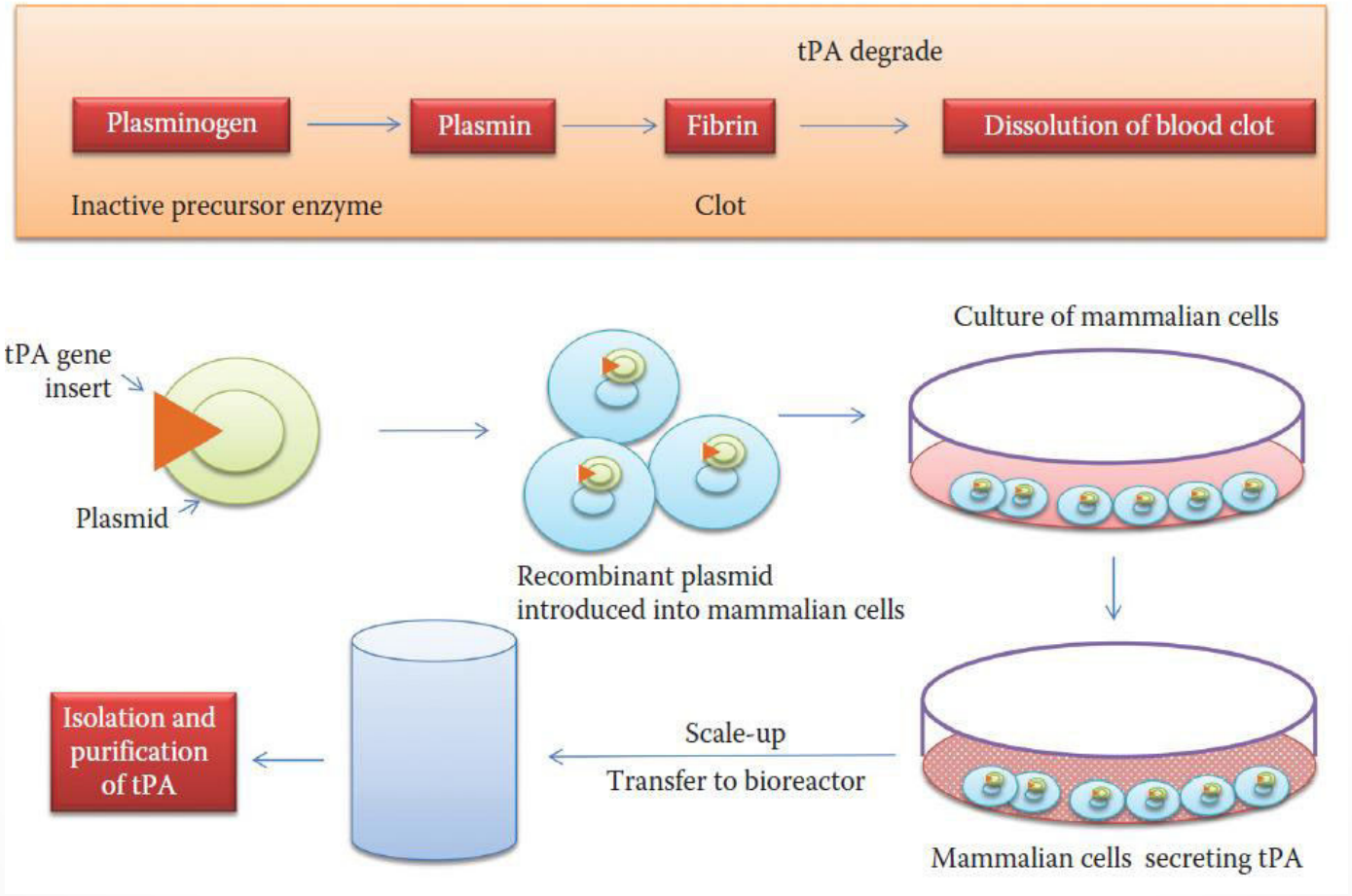
- Tissue plasminogen activator (tPA) is another kind of serine protease protein that is involved in the breakdown of blood clots in humans (Figure).

- In the human body, tissue plasminogen is found on endothelial cells, the cells that line the blood vessels. As an enzyme, tPA catalyzes the conversion of plasminogen to plasmin, the major enzyme responsible for clot breakdown.

- It has been reported that tPA is used as clinical medicine to treat only embolic or thrombotic stroke.
- Patients who have a problem related to blood clotting need a medicine to clot the blood immediately to avoid further releases of the blood from the body and tPA is considered to be an important medicine to promote blood clot.
- The tPA can be manufactured by using r-DNA technology and the production of tPA created this way is referred to as recombinant tissue plasminogen activator (rtPA).

- The basic principle behind blood clotting is that tPA catalyzes the conversion of plasminogen into plasmin, which is done by cleaving the single-chain plasminogen into two chains.
- These two chains are connected by a disulfide bond and the resulting molecule is called plasmin. In the patients, an increased enzymatic activity can cause hyper fibrinolysis, which demonstrates as an excessive bleeding.

Making of recombinant tissue plasminogen activator



- There are various commercially available rtPAs in the market, which include Alteplase, Reteplase, and Tenecteplase.
- There are other types of tPAs are under trial phases, including Desmoteplase.
- One example is streptokinase, which is a cheaper alternative that can be used as a thrombolytic in acute treatment.

Recombinant Erythropoietin

Lecture no 85

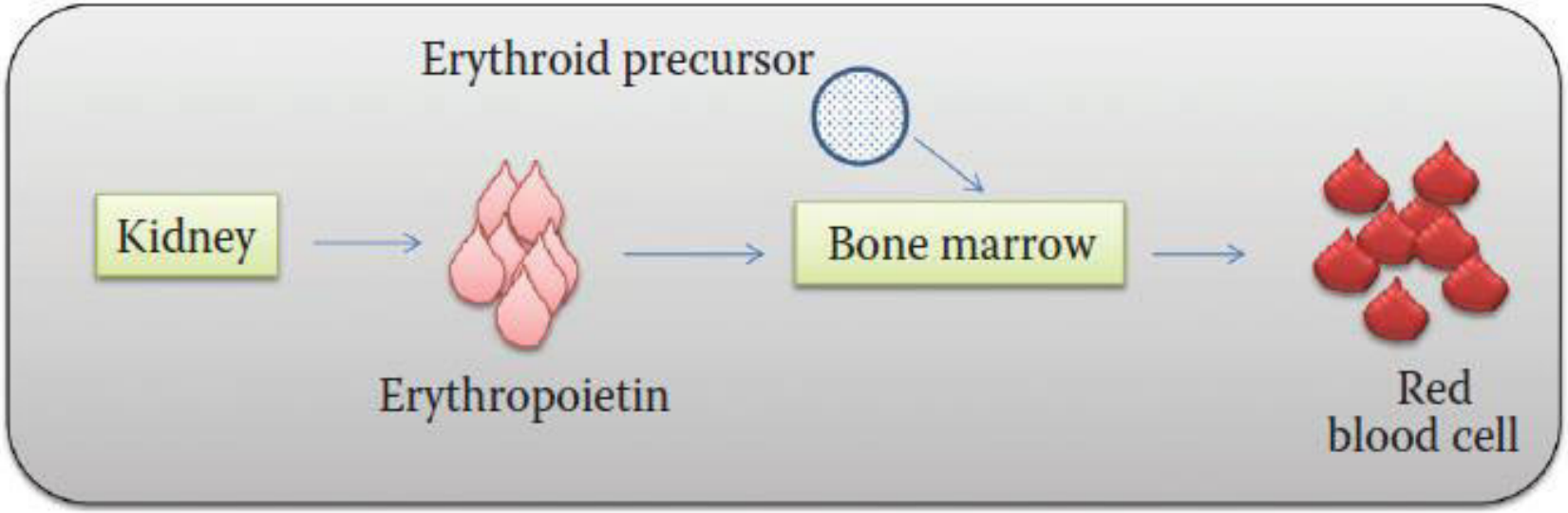
- Like insulin, erythropoietin (EPO) is a glycoprotein hormone that controls red blood cell production in the body (Figure 1).

- EPO, also called hemopoietin, is known to be located as precursors in the bone marrow.

- It is produced by the peritubular capillary endothelial cells in the kidney and liver.

- EPO also plays an important role in neuronal injury and in the wound healing process.
- It has been reported that EPO has its primary effect on red blood cells by protecting the blood cells from going to the apoptosis pathway.
- It also conjoins with various factors involved in the growth of precursor red cells.
- EPO has a range of actions in hypertension, stimulating angiogenesis, and inducing proliferation of smooth muscle fibers. It has also been shown that EPO can upsurge iron absorption by quashing the hormone hepcidin.

Normal synthesis of red blood cells



Applications of erythropoietin

- EPO can be used as a therapeutic agent to treat various disease conditions.
- It can be used in the treatment of anemia resulting from chronic kidney disease and myelodysplasia, and in the treatment of cancer.
- There are various types of EPO produced for patients who are suffering from anemia.
- These are epoetin, darbepoetin, epoetin delta, PD poetin, and methoxy polyethylene glycol-epoetin beta. The list of erythropoietin produced by using r-DNA technology is given in Table-1

List of Erythropoietin Produced by r-DNA Technology

Trade name of erythropoietin	Name of manufacturer
Epogen	Amgen
Espogen	LG Life Sciences
Epotin	Gulf Pharmaceutical Industry (Julphar)
Betapoietin	CinnaGen and Zahravi
Erykine	Intas Biopharmaceutical Pvt. Ltd.
Shanpoietin	Shantha Biotechnics Ltd.
Zyrop	Cadila Healthcare Ltd.

Significance of vaccines

Lecture no 86

- A vaccine is a biological preparation that improves human immunity to a particular disease.
- A vaccine basically contains an agent that resembles a disease-causing microorganism, and is often made from weakened or killed forms of the microorganism or its toxins.
- Upon injecting these vaccines in humans, the vaccine molecules stimulate the body's immune system to recognize the agent as foreign, destroy it, and memorize it, so that the immune system can more effortlessly familiarize with it and destroy any of these microorganisms that it later encounters.

- The story behind vaccine development started with the research work done by Edward Jenner in the late 1700s

- Louis Pasteur generalized Edward Jenner’s idea by developing what he called a rabies vaccine, now termed an antitoxin.

- Maurice Hilleman was the most successful developer of vaccines in the twentieth century

Type of vaccines

- Vaccines are produced by either dead or inactivated organisms and there are four main ways to develop vaccines:
- (i) live attenuated vaccines contain bacteria or viruses that have been altered so they cannot cause disease
- (ii) killed vaccines contain killed bacteria or inactivated viruses
- (iii) toxoid vaccines contain toxins (or poisons) produced by the germ that have been made harmless
- (iv) component vaccines contain parts of the whole bacteria or viruses.

Live attenuated vaccines

Lecture no 87

- The live attenuated or weakened vaccines can be created from the naturally occurring microorganisms
- Viruses are weakened by growing them over and over again in cell culture conditions.
- The procedure of growing a virus under cell culture conditions is known to decrease their disease-causing capabilities

Examples

- There are various examples of live attenuated vaccines that include:
- measles vaccine (as found in the MMR vaccine)
- rubella (German measles) vaccine (MMR vaccine)
- oral polio vaccine (OPV),
- mumps vaccine (MMR vaccine)
- and varicella (chickenpox) vaccine.

The advantages of attenuated vaccines

- Single dose is effective, and boosters are not essentially required because immune system is exposed to the live organism for prolonged period.

- Efficiently activates adaptive immune responses.

Disadvantages

- Storage conditions are to be strictly maintained.
- Reversion is one of an important risk.
- Vaccines can cause serious complications in immunocompromised patients

Inactivated vaccines

Lecture no 88

- Due to problem of reversion of pathogenic agent in attenuated vaccines, the live virus/bacteria were inactivated by heat or chemical treatment.

- Inactivated vaccines cannot cause an infection, but they still can stimulate a protective immune response

- These inactivated vaccines have microorganisms incapable of replication but is able to activate immune system for memory response.

- Controlled inactivation is done for production of vaccine where microorganism is properly inactivated (that is killed), but at the same time, care is taken to maintain the surface structure of epitope (for specificity and memory).

- Heat inactivation is not preferred because it causes denaturation and thus loss of native structure of the epitopes.
- Chemical inactivation using formaldehyde or various alkylating agents has been successfully used.
- In contrast to attenuated vaccine where one dose is sufficient for long term immunity, inactivated killed vaccines require boosters with appropriate adjuvants for activating immune system of the host.

Example

- Inactivated vaccine is the inactivated polio vaccine (IPV), which is the shot form of the polio vaccine

- Inactivated influenza vaccine.

Disadvantage

- The complications of inactivated vaccines were observed when Salk vaccine for polio was prepared and formaldehyde failed to kill the entire virus in two lots resulting in outbreak of paralytic polio in a high percentage of recipients.

Toxoid and Component vaccines

Lecture no 89

Toxoid vaccines

- There are some kinds of vaccines that can be made by treating them with toxins (formalin) produced by microorganisms, to destroy their ability to cause illness.

- Additionally, toxoids do not cause disease; in fact, they stimulate the body to produce protective immunity.

- There are several examples of toxoid-based vaccines such as diphtheria toxoid vaccine, which may be given alone or as one of the components in association with DTP, and furthermore, tetanus toxoid vaccine can be given alone or as part of the DTP.

Component vaccines

- Some vaccines are made by using only parts of the viruses or bacteria. These vaccines cannot cause disease; nevertheless, these vaccines can stimulate the body to create an immune response that protects against viral infection with the whole germ.
- So far, four vaccines have been produced by using this method that includes hemophilic influenza type b (Hib) vaccine, hepatitis B (Hep B) vaccine, hepatitis A (Hep A) vaccine, and pneumococcal conjugate vaccine.

DNA Vaccine

Lecture no 90

- DNA vaccination is a technique for protecting an organism against disease by injecting it with genetically engineered DNA to produce an immunological response

- DNA vaccines consist of plasmids engineered to contain a protein-encoding gene and all the necessary requisites

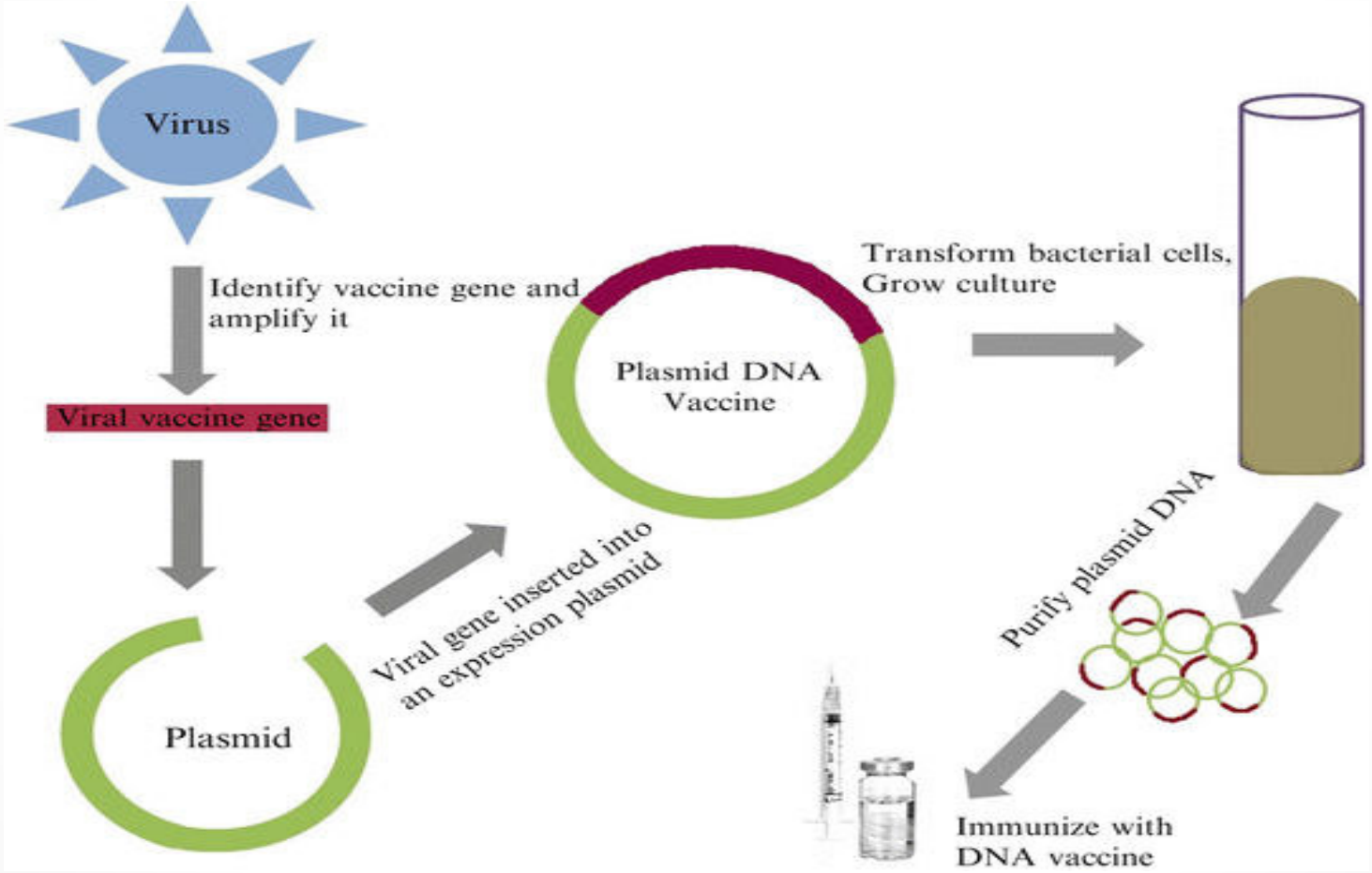
A single plasmid might contain:

- a promoter site
- a convenient cloning site for the gene of interest
- a poly-A tail used as a termination sequence
- an origin of replication
- a selectable marker sequence such as an ampicillin-resistant gene

- Animals can be immunized against disease merely by injecting plasmids suspended in saline (salt) solution.

- And delivery of the plasmids to target cells is relatively easy: researchers identified five effective ways for delivering influenza virus plasmids to the animal body: by nasal spray; by injection into the muscle, vein, and under the skin; and by a so-called gene gun.

- DNA vaccines has the added advantage of stimulating both antibody-mediated immunity (related to B-lymphocytes) and cell mediated immunity (related to T-lymphocytes).
- This is because DNA vaccines encode proteins that are released from the cell and taken up by phagocytes for delivery to the B-lymphocytes for antibody production.
- At the same time, some of the proteins fix themselves to the target cell surface and stimulate a response by T lymphocytes.


Virus
Identify vaccine gene and amplify it
Viral vaccine gene
Plasmid
Viral gene inserted into an expression plasmid
Plasmid DNA Vaccine
Transform bacterial cells, Grow culture
Purify plasmid DNA
Immunize with DNA vaccine

The advantages

- Protein is expressed in its native conformation.
- Long-term exposure of immune system to the antigen.
- Efficient activation of cell-mediated and humoral immune systems leading to memory response.
- There is no requirement of specific storage condition like refrigeration of plasmid DNA, and it reduces the cost and is easy to handle.
- The genes expressing cytokines can be ligated to the vector, and the cytokine produced at the site may activate the desirable branch of immune system.
- Vaccine is safe, without any infectious agent.
- Easier to produce in large quantity.

- The disadvantage of DNA vaccine is that they can be used for only protein antigens, and for the non-protein antigens (polysaccharide) they cannot be used

Nano Vaccine



Introduction

- A vaccine is a biological preparation that improves immunity to a particular disease.
- A vaccine typically contains an agent that resembles a disease-causing microorganism, and is often made from weakened or killed forms of the microbe or its toxins.
- The agent stimulates the body's immune system to recognize the agent as foreign, destroy it, and "remember" it, so that the immune system can more easily recognize and destroy any of these microorganisms that it later encounters.

Nano vaccine

- Nano is very vast field and it can be applied to any area, one of such area is vaccine.
- It provide a different routes of administration of vaccine.
- Nano vaccine can be designed, manufactured and introduced into the human body to improve health, including cellular repairs at the molecular level.
- The nanomaterial is so small that it can easily enter the cell; therefore, nanomaterials can be used in vivo or in vitro for biological applications.

Delivery of Nano-vaccine

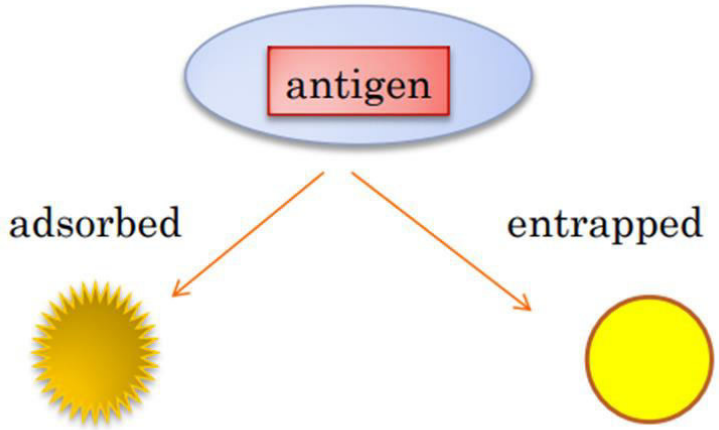
- Following are the different way to deliver vaccine at nano level
- **Nanobead:** Inert solid bead Size range 20-200nm.
- When antigens are adsorbed on the surface of bead it has been shown to stimulate CD8 -T cell response.
- The size of the bead play a major role in eliciting a combined response of humoral and cell-mediated immunity.
- Antigen covalently linked to inert nano-beads with a size of ~50 nm is preferentially taken up by dendritic cells, thus inducing humoral as well as cell-mediated immune responses.

Delivery of Nano-vaccine

- **2. Polymeric nanoparticles**

- Biodegradable, biocompatible polymers have been approved for use in humans.

- Poly(D,L-lactide-co-glycolide) (PLG) and polylactide (PLA).



Delivery of Nano-vaccine

- Polyglycolide (PLG) have been extensively used to encapsulate antigens.
- PLG forms lactic and glycolic acids, After hydrolysis of α-hydroxyl acids, yielding small spherical polymeric particles 1–100 nm in size.
- Adsorbed antigen offer improved stability and activity over encapsulated antigen by avoiding exposure to organic solvents used during formulation and acidic pH conditions caused by degradation of the polymer.

Delivery of Nano-vaccine

- 3. **Nano-emulsion**
- Size of globule (100-400nm)

- Nano-emulsion vaccine does not require refrigeration and is stable for 6 months.

- Nano-emulsion is non-toxic, pain free and avoids the risk of spreading needle-borne infections.

- Nano-emulsion of hepatitis B antigen, has been reported to be a safe and effective hepatitis B vaccine.

Nano Vaccine



Adjuvants

- These substance used in combination with a specific antigen that produced a more robust immunity than the antigen alone.

- It improve the immune response to vaccine antigen by different way:

1. Increasing immunogenicity of weak antigen.
2. Enhancing the speed and duration of the immune response.
3. Modulating antibody avidity, specificity, isotope or subclass distribution.

Adjuvants

- Examples
- MF59
- Monophosphoryl lipid A
- Montanide™
- Calcium phosphate nanoparticles.
- Immunostimulating complexes
- Cholesterol-bearing hydrophobized pullulan
- Nanoparticles (CHP)

MF59

- MF59 is the only nano-sized vaccine adjuvant approved for human use.
- MF59 is an oil-in-water emulsion ($\leq$ 250 nm droplets)
- Its showed a 34-fold increase in antibody titers when immunized with glycoprotein D of herpes simplex virus (HSV) in guinea pigs.
- The mechanism of adjuvanticity of MF59 is believed to be through direct stimulation of cytokine production.

Monophosphoryl Lipid a

- Monophosphoryl lipid A (MPL®) is an immunostimulant TLR-4 receptor agonist composed of detoxified lipopolysaccharide (LPS) from Salmonella Minnesota R595.

- It is a versatile vaccine adjuvant that may be included in aqueous formulations or in an oil-in water emulsion for a more dynamic response.

Montanide™

- There are several different types of Montanide™, including ISA 50V, 51, 206 and 720.
- ISA 50V, 51 and 720 are water-in-oil emulsions while ISA 206 is a oil-in-water emulsion.
- ISA 206 and 50V only used in veterinary vaccine formulations.
- SA 51 & ISA 720 are under investigation for use in humans.
- Emulsions of Montanide™ ISA 51 and 720 are composed of a metabolizable squalene-based oil with a mannide monooleate emulsifier.

Montanide™

- The immune enhancement produced by the Montanide™ emulsions is believed to be due to the formation of a depot at the site of injection.

- The emulsion vaccines against malaria, HIV and various cancer have been in phase I and/or II clinical trials.

- A phase I trial of a trivalent malaria vaccine containing ISA 720 induced both humoral and cellular immune responses

Calcium Phosphate Nanoparticles

- Calcium phosphate nanoparticles are less than 1000nm in diameter.
- Nanoparticles can be generated by combining (while stirring) calcium chloride, sodium phosphate and sodium citrate.
- It produces longer duration of response as compare to Al salt.
- Phase I study showed that CaP is safe and non-toxic when administered subcutaneously.
- Vaccines utilizing CaP in preclinical studies include anthrax, HBV, flu (H5N1 avian and seasonal) and HSV-2.

Advantages of Nano-vaccine

- Nano-vaccine have potential to deliver safe and more effective vaccine.
- Nanobead covalently coupled with antigen offer distinct advantages – a low dose of antigen is required, efficient processing by antigen-presenting cells and stability during storage.
- Encapsulated nanoparticles easily deliver antigen, protects the antigen from degradation and is found to be effective with a single dose due to slow release of the antigen.
- Many of the nano-vaccines are non-invasive, delivered by the oral or nasal route, diffusion patches or microneedle arrays, thus allowing pain-free delivery with minimal damage. This is an advantage over conventional vaccines, which are usually multi injection, multi-dose delivery systems.

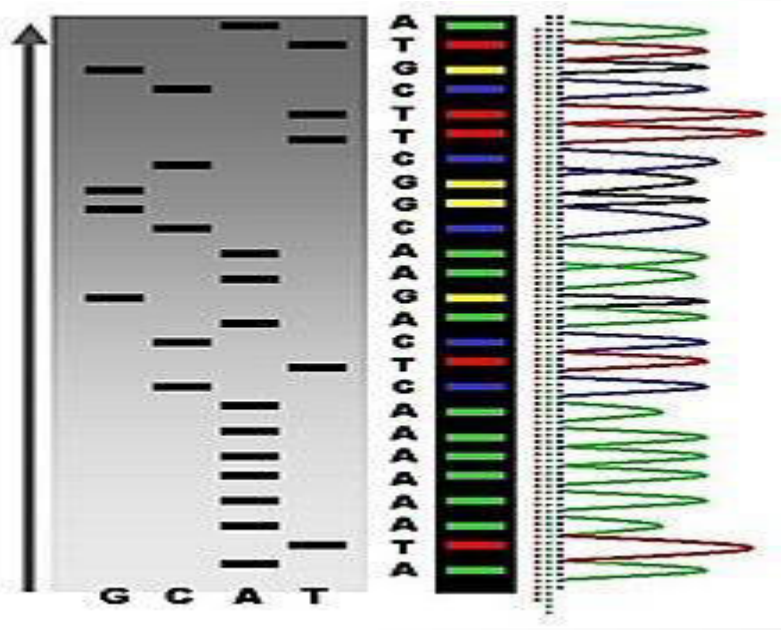
Disadvantages

- Cost of production.
- Nanomaterials can change size, shape but not composition, which may change their toxicity.
- Small nanoparticles are cleared quickly from the body, large counterparts may accumulate in vital organs causing toxic problems.

Nanopore Sequencing

DNA Sequencing

- DNA sequencing is the process of determining the precise order of the nucleotides within a DNA molecule
- DNA sequencing may be used to determine the sequence of individual genes, larger genetic regions, full chromosome or entire whole genome.
-



What is Nanopore?

- Nanopore = Very small hole
- Electrical current flows through the hole
- Introduce analyte of interest in the hole
- Identify analyte by the disruption or block to the electric current.

Introduction

- The fourth-generation DNA sequencing technology
- Studies the interaction between DNA and protein, as well as between protein and protein.
- Have the potential to quickly and reliably sequence the entire human genome for less than $1000, and possibly for even less than $100.
- The detection principle is based on monitoring the ionic current passing through the nanopore as a voltage is applied across the membrane.
- When the nanopore is of molecular dimensions, passage of molecules (e.g., DNA) cause interruptions of the current level, leading to a signal.

Introduction

- Since DNA bases (adenine, thymine, cytosine and guanine) are different from each other in atomic scale, it is essential to collect base-specific information at atomic level to correspond the DNA sequence with the measured signals.
- According to the different types of the signals, the detection methods can be roughly classified into two categories:
  - The electrical detection methods and the optical readout method.
- In the 1990s, Church et al. And Deamer and Akeson separately proposed that it is possible to sequence DNA using nanopore sensors

Types of Nanopores Sequencing

- Nanopore technologies can be broadly divided into two categories:
- 1. Biological nanopore
- 2. Solid-state nanopore
- More recently, hybrid nanopores have been proposed to take advantage of the features of both biological and solid-state nanopores.

Biological Nanopore Sequencing

Biological Nanopores

- Also called transmembrane protein channels
- Are usually inserted into a substrate, such as planar lipid bilayers, liposomes, or other polymer films
- Nanopores formed by pore-forming proteins
- Examples:
  - alpha-hemolysin
  - MspA
  - phi 29

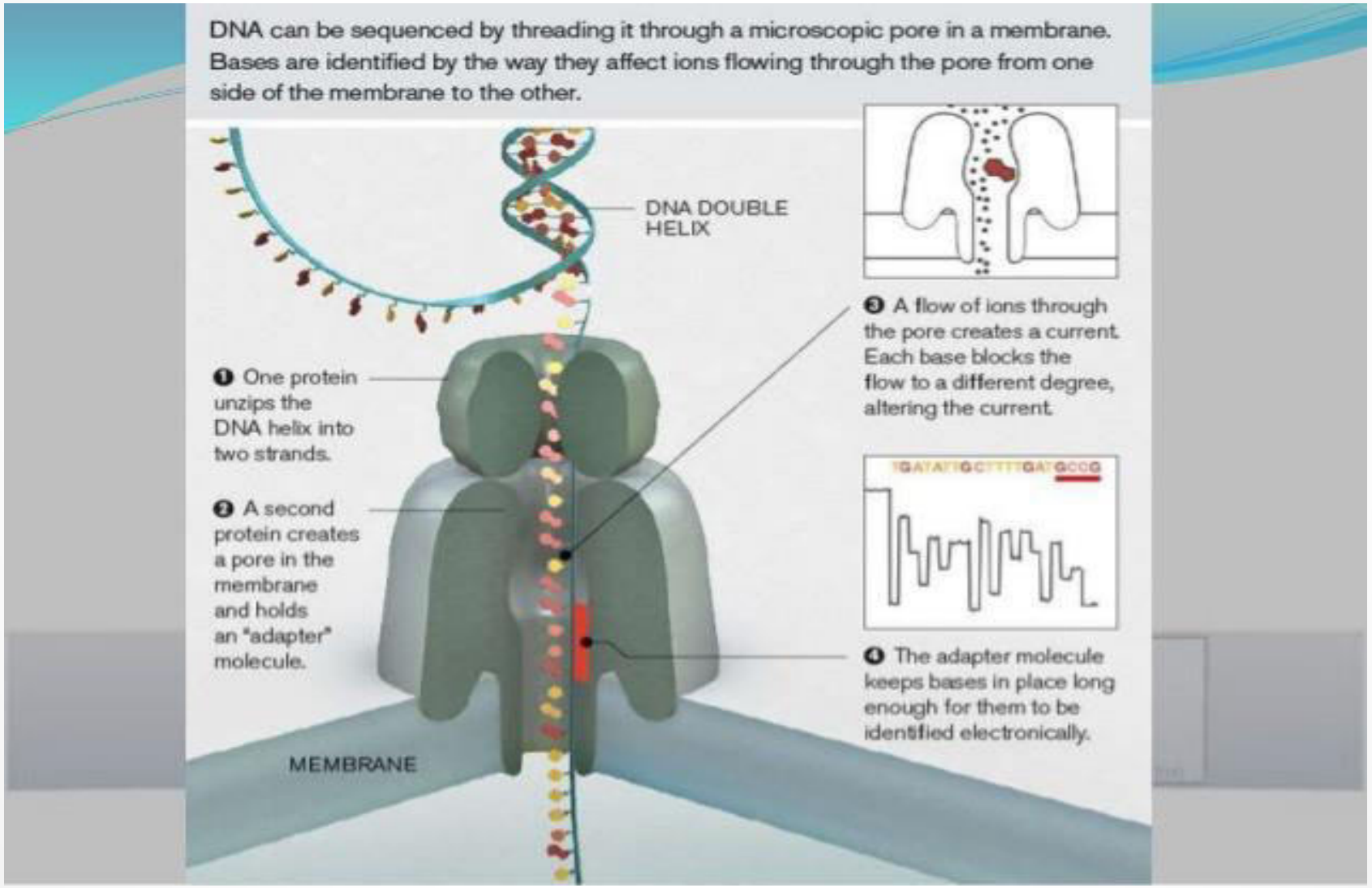
ALPHA-HEMOLYSIN

- a-Hemolysin (a-HL, also called a- toxin)
- a-HL is an exotoxin secreted by the bacterium Staphylococcus aureus, a human pathogen.
- This mushroom-shaped heptameric protein normally forms pores in the membranes.
- The limited pore size (1.4 nm) has restricted its application in the analysis of ssDNA, RNA, or small molecules.

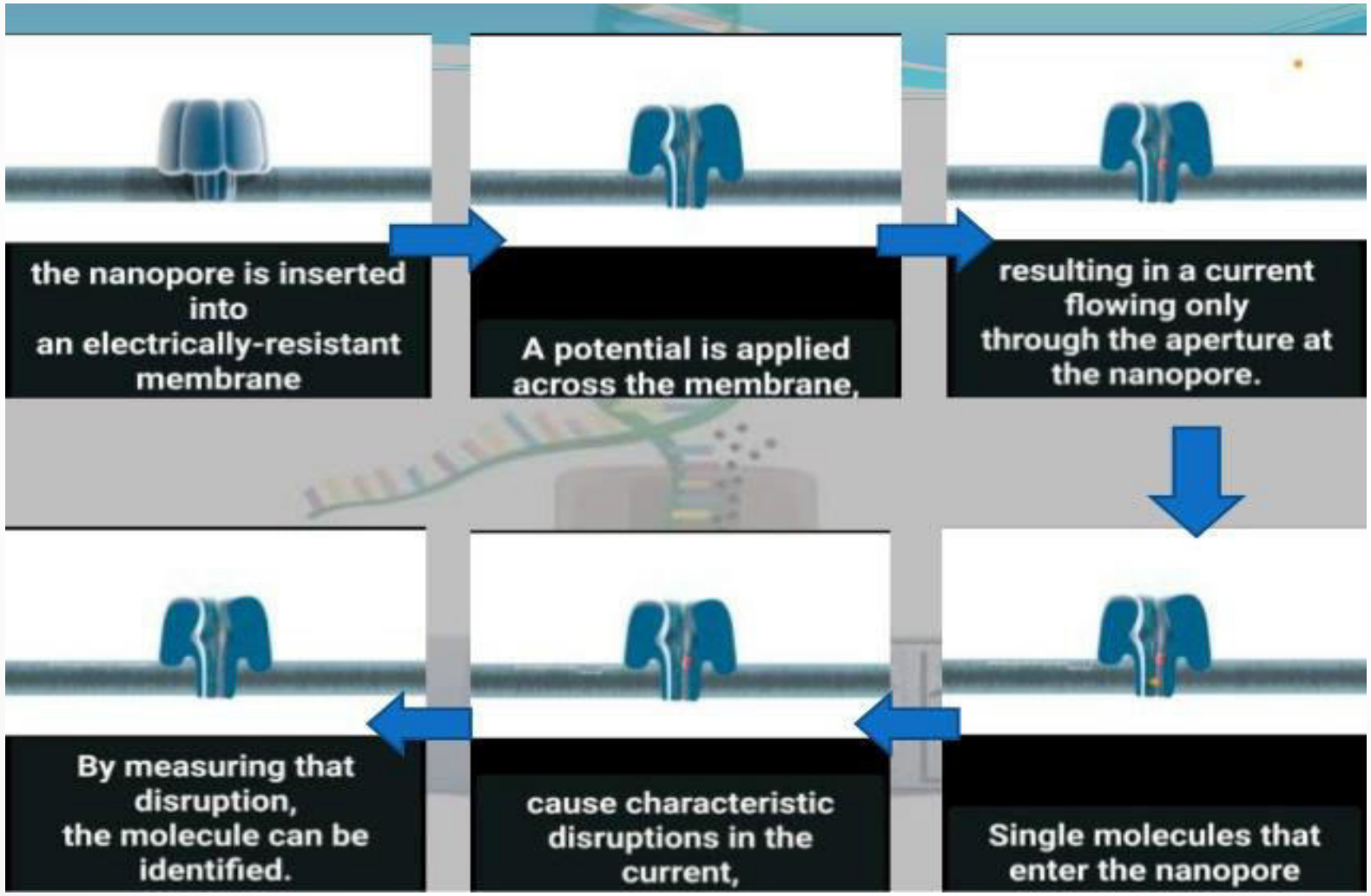
Mycobacterium smegmatis porin A (MspA)

- Powerful nanopore for reading information from four nucleotides simultaneously.
- By Butler et al. in 2008.
- The channel of the MspA octamer is 1 nm in diameter at the minimal point, which is relatively small and narrow, compared to that of alpha-HL.
- Thus, it can improve the spatial resolution of ssDNA sequencing

Principle


DNA can be sequenced by threading it through a microscopic pore in a membrane. Bases are identified by the way they affect ions flowing through the pore from one side of the membrane to the other.
DNA DOUBLE HELIX
❶ One protein unzips the DNA helix into two strands.
❷ A second protein creates a pore in the membrane and holds an "adapter" molecule.
MEMBRANE
❸ A flow of ions through the pore creates a current. Each base blocks the flow to a different degree, altering the current.
❹ The adapter molecule keeps bases in place long enough for them to be identified electronically.

Working Process



Advantages of Biological Nanopores

- Their well-defined and highly-reproducible nanopore size and structure.
- Biological nanopores can be modified easily with modern molecular biology techniques, such as mutating the nucleotide sequence to change the amino acid residue at a specific site

Disadvantages of biological nanopores

- The environmental demands (like temperature, electrolyte concentration and Ph) of nanopores to keep there biological activities.
- The fragility of lipid bilayer makes the biological nanopore breakdown easily

BIOPHARMACEUTICALS



Biopharmaceuticals

- Biopharmaceuticals are defined as pharmaceuticals manufactured by biotechnology methods, with the products having biological sources, usually involving live organisms or their active components
- A protein or nucleic acid used as a drug, which has been produced by living organisms as a result of biotechnology

Included products like

- Monoclonal antibodies
- Recombinant proteins or oligonucleotides used as vaccines
- Nucleotide sequences and vectors used in gene therapy
- Wide range of recombinant proteins used as drugs.

Excluded Products

- Proteins and nucleic acid extracted from sources that have not been manipulated using biotechnology e.g.
- Recombinant human growth hormone
- Growth hormone extracted from pituitary gland.

Pharmaceutical Technology

- The methods and techniques that involve the use of living organisms (such as cells, bacteria, yeast and others) are tools to perform specific industrial or manufacturing process are called biotechnology
- Pharmaceutical Technology will continue to provide new breakthroughs in medical research in the years to come, leading to treatment in field which have previously eluded us (including AIDS, cancer asthma, Parkinson's disease, Alzheimer disease)

The range of Biopharmaceuticals

- First human recombinant protein somatostatin expressed in E.coli…..1970s
- True biopharmaceutical produced was recombinant protein insulin…1982
- First therapeutic use of monoclonal antibody was in 1986.
- Hormones
- Clotting factors
- Cytokines
- Growth factors
- Thrombolytic factors and enzymes etc.

- Rapid increase in growth of biopharmaceutical products is due to combination of
  - Increase in our understanding of disease processes
  - The relative ease with which same technique can be used to produce biologically active molecules.

Advantages of Biopharmaceuticals

- BP can be engineered to be identical to human proteins e.g. insulin
- Molecularly engineered biopharmaceuticals avoid the need to extract human protein thus avoiding transmission of diseases.
- BP are easy to produce (in lab) and less amount is sufficient.
- High degree of specificity i.e. limited to a certain species

Disadvantages of Biopharmaceuticals

- Large scale production is expensive that's why these are expensive drugs.
- Generic products are as expensive as new conventional drugs.
- Risk of contamination from infectious agents, cells or products
- Very small changes can produce significant changes in the product.
- An ideal drug is one which is given orally and one dose is enough unlike BP products.
- Despite their extreme selectivity these drugs do produce side effects.

BIOPHARMACEUTICALS



Biopharmaceuticals

- Biopharmaceuticals are defined as pharmaceuticals manufactured by biotechnology methods, with the products having biological sources, usually involving live organisms or their active components
- A protein or nucleic acid used as a drug, which has been produced by living organisms as a result of biotechnology

Large scale Production of Biopharmaceuticals

- Most commercial production of biopharmaceuticals uses cultured mammalian cells. Other expression systems are
- Bacteria
- E.coli and Bacillus subtilis
- Simple physiology
- Grow rapidly
- Large product yields
- Made to excrete product into the medium to ease the recovery process.
- However not suitable for mammalian protein because of no glycosylation
- Insulin, growth hormone and some other are produce through bacterial system

Yeast

- Widely used in BP industry
- Post translational modifications occurs (to some extent).
- Made to excrete product into the medium to ease the recovery process
- One disadvantage is the active production of proteases that destroys proteins which can be engineered.

Transgenic Animals

- First transgenic animal was a mouse produced in yale 1980.
- 1996 Dolly was cloned
- Several companies are producing recombinant protein from transgenic sheep and goats from their milk.
- This technique has several advantages over other systems while making original transgenic animal is complex.
- Yield are comparable to large bioreactors.
- But still potential risk of animal disease transmission.
- Main drawback is the animal welfare.

Transgenic Plants

- Number of therapeutic medicines are produced through tobacco and potato plants.
- Transgenic plants are much cheaper than animals.
- Product purification is costly in plants.
- Use of raw eating plants to avoid extraction and purification step.
- Concentration of products vary with different environment.
- Cross pollination and cross contamination cannot be avoided even with good manufacturing practices.

Mammalian Cell Expression Systems

- First used for the production of inactivated polio vaccine.
- Nowadays continuous immortalized cells lines are used instead of primary cell lines.
- General method is culturing transfected cells on large scale and then target protein is recovered.
- Practicing good manufacturing practices.
- Any small change require change in license product

Good Manufacturing Practice

(GMP)

- Adherence to GMP is requirement of the licensing of any drug.
- All the components of manufacturing process must be strictly defined, to make sure that the product is made to the highest levels of safety and efficacy.
- Any change in the system i.e. even to increase the manufacturing product is likely to necessitate a change in the license.

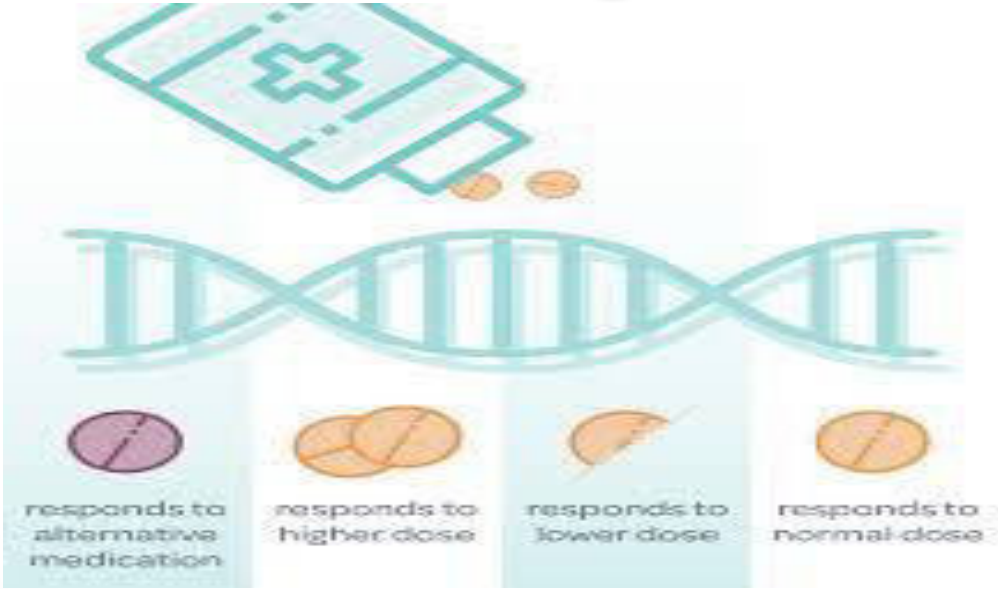
Cell Characterization and Banking

- It is necessary to validate the nature and source of all cell lines in the production of BP.
- Several banks are maintained to ensure the safety and validity of this valuable resource. There would be:
  - Two master banks where original stocks are stored
  - One or more working banks, where cells are withdrawn for working.
  - Banked cells are continuously checked to ensure that they are of right type i.e. genetically stable and not infected or contaminated.
  - Age of cells are also monitored.

Product Characterization and Validation

- Careful characterization of the product is essential as small change can lead to change in product activity. A single amino acid change can effect activity to 50%.
- Cells used also effect the nature of product yielded.
- Changes in the conditions under which cells are cultured can affect the product yield to considerable extent.

Pharmacogenetics



Introduction

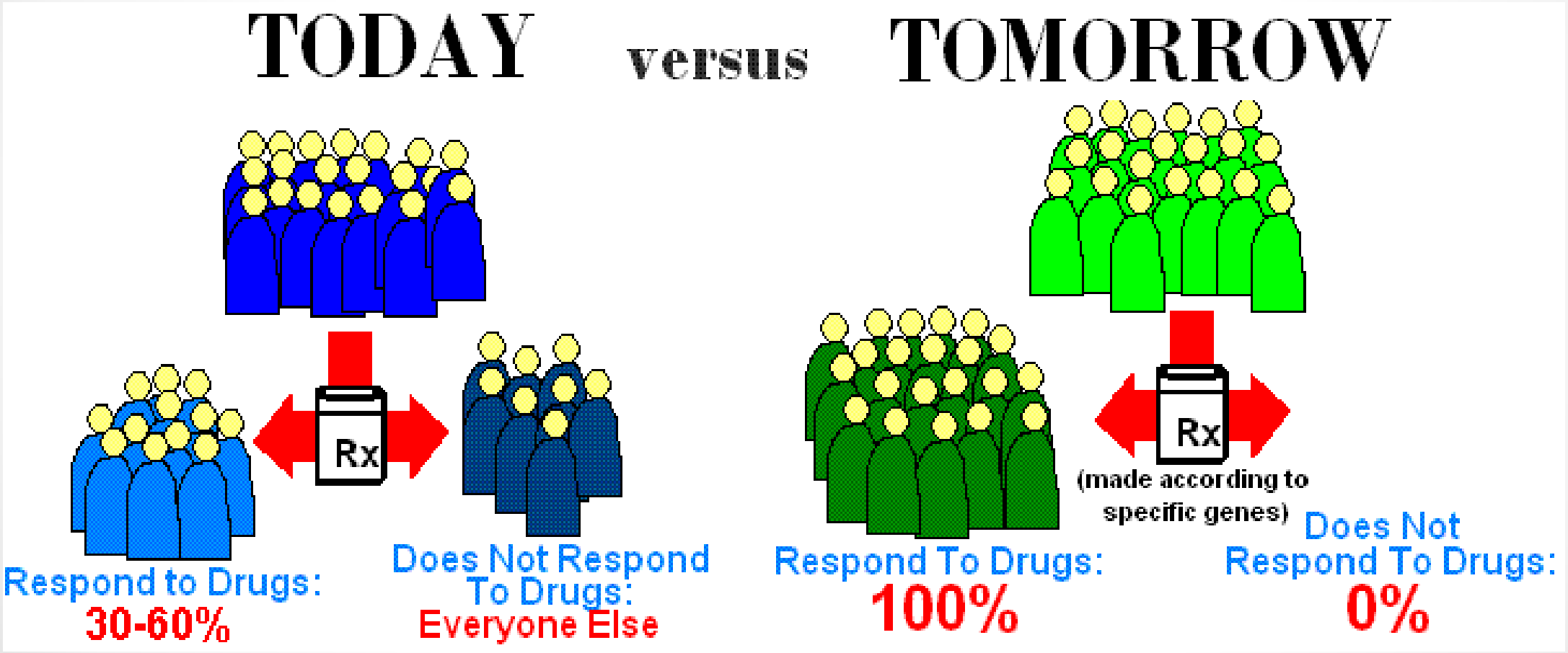
- **Pharmacogenomics:** is the science of how genes affect the way people respond to drugs
- How genes affect, the way our body processes drugs (pharmacokinetics)
- The interaction of drugs with receptors (pharmacodynamics)
- The treatment efficacy and adverse side effects
- **Pharmacogenetics**
- A subset of ‘pharmacogenomics’
- The study of how inherited variation affects drug response and metabolism

Why is this a good approach?

- Drugs can be dangerous
- Many people have severe adverse reactions to drugs
- Many people respond to drugs at different doses
- Many drug treatments are horribly unpleasant, painful
- Drugs are expensive (to take and to make)
- Ineffective drugs are a waste of money to take

Why is this a good approach?

- Drug development needs to account for response variability
- Genetics provide a prior information
- Genetics don’t change (except in cancer)
- Genetics can point to the cause not just the symptom


TODAY
versus
TOMORROW
Rx
Respond to Drugs:
30-60%
Does Not Respond
To Drugs:
Everyone Else
Rx
(made according to
specific genes)
Respond To Drugs:
100%
Does Not
Respond To Drugs:
0%

Early Studies

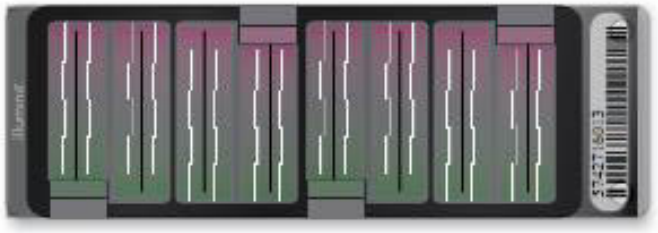
- Fast increase in awareness of the interaction of drug and drug response
- Many family studies, twin studies and ‘top-down’ genetic studies
- Very laborious experimental work through the study enzyme actions and clinical observation
- PCR sped things up a bit

CYP2D6

- 1975 Smith and colleagues ingest a drug they are testing
- He had a bad reaction but his colleagues did not
- Family studies revealed genetic inheritance
- Enzyme discovered and characterized
- Enzyme cDNA sequenced and variants found (1990)
- This family of gene important for many drugs

The Technology: Genotyping

- Uses a microarray to measure a limited predefined set of SNPs
- Very high throughput (fast)
- Very inexpensive (cheap)
- Excellent coverage of common variation (up to 5,000,000 SNPs)
- Relies on Linkage Disequilibrium



Microarray

A T C G A A A T G C A T G A C C T T T G A T A T G A T C G G C T G C A G T C A G C
T T C G A A G T G C A T G A C T T T T G A C A T G A G C G G C G G C C C A C A G C

Common Variation Rare Variation No Recorded Variation

The Technology: Deep Sequencing

- Captures every base pair in the genome (3,000,000,000)
- (Currently) low throughput (slow)
- (Currently) Very expensive (> 10k)
- Captures common, rare, and personal variation
- New and hard to analyze



Sequencer

A T C G A A A T G C A T G A C C T T T G A T A T G A T C G G C T G C A G T C A G C
T T C G A A G T G C A T G A C T T T T G A C A T G A G C G G C G G C C C A C A G C

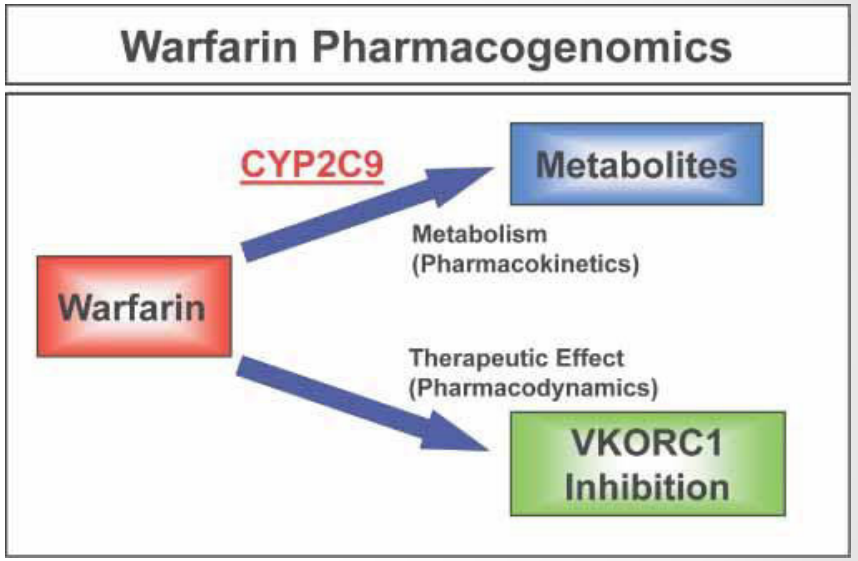
Common Variation **Rare Variation** **No Recorded Variation**

Utility of Drugs

- The utility of pharmacogenetics:
  - Determining appropriate dosing
  - Avoiding unnecessary toxic treatments
  - Ensuring maximal efficacy
  - Reducing adverse side effects
  - Developing or choosing novel treatments
  - Can also explain variable response to illicit drugs

Warfarin: A dosage story

- Most widely used anticoagulant in the world
- A “blood thinner”
- Prescribed doses vary widely (1-40mg / daily)
- Therapeutic index is very low
- High risk of bleeding early in treatment
- Two genes involved in metabolism: CYP2C9 and VKORC1



Plavix: A story about effectiveness

- Anti-clotting drug
- Prescribed for coronary artery disease and those who have suffered a heart attack or stroke or have a stent
- A “pro-drug”
- Converted to active form in the liver by CYP2C19

Pegasys: A toxic treatment story

- Pegylated Interferon α-2a
  - Interferons are proteins made in response to virus
- Treatment for Hepatitis B and C Virus
- Highly toxic treatment
- Highly variable response, especially in African Americans
- Very expensive

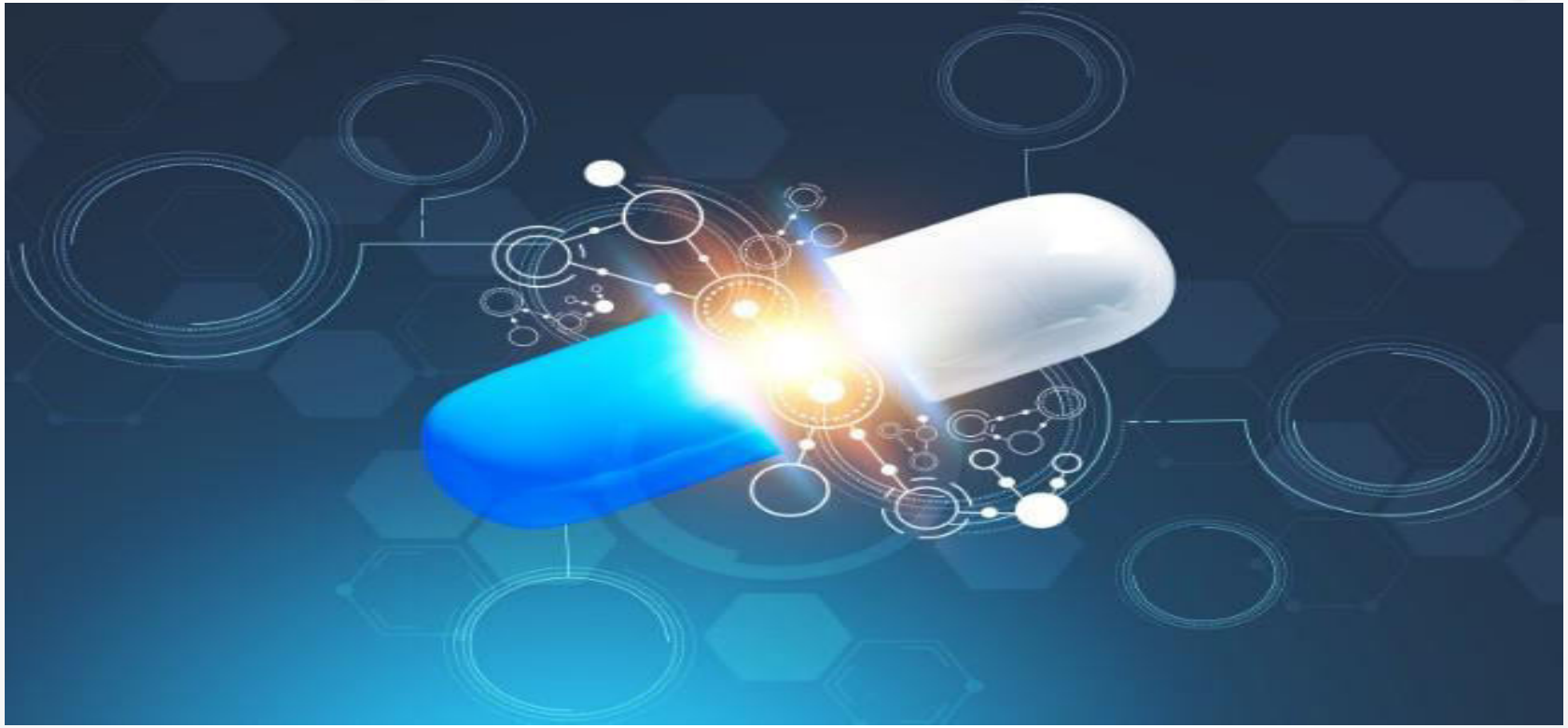
Personalized Medicine

- There is an emerging goal among ‘translational scientists’ to make medical practice more personalized
- Pharmacogenetics is an important step towards that goal
- Personalized medicine is an emerging practice of medicine that uses an individual's genetic profile to guide decisions made in regard to the prevention, diagnosis, and treatment of disease.
- Knowledge of a patient's genetic profile can help doctors select the proper medication or therapy and administer it using the proper dose or regimen.

Concerns with this Approach

- How reliable are the tests?
- Are health care providers prepared to use this information?
- Will Insurance companies pay for the tests?
- Will tailor made medicine lead to discrimination?
- Will ethnic biases in science bias treatment developments?
- Will this affect people's privacy?

Principles of Drug Delivery



Drug Delivery

- The appropriate administration of drugs through various routes in the body for the purpose of improving health
- It is highly interdisciplinary
- It is not a young field
- It has recently evolved to take into consideration

Oral Administration

- **Advantages**
  - Patient: Convenience, not invasive, higher compliance
  - Manufacture: well established processes, available infrastructure
- **Disadvantages**
  - Unconscious patients cannot take dose
  - Low solubility
  - Low permeability
  - Degradation by GI enzymes or flora
  - First pass metabolism
  - Food interactions
  - Irregular absorption

Oral Administration

- Traditional oral delivery systems
- Tablets
- Capsules
- Soft gelatin capsules
- Suspensions
- Elixirs

Buccal/Sublingual

- **Advantages**
  - By-pass First pass metabolism
  - Rapid absorption
  - Low enzymatic activity
- **Disadvantages**
  - Discomfort during dissolution
  - Probability of swallowing- lost of effect
  - Small doses

Rectal

- **Advantages**
  - By-pass first pass metabolism
  - Useful for children
- **Disadvantages**
  - Absorption depends on disease state
  - Degradation by bacterial flora
  - Uncomfortable

Intravenous (IV)

- **Advantages**
  - Drug 100% bioavailable
  - Rapid response
  - Total control of blood concentration
  - Maximize incorporation of degradable drugs

- **Disadvantages**
  - Invasive
  - Trained personnel
  - Possible toxicity due to incorrect dosing
  - Sterility

Subcutaneous

- **Advantages**
  - Patient self-administration
  - Slow, complete absorption
- **Disadvantages**
  - Invasive
  - Irritation, inflammation
  - Maximum dose volume - 2mL

Intramuscular

- **Advantages**
  - Patient can administer the drug himself
  - Larger volume than subcutaneous
  - By-pass first pass metabolism
- **Disadvantages**
  - Invasive – patient discomfort
  - Irritation, inflammation
  - May require some training

Inhalers

- **Advantages**
  - Gases are rapidly absorbed
  - Easy to use
- **Disadvantages**
  - Solids and liquids can be absorbed if size is below 0.5um

Factors Influencing the Selection of the Delivery Route

- pH
- pKa - ionization
- Temperature
- Concentration
- Crystallinity
- Particle size
- State of hydration

Factors Influencing Bioavailability

- Delivery route
- The site of measurement
- Type of animal employed
- Physiological state of the animal/human

Synthetic Pharmaceuticals



Drug & Medicine

- Drug is a chemical substance that interacts with a living system and produces a biological response
- **Drugs** are considered to be such things as stimulants, narcotics, and hallucinogens, and often are used in the medical field to treat or cure disease. **Medicine**, however, is a term for a drug when that drug is used to cure, mitigate, treat, or prevent a disease.

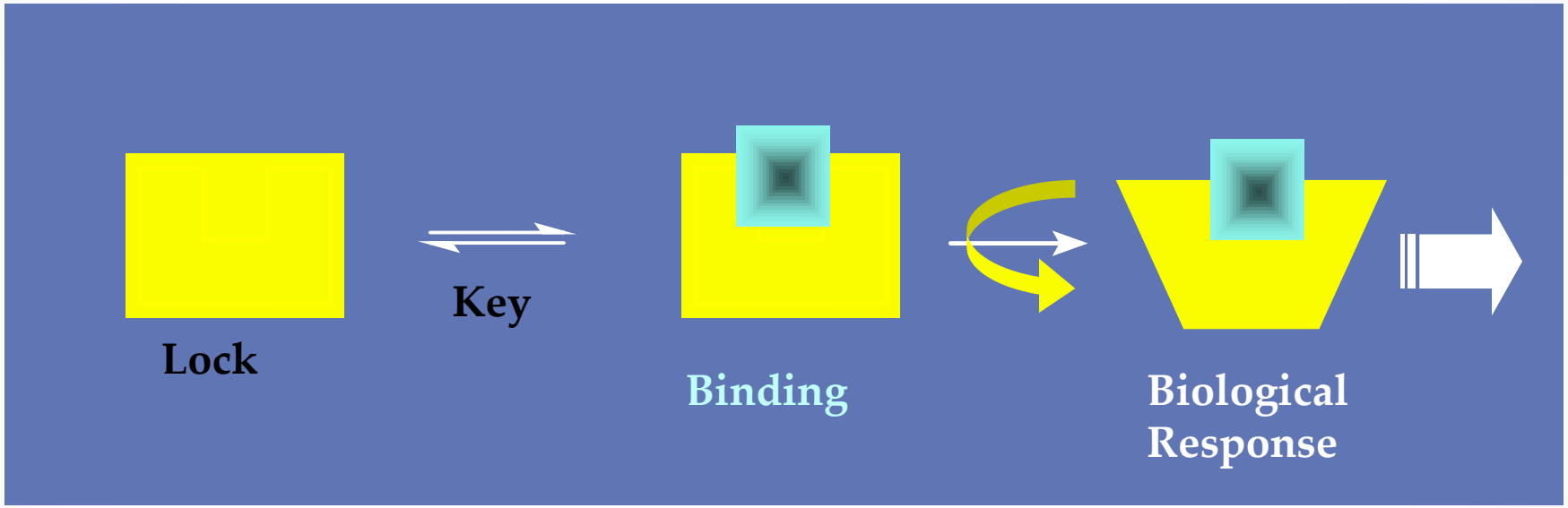
What criteria MUST new drugs meet?

- Drugs must address a new need or provide a significant “added benefit” over an existing medicine
- Drugs must also meet these criteria:
  - Must be safe
  - Effective
  - Of high quality
  - Cost effective

Major Therapeutic Targets

- **Infectious disease – anti-infectives**
  - Anti-bacterial, anti-viral, anti-parasitic drugs
- **Metabolic disease**
  - cancer, cardiovascular, diabetes,
    inflammation, high blood pressure, neurological disease, pain
- **Other aspects of health care**
  - Hormonal treatments, contraception, vaccines, immunosuppressant,
    anaesthetics, nutraceuticals,

The “Lock and Key” analogy

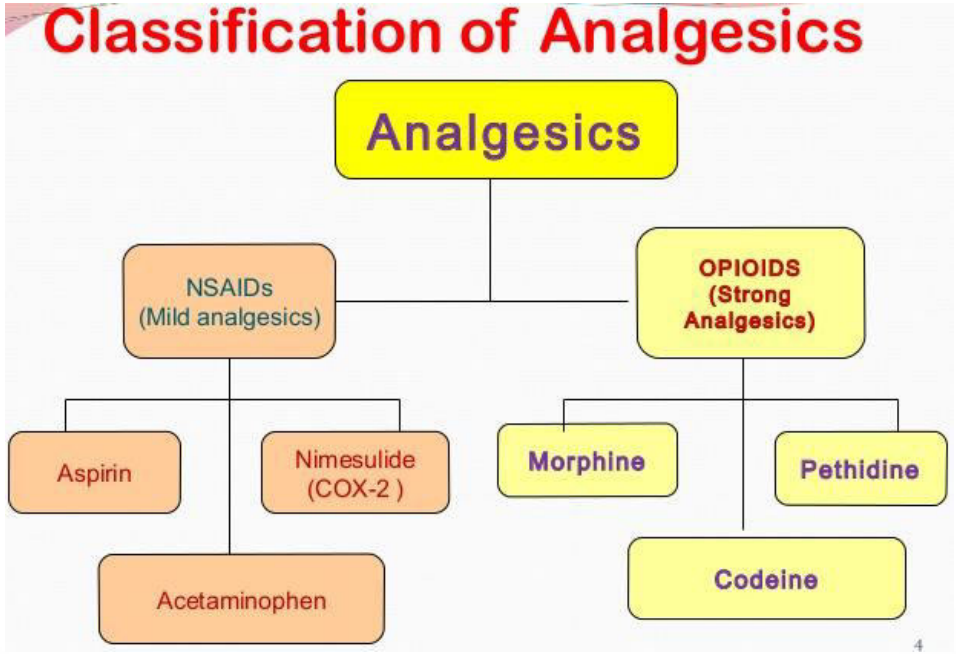


- Here the KEY is the natural substrate
- Binding of the KEY to the LOCK (an enzyme or a receptor) then causes a response – a shape change in the protein/receptor

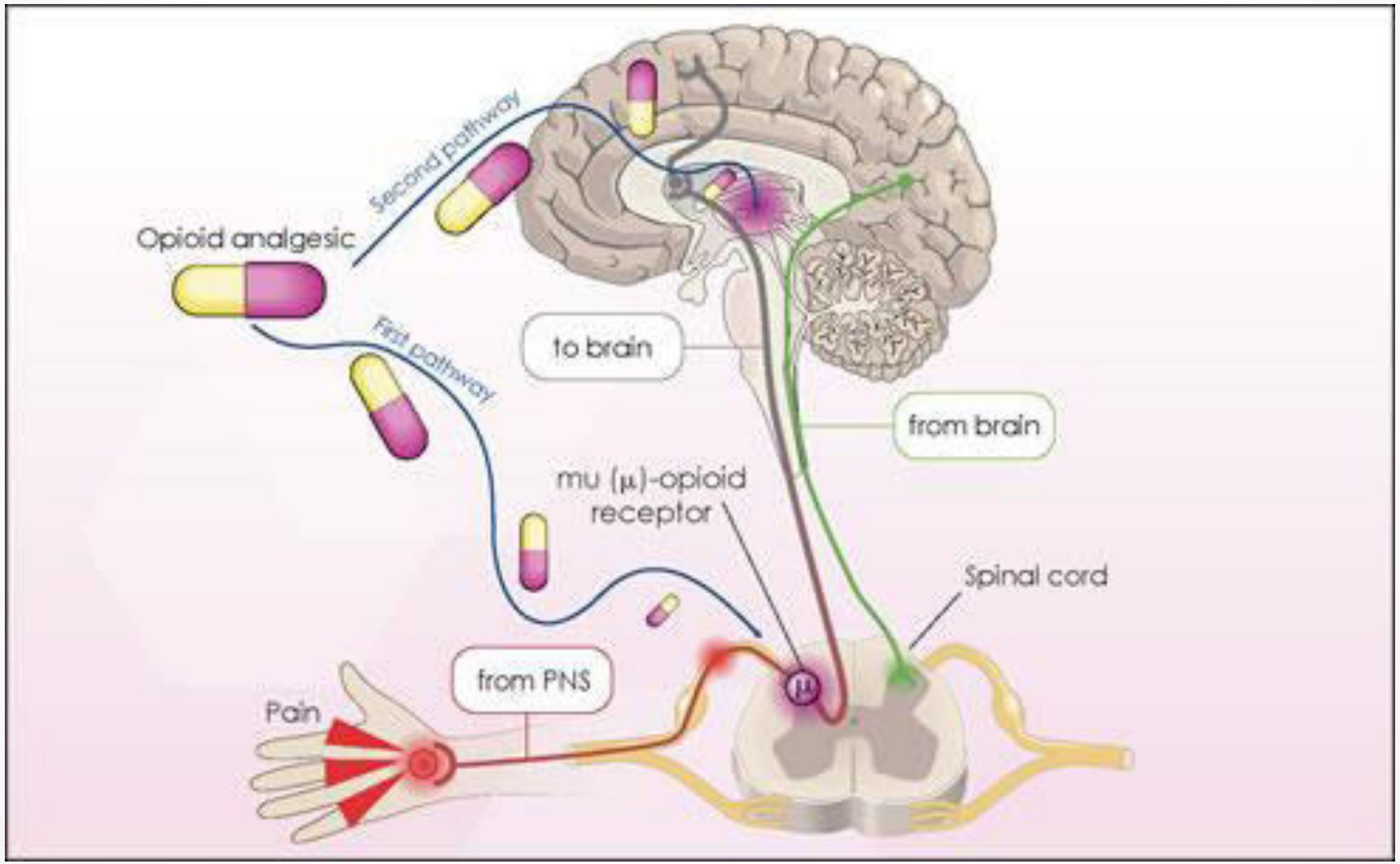
Analgesics

- Analgesic drugs act in various ways on the peripheral and central nervous systems.

- They are distinct from anesthetics, which reversibly eliminate sensation.



Mechanism of action



Paracetamol and NSAIDs

- The exact mechanism of action of paracetamol/acetaminophen is uncertain but appears to act centrally in the brain rather than peripherally in nerve endings. Aspirin and the other non-steroidal anti-inflammatory drugs (NSAIDs) inhibit cyclooxygenases, leading to a decrease in prostaglandin production. In contrast to paracetamol and the opioids, this reduces not only pain but inflammation as well.

Opioids

- Morphine, the archetypal opioid, all exert a similar influence on the cerebral opioid receptor system.
- Opioids, while very effective analgesics, may have some unpleasant side-effects. Patients starting morphine may experience nausea and vomiting, itching, constipation etc.

- When used appropriately, opioids and similar narcotic analgesics are otherwise safe and effective, however risks such as addiction and the body's becoming used to the drug (tolerance) can occur.
- **The effect of tolerance means that frequent use of the drug may result in its diminished effect so, when safe to do so, the dosage may need to be increased to maintain effectiveness**

Anti-inflammatory

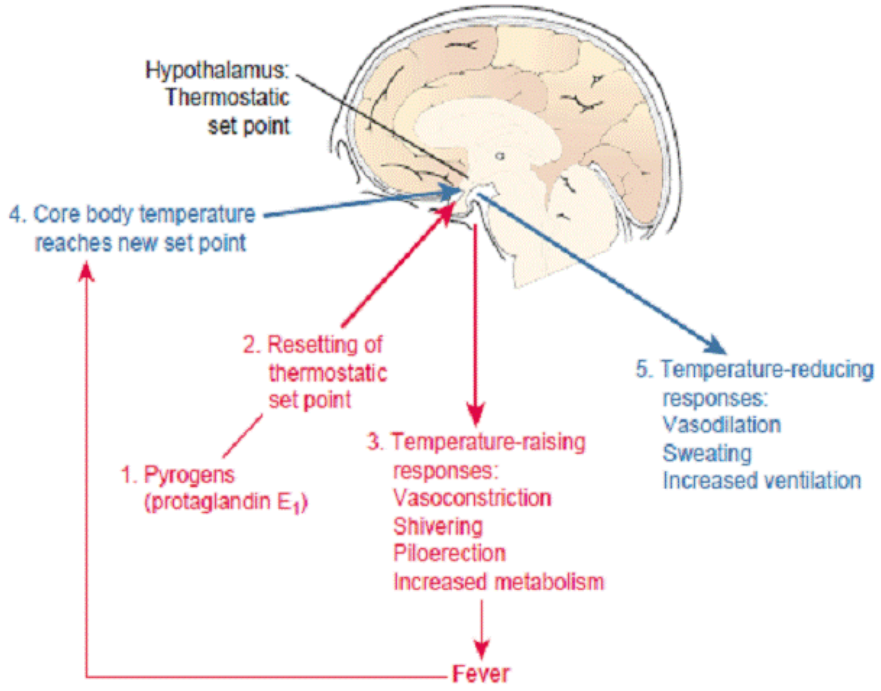
- Anti-inflammatory refers to the property of a substance or treatment that reduces inflammation or swelling.
- COX-2 inhibitors
- Prostaglandins inhibitors

Inflammation

Cardinal Signs	Physiological rationale
1. Rubor (Redness)	Increased Blood flow
2. Tumor (Swelling)	Exudation of fluid
3. Calor (Heat)	Increased Blood flow , Exudation of fluid, Release of inflammatory mediators
4. Dolor (Pain)	Stretching of pain receptors and nerves by inflammatory exudates , chemical mediators
5. Functio laesa (Loss of function)	Pain, Disruption of tissue structure, Fibroplasia and metaplasia

Antipyretic

- Antipyretics cause the hypothalamus to override an interleukin-induced increase in temperature.



Antiemetic

- Antiemetic is a drug that is effective against vomiting and nausea.
- Block serotonin receptors in the central nervous system and gastrointestinal tract

Antihistamine

- Antagonists of the histamine H_1-receptor are used to treat allergic reactions.
- Histamine produces increased vascular permeability, causing fluid to escape from capillaries into tissues, which leads to the classic symptoms of an allergic reaction — a runny nose and watery eyes.

Steroid (Life Saving Drug)

- Steroids reduce the production of chemicals that cause inflammation. This helps keep tissue damage as low as possible. Steroids also reduce the activity of the immune system by affecting the way white blood cells work.
- Given when,
    - Body defense is poor
    - Extensive tissue damage

Anesthetics

- An **anesthetic** is a drug that causes anesthesia, which is a reversible loss of sensation.

1. **General anesthetics**

2. **Local anesthetics**

Antibiotics

- Medications used to treat bacterial infections

1. Sulfonamides
2. Penicillins
3. Cephalosporins
4. Tetracyclines
5. Aminoglycosides
6. Quinolones
7. Macrolides

How Antibiotics Work

- Block protein formation
  - Macrolides, Tetracyclines, Aminoglycosides
- Inhibit cell wall formation
  - Pencilline, Cephlosporins
- Interfere with DNA formation
  - Nalidixic acid (synthetic quinilone) , Quinilone
- Prevent folic acid synthesis
  - Sulfonamides

Others

Diuretics

Anti- depressants

Contraceptives

Anti-hypertensive

Athletic steroids

Anti-coagulants

Drug Abuse



What are drugs and What is their normal use?

- A chemical substance that is used in treatment, cure, prevention, or diagnosis of disease or used to otherwise enhance physical or mental well-being.

- They altering biochemical and physiological process of tissue or organism.

Street Drug

- Drug that are taken for non medical reasons(usually for mind altering effects)

- Can lead to
  - Physical damage
  - Mental damage
  - Dependence and addiction

- Include alcohol, heroin, methamphetamine, crack, cocaine and marijuana (Cannabis)

Drug Abuse

- A form of substance-related disorder
- Patterned use of a drug in which the user consumes the substance in amounts or with methods which are harmful to themselves or others
- Drugs most often associated with this term include:
  - Alcohol, substituted amphetamines, barbiturates, benzodiazepines, cocaine, methaqualone, and opioids.

Drug Addiction

- Chronic, often relapsing brain disease that causes compulsive drug seeking and use, despite harmful consequences to the addicted individual and to those around drug addicts.

Difference between drug abuse and addiction

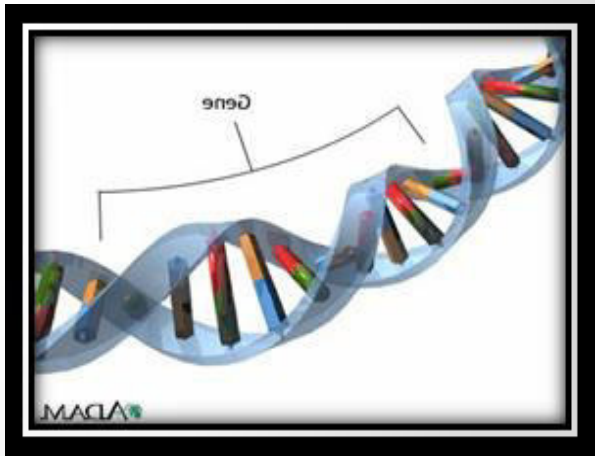
- **Drug abuse** refers to using drugs even though it has become a problem in life.
- **Drug addiction** refers to the substance, taken in higher amounts or taken over a longer period of time than the user intended.

Cause of Substance Abuse

- Not clear, with theories including:
  - A genetic disposition
  - Learned from others
  - A habit which if addiction develops, manifests as a chronic debilitating disease

Genetics

- Factors that make it harder to become addicted also may be genetic.
- Example: an individual may feel sick from a drug that makes other people feel good.
- There is no single addiction gene but many interacting genes.
- Social and environmental factor contribute to risk of addiction



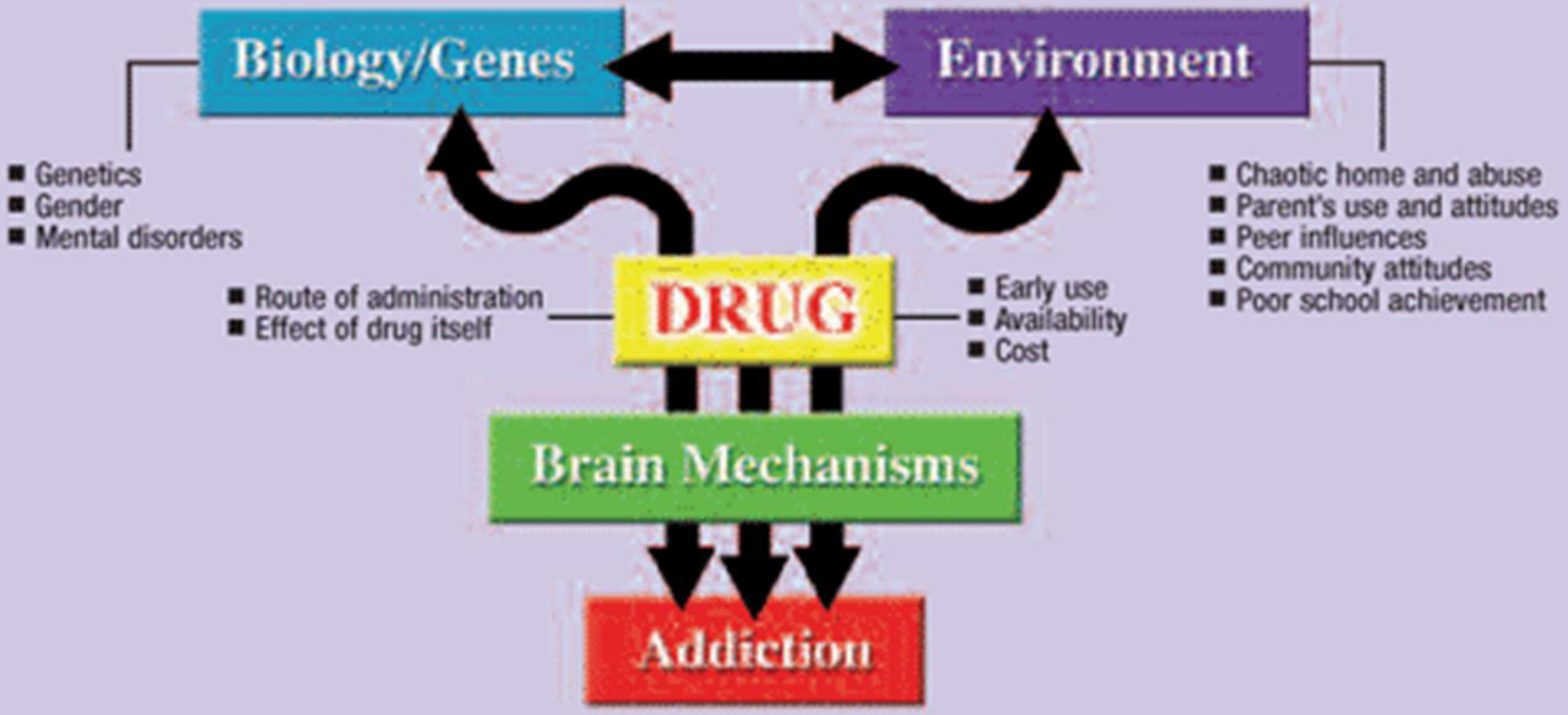
Other Factors

- 1. **Gender**
- Men are twice as likely to have problems with drugs
- 2. **DRUG**
- Cause temporary, pleasurable feelings
- Dependent
- 3. **Mental condition**
- Anxiety, depression and other mental conditions

Individual factors	Environmental factors	Constitutional factors
Personality	Socioeconomic status	Biological/genetic predisposition to use/misuse drugs
Gender	Poverty	
Anxiety	Delinquency	
Stress	Family background/disturbance	
Power needs	Peer pressure	
Age	Ideology/religion	
Intelligence	Educational opportunities	
Psychological health	Educational disturbance	
Life events	Truancy	
Predisposition to take risks	Drug availability	
Hedonism	Drug price	
Self-destructiveness	Unemployment/job opportunities	
Curiosity	Anomie (lack of social/moral standards)	
	Alienation (rebelliousness)	
	Tradition	
	Legal arrangements	
	Historical factors	

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RISK FACTORS



Conclusion

- Just because you are prone to addiction doesn't mean you're going to become addicted. It just means you've got to be careful.

- There are many ways that genes could cause one person to be more vulnerable to addiction than another.

Blood Transfusion

Introduction

- Blood transfusion is defined as the process of receiving blood products into one's circulation intravenously.
- This is usually done as a life saving maneuver to replace blood cells or blood products lost through severe bleeding, during surgery when severe blood loss occurs or to increase the blood count in an anemic patient.
- Blood transfusions involves the use of whole blood, red blood cells, white blood cells, plasma, clotting factors and platelets.

History of Transfusion

- Blood transfused in humans since mid- 1600's
- 1828 – First successful transfusion
- 1900 – Landsteiner described ABO group
- 1916 – First use of blood storage
- 1939 – Levine described the Rh factor

Purpose

- To replace blood lost during surgery or a serious injury.
- To restore oxygen-carrying capacity of the blood.
- To provide plasma factors to prevent or treat bleeding.
- Done if patient's body is not capable of making blood properly because of an illness.

Blood Types and Compatibility

- Each person has one of the following blood types:
- A, B, AB, or O.
- O can be given to anyone but can only receive O.
- AB can receive any type but can only be given to AB.
- Also, every persons blood is either Rh-positive or Rh-negative.

Transfusion



A Better way to Say " Get Well Soon"

✚Donate Blood Save Life

Type	You Can Give Blood To	You Can Receive Blood From
A+	A+, AB+	A+, A-, O+, O-
O+	O+, A+, B+, AB+	O+, O-
B+	B+, AB+	B+, B-, O+, O-
AB+	AB+	Everyone
A-	A+, A-, AB+, AB-	A-, O-
O-	Everyone	O-
B-	B+, B-, AB+, AB-	B-, O-
AB-	AB+, AB-	AB-, A-, B-, O-

Transfusion

- The blood used in a transfusion must be compatible with the patients blood type.
- Type O blood is called the universal donor
- People with type AB blood are called universal recipients
- People with Rh-positive blood can get Rh- positive or Rh-negative blood. But people with Rh-negative blood should get only Rh- negative blood.

Blood Banks

- Blood banks collect, test, and store blood.
- Autologous transfusion - If surgery is scheduled months in advance, patients may be able to donate their own blood and have it stored.

Typing and Cross Matching

- Before any blood can be given to a patient, it must be determined that the blood of the donor is compatible with the patient. The laboratory examination to determine a person's blood group and Rh factor is called Blood Typing.
- The process of determining compatibility between blood specimens is Cross matching. RBCs from the donor blood are mixed with serum from the recipient, a reagent (Coombs'serum) is added, and the mixture is examined for visible agglutination. If no antibodies to the donated RBCs are present in the recipient's serum, agglutination does not occur and the risk of transfusion reaction is small.

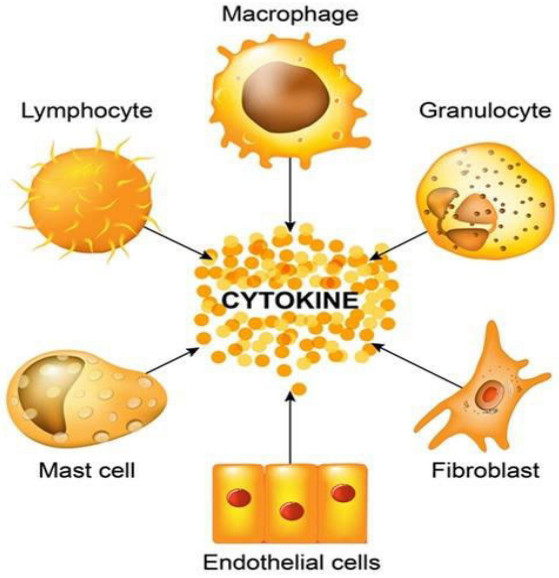
Blood products

- Components of the blood which are collected from a donor for use in blood transfusion.
- Packed red Blood Cells (*PRBCs*)
- Fresh Frozen Plasma (FFP)
- Platelets

Side Effects

- Hemolytic reactions: the patient's own blood destroys the transfused blood. This is due to a human error. Careful patient identification steps are followed to make sure the correct blood is given.
- Bacterial infection (highest with platelet transfusions)
- Itching, skin rash, fever, or feeling cold
- More serious side effects such as trouble breathing are very rare
- HIV (AIDS)
- Hepatitis C
- Hepatitis B

Cytokines


Macrophage
Lymphocyte
Granulocyte
CYTOKINE
Mast cell
Fibroblast
Endothelial cells

Cytokines

- Derived from the Greek word “cyto” means cell and “kinos” means moment.

- Cytokines are low molecular weight regulatory proteins or glycoprotein's (8~80 KD) secreted by white blood cells and various other cells in the body in response to a number of stimuli

General properties of cytokines

- Cytokines are produced by a variety of cell types
- Many cell types can produce more than one cytokine
- Many cytokines play a regulatory role in processes of immunity and inflammation
- Most cytokines are pleiotropic
- Most cytokines are inducible and secreted by their producer cells
- Many cytokines exhibit redundancy

Classification of cytokines

Table The major proteins/protein families that constitute the cytokine group of regulatory molecules

The interleukins (IL-1 to IL-33)
The interferons (IFN-α, -β, -γ, -τ, -ω)
CSFs (G-CSF, M-CSF, GM-CSF)
TNFs (TNF-α, -β)
The neurotrophins (NGF, brain-derived neurotrophic factor (BDNF), NT-3, NT-4/5)
Ciliary neurotrophic factor (CNTF)
Glial cell-derived neurotrophic factor (GDNF)
EGF
EPO
Fibroblast growth factor (FGF)
Leukaemia inhibitory factor (LIF)
Macrophage inflammatory proteins (MIP-1α, -1β, -2)
PDGF
Transforming growth factors (TGF-α, -β)
TPO

[a]M-CSF: macrophage-colony-stimulating factor; NT: neurotrophin.

Interferon

- Interferon's (IFNs) are proteins made and released by host cells in response to the presence of pathogens such as viruses, bacteria, parasites or tumor cells. In a typical scenario, a virus-infected cell will release interferon causing nearby cells to ramp up their anti-viral defenses.
- There are three basic families of interferon
- IFN-α –consist of 166 to 172 A.A having 19-20 kDa mass
- IFN-β- consist of 166 A.A having 20 kDa mass
- IFN-γ- consist of 143 A.A having 17 kDa mass

Important Function of INFs

- Induction of cellular resistance to viral attack;
- Regulation of most aspects of immune function;
- Regulation of growth and differentiation of many cell types;
- Sustenance of early phases of pregnancy in some animal species.

Clearance of Viral Infected Cell

- Exposure of cells to interferon normally results in the induction of a protein kinase termed eIF-2α protein kinase.
- The enzyme, which is synthesized in a catalytically inactive form, is activated by exposure to dsRNA. The activated kinase then phosphorylates its substrate, i.e. eIF-2α, which is the smallest subunit of initiation factor 2 (eIF2).
- This, in turn, blocks construction of the smaller ribosomal subunit, thereby preventing translation of all viral (and cellular) mRNA

Uses of interferon

- iNF-∝2a and 2b are effective against leukemia and malignant tumor of bone marrow
- iNF-∝2b is approved and effective against hepatitis B and C
- iNF-∝3n and rh-iNF are approved for sexually transmitted genital warts caused by papilloma virus
- iNFβ is effective against the chronic disease of NS
- iNFγ effective to treat lieshmania disease caused by parasitic protozoa stay inside the macrophages

Interleukins

- Interleukins represent large family of cytokines, with at least 33 different constituent members (IL-1 to IL-33)
- Only two interleukin-based products have gained approval for general medical use (Proleukin an IL-2 and Neumega an IL-11)
- Many cell types are capable of producing a whole range of interleukins.

Interleukins 2

- IL-2, also known as T-cell growth factor.
- Human IL-2 is a single-chain polypeptide containing 133 amino acids.
- The mature molecule displays a molecular mass ranging from 15 to 20 kDa.
- Initially expressed in E. coli.
- Large-scale IL-2 production was made possible by recombinant DNA technologies.

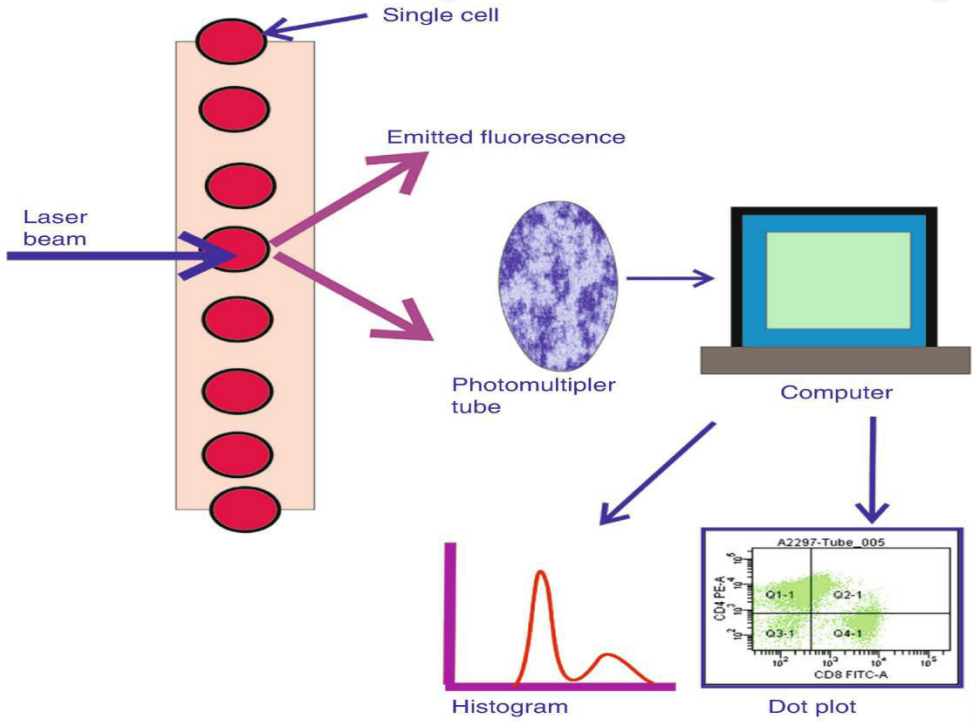
Therapeutic Aspects of Tumor Necrosis Factor

- Many tumors are not susceptible to destruction mediated by TNF (indeed, some tumors produce TNF as an autocrine growth factor)
- Tumor cell necrosis is not TNF's major biological activity
- Severe side effects usually accompany systemic administration of therapeutically relevant doses of this cytokine

Some diseases in which TNF is known to mediate many of the symptoms

Disease	Symptoms induced by TNF
Cancer	Cachexia Stimulation of growth of some tumours
Septic shock	Vascular leakage Tissue necrosis Hypotension Activates blood clotting
Rheumatoid arthritis	Tissue inflammation Possible role in destruction of joints
MS	Inflammation
Diabetes	Death of pancreatic islet cells induces insulin resistance

Flow Cytometry

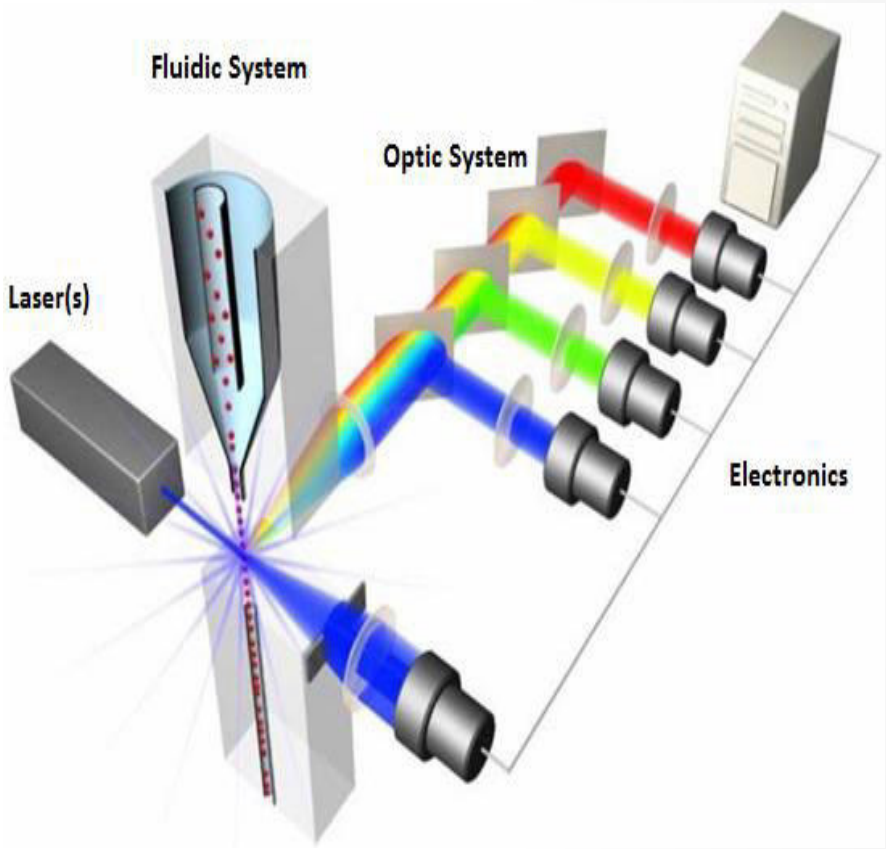


Introduction

- Flow cytometry is a technique of quantitative single cell analysis.
- This technique was first described by Wallace Coulter in the 1950s.
- The flow cytometer was developed in the 1970's and applied to automated cell counting.
- Flow cytometer count, examine and sort cells based on their optical properties (Scattering and fluorescence).
- The present "state-of-the-art" flow cytometers are capable of analyzing upto 13 parameters (forward scatter, side scatter, 11 colors of immunofluorescence).

Components of Flow Cytometer

1. Lasers
2. Dichroic mirrors
3. Filters
4. Detectors



Basic Principle of Flow Cytometry

- Prepared single cell or particle suspension is necessary for flow cytometric analysis.
- The suspension of cells or particles is aspirated into a channel surrounded by a narrow fluid system.
- They pass one at a time through a focused laser beam.
- The light is either scattered or absorbed when it strikes a cell.

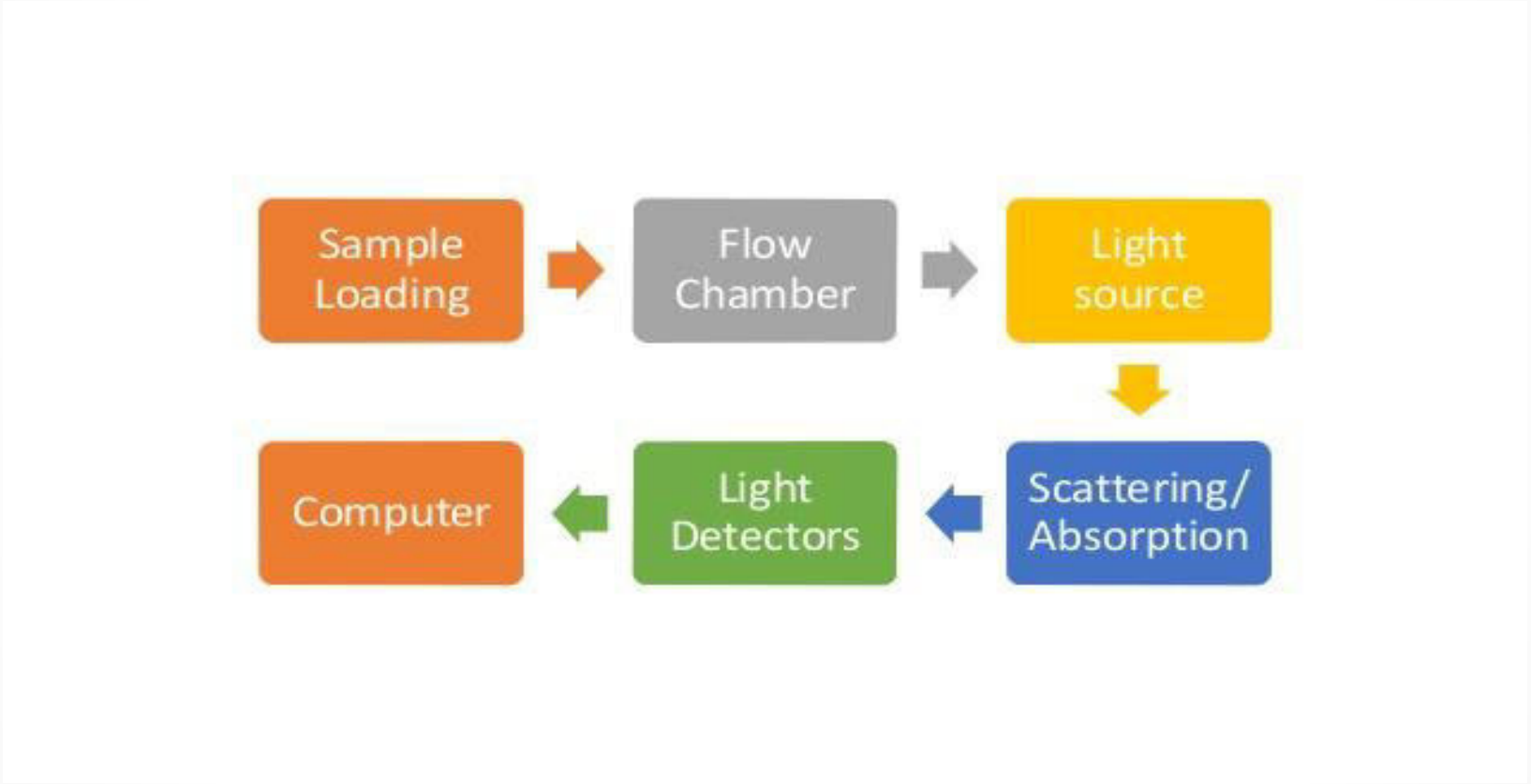
Principle

- Light scattering is dependent on the internal structure of the cell and its size and shape.
- Absorbed light of the appropriate wavelength may be re-emitted as fluorescence. (The cell may have a naturally fluorescent substance or one or more fluorochrome-labelled antibodies are attached to surface or internal cell structures)

Principle

- Light and/or fluorescence scatter signals are detected by a series of photodiodes and amplified.
- Optical filters are essential to block unwanted light and permit light of the desired wavelength to reach the photodetector.
- Fluorescein isothiocynate (FITS), Texas red and phycoerythrin (PE) are the most common fluorescent dyes used in the biomedical sciences.
- Large number of cells are analyzed in a short period of time (>1,000/sec).

Working Process



Working of Flow Cytometer

- A flow cytometer is composed of three main systems:
- **Fluidics** – Transport cells in a stream to the laser beam for interrogation.
- **Optics** – Consist of lasers to illuminate the cells in the sample stream and optical filters to direct the resulting light signals to the appropriate detectors.
- **Electronics** – Converts the detected light signals into electrical signals that can be processed by computer.

Fluidics System

- Flow cytometers use the principle of hydrodynamic focusing for presenting cells one at a time to a light source.
- The fluidics system consist of a central channel through which the sample is injected, enclosed by an outer sheath that contains faster flowing fluid.

Optics

- Light scattering or fluorescence emission provides information about the cell's properties.
- Light that is scattered by an object is detected by different detectors.
- One detector is placed in line with the beam to measure the forward scatter (FSC) from the objects.
- Detectors perpendicular to the beam measure side scatter (SSC) and fluorescence.

Optical Detectors

- Scattered and emitted light from cells are converted to electrical pulses by optical detectors.
- Once a cell or particle passes through the laser light, the scattered and fluorescence signals are diverted to the detectors.
- Detectors are either silicon photodiodes or photomultiplier tubes (PMTs).
- The photodiode is less sensitive to light signals than the PMTs and thus is used to detect the stronger FSC.
- PMTs are used to detect the weaker signals generated by SSC and fluorescence.

Types of Filter

- There are three major type of filters:
- **Long pass Filter:** Transmits wavelength of light equal to or greater than the spectral band of the filter.
- **Short pass Filter:** Transmits wavelength of light equal to or shorter than the spectral band of the filter.
- **Bandpass Filter:** Transmits wavelength of light within a specific range of wavelengths.

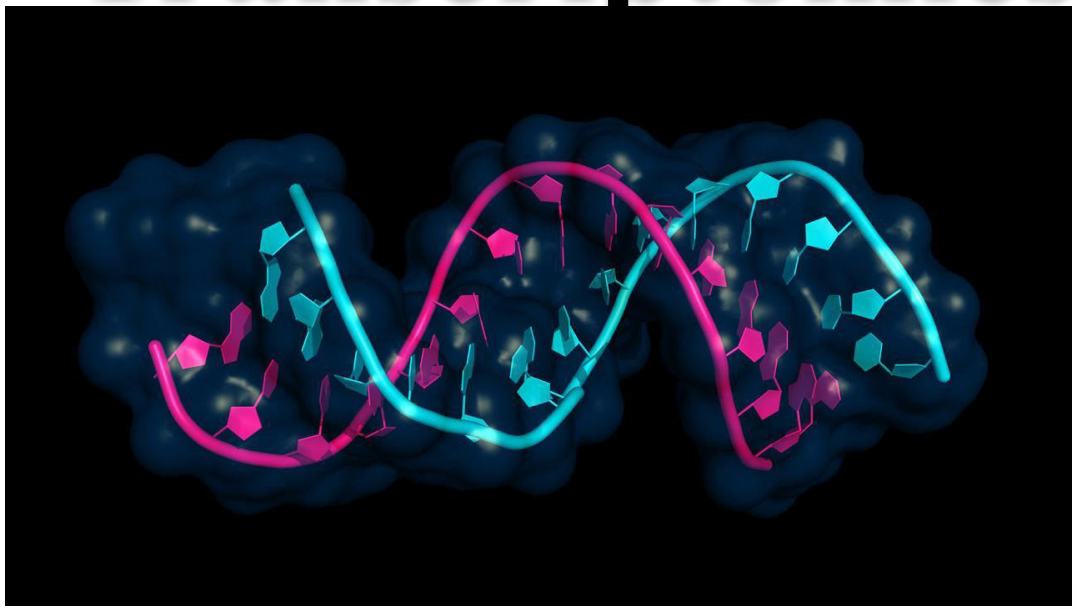
Electronics (Signal Processing)

- Scattered and emitted light data can be converted to electrical pulses by optical detectors
- Flow cytometry data may be represented as histograms or dot plots

Clinical Applications of Flow Cytometry

- Diagnosis of Hematologic Malignancies
- Detection of Minimal Residual Disease
- Lymphocyte Subset Enumeration (HIV)
- Efficacy of Cancer Chemotherapy
- Reticulocyte Enumeration
- Cell Function Analysis
- Application in Organ Transplantation

Transcriptomics



Transcriptome

- The transcriptome is the complete set of transcripts in a cell and their quantity, for a specific developmental stage or physiological condition.
- Understanding the transcriptome is essential for interpreting the functional elements of the genome and revealing the molecular constituents of cells and tissues, and also for understanding development and disease.

Transcriptomics

- Transcriptomics, the study of RNA in any of its forms.
- The transcriptome is the set of all RNA molecules, including mRNA, rRNA, tRNA, and other non-coding RNA produced in one or a population of cells.

Transcriptomics scope

- The term can be applied to the total set of transcripts in a given organism, or to the specific subset of transcripts present in a particular cell type.
- Unlike the genome, which is roughly fixed for a given cell line (excluding mutations), the transcriptome can vary with external environmental conditions.

Transcriptomics scope

- Because it includes all mRNA transcripts in the cell, the transcriptome reflects the genes that are being actively expressed at any given time, with the exception of mRNA degradation phenomena such as transcriptional attenuation (Premature termination of transcription).
- The study of transcriptomics, also referred to as expression profiling, examines the expression level of mRNAs in a given cell population, often using high-throughput techniques based on DNA microarray technology.

Transcriptomics Aims

I. To catalogue all species of transcripts, including mRNAs, noncoding RNAs and small RNAs.

II. To determine the transcriptional structure of genes, in terms of their start sites, 5′ and 3′ ends, splicing patterns and other post-transcriptional modifications.

III. To quantify the changing expression levels of each transcript during development and under different conditions.

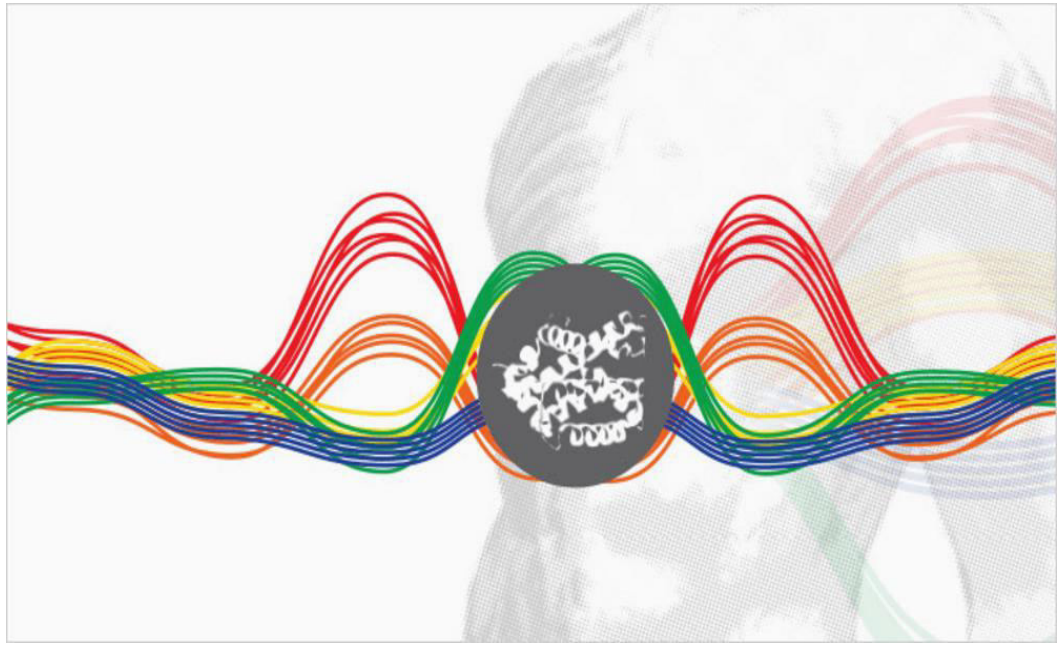
Technologies

- **Hybridization-based approaches**
  - Fluorescently labelled cDNA with custom-made microarrays
  - Commercial high-density oligo microarrays
- **Sequence-based approaches**
  - Sanger sequencing of cDNA or EST libraries
  - Serial analysis of gene expression (SAGE)
  - Cap analysis of gene expression (CAGE)
  - Massively parallel signature sequencing (MPSS)

Applications

- One way to directly or indirectly identify biomarkers is with the use of transcriptomics.
- This technique enables the analysis of the entire set of RNA transcripts in an organism.
- The transcriptome consists of many different RNA species that are categorized into messenger RNAs (mRNA) and non-coding RNAs
- Bacterial transcriptome analysis could lead to the identification of novel antigens

Proteomics



Introduction

- The terms proteome and proteomics were coined by Mark Wilkins and colleagues in the early 1990's
- Proteome : is the complement protein found in a single cell in a particular environment/ is complete collection of proteins encoded by genome of an organism.
- Proteomics : is the study of composition, structure, function and interaction of the proteins directing the activities of each living cell

Types of Proteomics

- Interaction proteomics
  - Protein-protein association
- Expression proteomics
  - Protein quantification

Life Cycle of Protein

- The level of any protein in the cell at any given time is controlled by
  - Rate of transcription of the gene
  - The efficiency of translation of m RNA into protein
  - The rate of degradation of protein in the cell

Protein structure

- **Primary structure:** is sequence of specific amino acid
- **Secondary structure:** the primary polypeptide chain gets properly folded In the form of alpha-helix and β-pleated sheets
- **Tertiary structure:** secondary structure interact with each other chemically to form the 3 dimensional shape of the proteins.
- **Quaternary structure:** interaction between different polypeptide unit helix, beta pleated sheet, random coils and turns

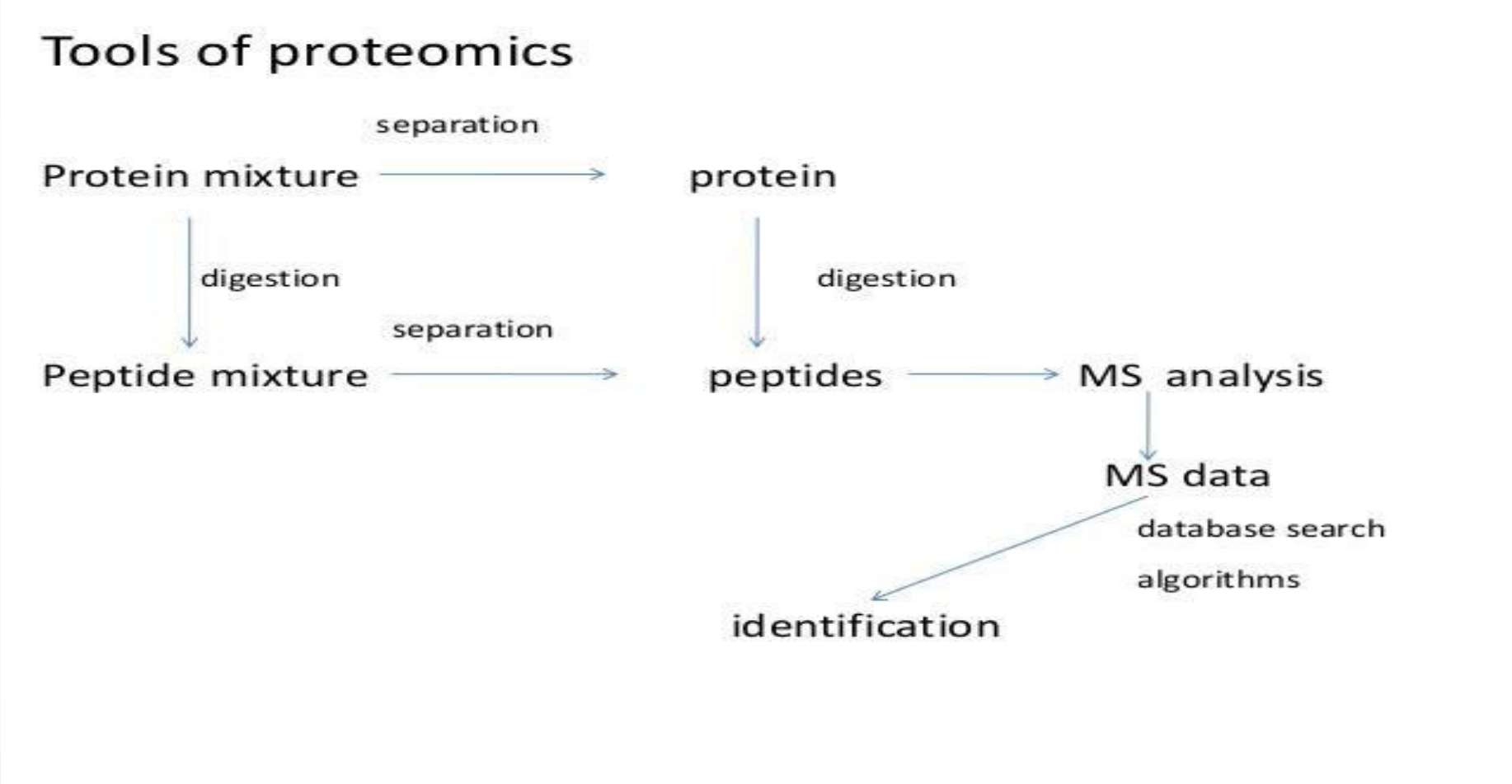
- **Domains:** discrete portions of the proteins that fold independently from the rest of protein and they have their own function and serve as one of the building blocks of that proteins.
- **Motif:** domains contain a region of conserved pattern of amino acids or a conserved combinations of the structural elements formed by folding of near by amino acid sequences.
- Determining the protein structure/ polypeptide sequence by:

i. x-ray crystallography
ii. Nuclear magnetic resonance
iii. Protein predicting programmes- computer based

Post Translational Modifications

- **Glycosylation:** structural components of cell surface helps in binding by the receptors and eliciting immune response. eg. ABO blood group antigen, Human IgG
- **Phosphorylation:** Play a role in the regulation of many cell processes such as growth and cell cycle control.
- **Sulfation:** Plays role in strengthening protein-protein interactions

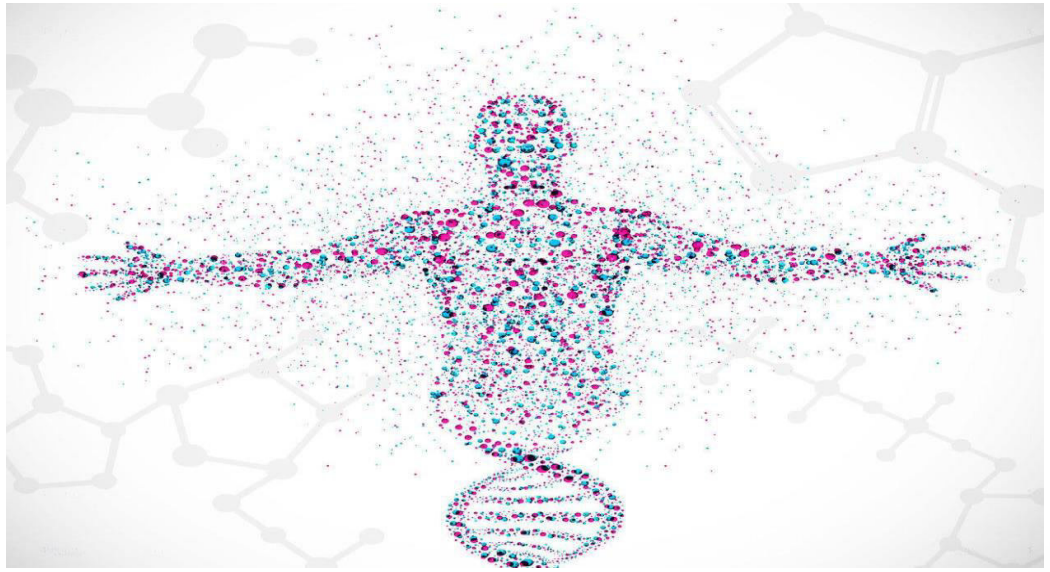
Proteomics Tools



Applications of Proteomics

1. **Mining**-Identification of all of the proteins in a sample
2. **Protein expression profiling**-Identification of proteins in a sample as a function of particular state of a cell i.e., two states of a particular systems are compared
3. **Protein network mapping**-To determine protein interaction with each other in a living system
4. **Mapping of protein modifications**-Task to identify how and where proteins are modified post translationally

Metabolomics



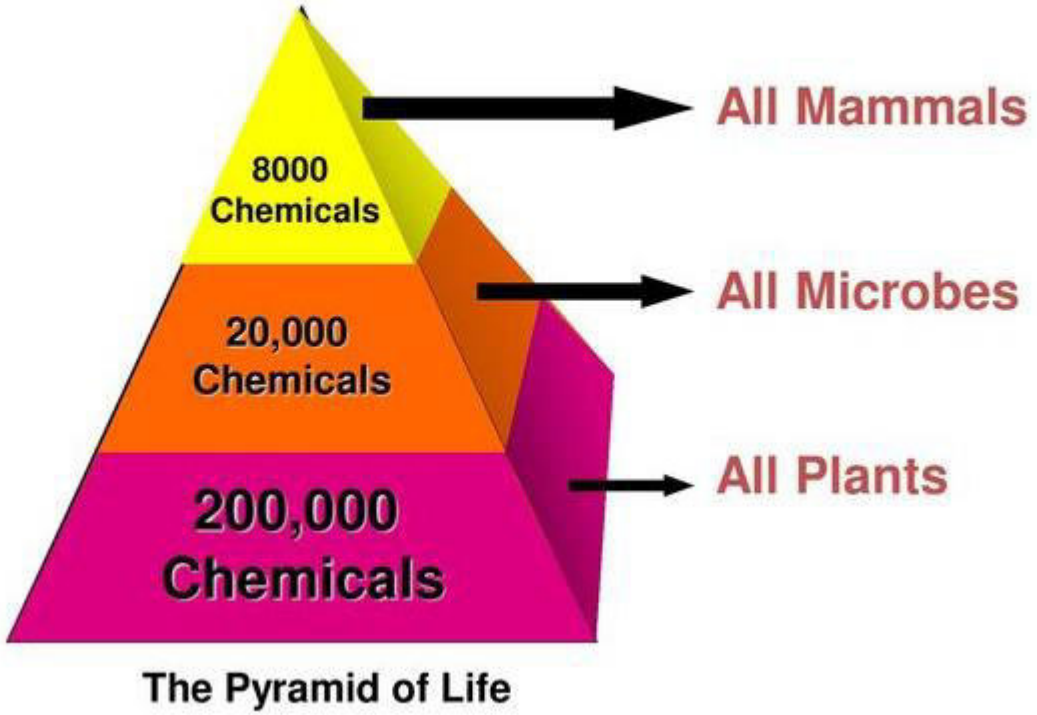
Introduction

- Metabolomes are group of small molecules- Metabolites
- **Metabolites**
  - Chemical molecules
  - Essential in metabolic processes
  - Example : proteins , lipids

Metabolome

- The complete collection of small molecule chemicals found within a biological sample
- The biological sample can be a cell, a cellular organelle, an organ, a tissue, a tissue extract, a biofluid or an entire organism

Different Metabolomes



Metabolites

- Intermediates and products of metabolism
- MW < 1500 Da
- Examples- antibiotics, pigments, carbohydrates, fatty acids and amino acids
- Primary and secondary metabolites

Metabolomics

- Newly emerging field of 'omics' research
- Field of Comprehensive and simultaneous systematic determination of metabolites
- It includes, study of levels in the metabolome, Types
- Deals with function, Structural changes, Effects
- Also known as metabonomics
- One of computational science among omics
- It involves in quantitative measurement of the dynamic multi- parametric metabolic response of living systems

Cont…

- Deals with, systematic identification and quantitation of all/ biologically relevant small molecules in a given compartment, cell, tissue or body fluid
- It represents the functional end-point of physiological and pathophysiological processes depicting both genetic predisposition and environmental influences like nutrition, exercise or medication

Aims

- Four main points in Analysis of metabolomics data
  - Efficient and unbiased
  - Separation of analytes
  - Detection
  - Identification and quantification

Detection Techniques

- **Separation Techniques**
  - Gas Chromatography (GC)
  - Capillary Electrophoresis (CE)
  - High Performance Liquid Chromatography (HPLC)
  - Ultra Performance Liquid Chromatography (UPLC)
- **Combination of Techniques**
  - GC-MS
  - HPLC-MS
- **Detection Techniques**
  - Nuclear Magnetic Resonance Spectroscopy (NMR)
  - Mass Spectrometry (MS)

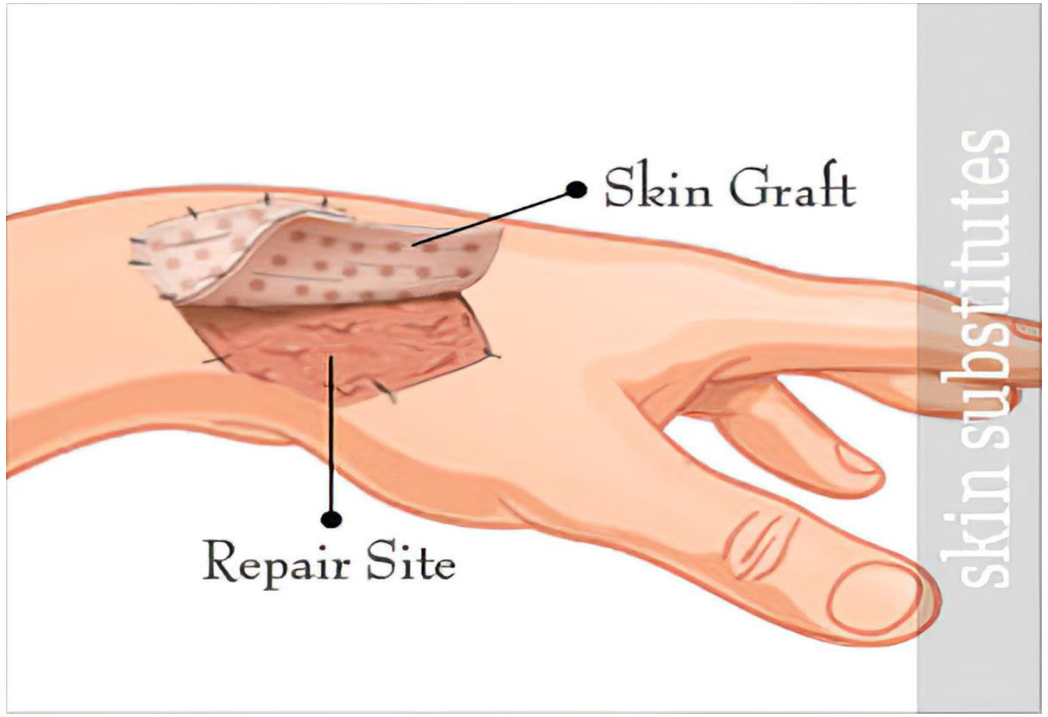
Importance of metabolites

- >95% of all diagnostic clinical assays test for small molecules
- 89% of all known drugs are small molecules
- 50% of all drugs are derived from pre-existing metabolites
- 30% of identified genetic disorders involve diseases of small molecule metabolism
- Small molecules serve as cofactors and signaling molecules to 1000's of proteins

Applications

- Toxicology Testing
- Clinical Trial Testing
- Fermentation Monitoring
- Food & Beverage Tests
- Nutraceutical Analysis
- Drug Phenotyping
- Water Quality Testing
- Petrochemical Analysis

Bioartificial Skin

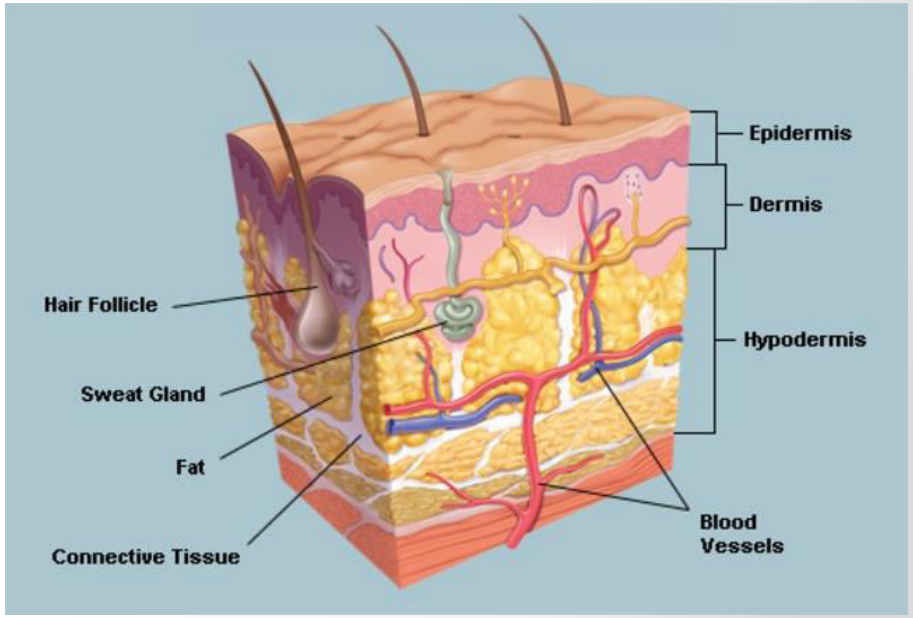


Bioartificial Skin

- The term artificial skin is used to describe any material used to replace (permanently or temporarily) or to mimic the dermal and epidermal layers of the skin
- The primary current application of artificial skin is for the treatment of skin loss or damage on burn patients
- Alternatively however, artificial skin is now being used in some places to treat patients with skin diseases, such as diabetic foot ulcers etc

Anatomical Overview of Skin

- Human skin is comprised of two primary layers, the dermis and the epidermis
- A diagram of a typical section of human skin is shown here
- The epidermis contains no blood vessels meaning transport of nutrients to the epidermis occurs from the dermis below



Historical Overview

- 1871, England: Pollock proposes skin grafts for burn treatment
- 1998, United States: First tissue engineered skin is approved by the FDA
- 2001, United States: Dermagraft, Orcel, Composite Cultured Skin Graft are FDA approved
- 2002, United States: Integra is FDA approved for treatment of severe burns

How Artificial skin is Made

- Skin is usually donated by other donors
- Fibroblasts are removed from the donated skin and are frozen until they are needed
- The fibroblasts are placed on a polymeric mesh scaffolding, gather oxygen, and grow new cells.
- The cells are then transferred to a culture system.

- After 4 weeks the polymer mesh dissolves and leaves behind a new layer of dermal skin
- When the growth cycle is completed, they add more nutrients
- Keratinocytes are added to the collagen and are exposed to air to form epidermal layers
- The skin is now completed and is stored in sterile contains until ready to use

How It Works

- Artificial skin is secured over wound during surgery
- The skin remains in place for several weeks and allows new tissue to grow in to bottom matrix layer
- Top layer provides protection from infection and dehydration

Companies

- Bioartificial skin is produced by many companies but the initial ones are

1. Integra
2. Advance tissue sciences (ATS)
3. Organogenesis

- These companies grow their skin and provide it to others.
- Ultimate source of skin is foreskin after infant circumcision.
- These infantile cells have the ability to divide and have long life as compared to adult cells

Areas of Applications

- Artificial Skins are primarily used for the Treatment of Skin loss or damage on burn Patients.
- Alternative Areas of Application of Artificial Skins includes
  - Treatment of patients with skin diseases, such as diabetic foot ulcers, and severe scarring.
  - Plastics and Cosmetic Surgery.

Advantages

- Chances of survival for burn patients
- Artificial skin seals the wound preventing fluid loss and bacteria from entering through the wound.
- The fear of Stigmatization of the Patient is eliminated

Disadvantages

- Risks of Infection and Rejection by the Patients
- Loss of Sensitivity
- Cut of Blood Supply
- Complication could arise due to Skin Adhesion and/or fluid buildup between the wounded site and the transplanted skin
- Artificial Skins are very expensive

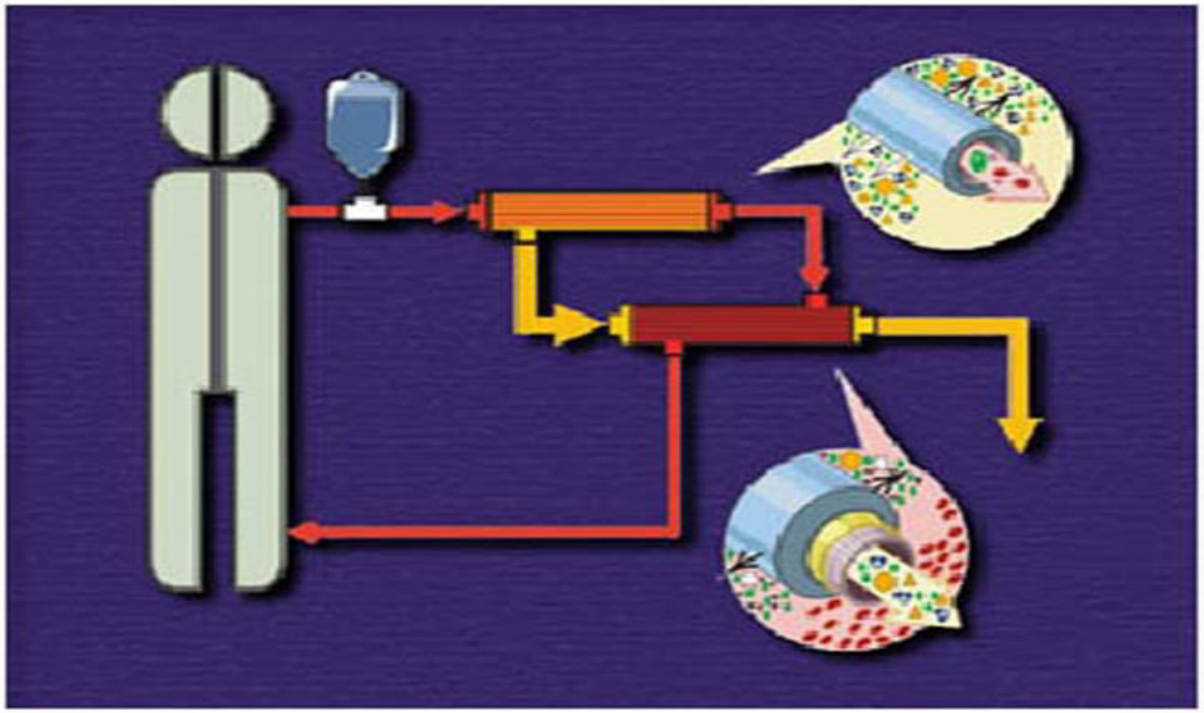
Current Research

- Though artificial skin has aided significantly in skin regeneration, there remain several areas for improvement.
- Ideally, this would allow in vitro replication of a patient's own genetically modified skin cells. These cells could then be put into the artificial matrix for bacteria-free growth.
- Another current trend in Artificial is the creation of Electronic Skin.
- Scientist are working towards the Incorporation of flexible pressure transducers and Bioreceptors to the Artificial Skin, these will give a sense of Touch to the Patients

Conclusion

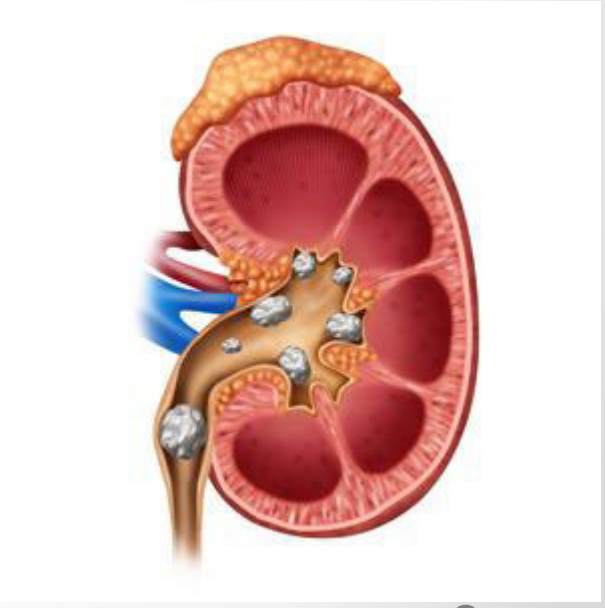
- The ultimate goals of current artificial skin technologies are to provide protection from infection, dehydration, and protein loss after severe skin loss or damage.

Bioartificial Kidney



Kidneys

- Kidneys are the main excretory organs of the body.
- Located near the back of the abdomen behind the intestines.
- The kidneys are two reddish-brown bean shaped organs 10cm long and 6cm wide.



Main Functions of Kidney

- Ion Balance - Especially Na+ and K+
- Osmotic Balance - Determine volume of urine produced
- Blood Pressure - Controls blood volume and ECF volume
- pH Balance - Retains or excretes H+ or HCO3-
- Excretion - of nitrogenous wastes & other hydrophilic toxins
- Hormone Production-Renin, Erythropoietin

Artificial Kidney (Hemodialyzers)

- Artificial kidneys or hemodialyzers are used in extracorporeal renal-therapies for removal of uremic solutes and excess plasma water from the blood of patients with kidney failure.

History of Artificial Kidney

- The first scientific description of hemodialysis principle was published by Graham in 1854
- The first prototype hemodialyzer was developed by Abel, Rowntree and Tunner in 1913
- The first hemodialyzer used on human was reported by Hass in 1923.
- Kolf developed the rotating drum artificial kidney in 1945 and is succeeded in treating ARF for the first time and it is modified in 1956

- In 1960,a plate and frame hemodialyzer called Killi dialyzer was developed by Killi.
- In 1964 ,the hollow-fiber hemodialyzers called the capillary kidney was First proposed by Stewart, Creny and Mahon.
- Hollow-fiber type hemodialyzers are the most widely used

How does Artificial Kidney works?

- Artificial kidney or hemodialyzer works on the principle of dialysis which is the diffusion of small solute molecules through a semi permeable membrane.
- Blood is removed from the body and pumped by a machine outside the body into a dialyzer (artificial kidney)
- The dialyzer filters metabolic waste products from the blood and then returns the purified blood to the person
- The total amount of fluid returned can be adjusted
- A person typically undergoes hemodialysis at a dialysis center
- Dialysate is the solution used by the dialyzer

- HD consists of perfusion of heparinized blood and physiologic salt solution on opposite sides of a semi permeable membrane
- Waste products (urea, creatinine,…etc) move from blood into the dialysate by passive diffusion along concentration gradient
- Diffusion rate depends on;
- 1. The difference between solute concentrations in the blood and dialysate
- 2. Solute characteristics
- 3. Dialysis filter composition
- 4. Blood and dialysate flow rate

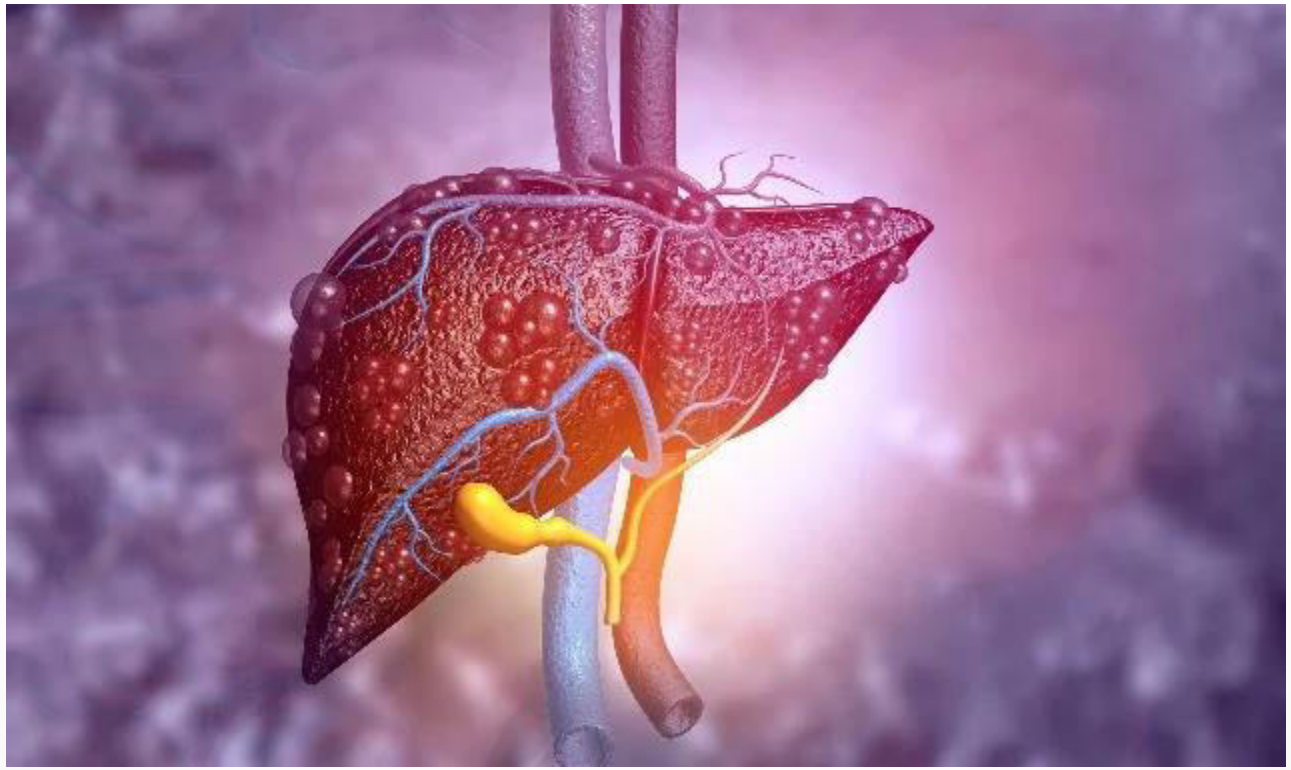
Dialysis membrane

- Dialysis membrane act as a semi permeable barrier for the removal of uremic solute and excess plasma water from the plasma water.
- They very according to their morphology and chemical composition
- The morphology determine transport properties of the uremic solute while chemical composition determines the biocompatibility of the molecule

Classification of Dialysis Membrane

- Dialysis membrane can be classified in to three based up on their polymeric composition and preparation process. They are:
- Regenerated cellulosic membrane
- Modified cellulosic membrane
- Synthetic membrane

Bioartificial Liver



Acute liver failure

- Acute liver failure is loss of liver function that occurs rapidly in days or weeks usually in a person who has no pre-existing liver disease.
- Less common than chronic liver failure, which develops more slowly.
- Also known as fulminant hepatic failure, it can cause serious complications, including excessive bleeding and increasing pressure in the brain. It's a medical emergency that requires hospitalization.
- Depending on the cause, acute liver failure can sometimes be reversed with treatment. Unfortunately, a liver transplant is often the only cure.

Liver and You

- What the liver does is filter your blood and removes harmful toxins that could damage other organs in your body
- There are still a few drugs and diseases that can lead to acute liver failure

Liver Physiology

- Among the most important liver functions are:
- Removing or excreting body wastes as well as drugs
- Synthesizing plasma proteins, including those necessary for blood clotting.
- Producing bile to aid in digestion
- Storing certain vitamins, minerals
- Excretion of Bilirubin

Requirements for Bioartificial Liver

- FDA has number of requirements that mainly deal with patients advocacy and safety.
  - If a cellular component is used, it must be purified, every component in it must be clearly identified.
  - It must not transmit any infectious disease.
  - It must stay viable and active.

Bioartificial liver device

- A bioartificial liver device (BAL) is an artificial extracorporeal supportive device for an individual who is suffering from acute liver failure.
- The purpose of BAL-type devices, currently, is not to permanently replace liver functions, but to serve as a supportive device, either allowing the liver to regenerate properly upon acute liver failure, or to bridge the individual's liver functions until a transplant is possible.
- BALs are essentially bioreactors, with embedded hepatocytes (liver cells) that perform the functions of a normal liver.
- Several types of BALs are being developed, including hollow fiber systems and flat membrane sheet systems.
- Currently two companies bioartificial liver are in clinical trial which are of hollow fiber system e.g., LIVERAID by Arbios systems Inc. and ELAD from Vital therapies

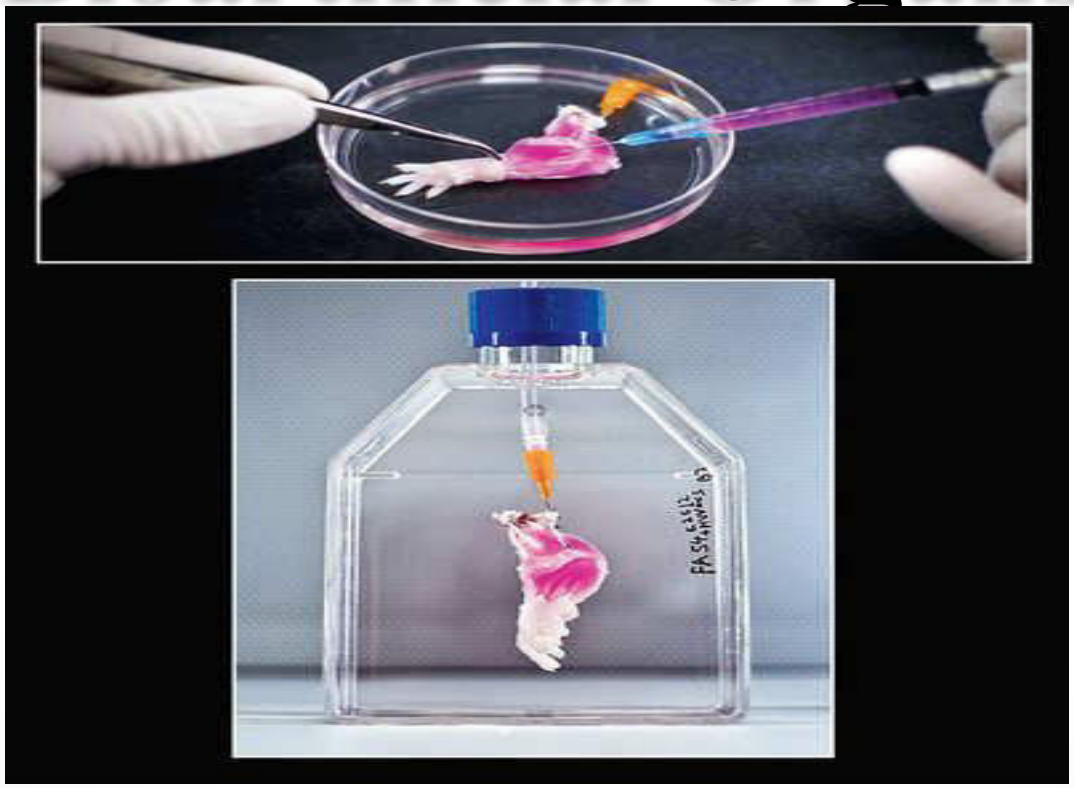
Hollow Fiber System

- One type of BAL is similar to kidney dialysis systems that employ a hollow fiber cartridge.
- Hepatocytes are suspended in a gel solution, such as collagen, which is injected into a series of hollow fibers. In the case of collagen, the suspension is then gelled within the fibers, usually by a temperature change.
- The hepatocytes then contract the gel by their attachment to the collagen matrix, reducing the volume of the suspension and creating a flow space within the fibers. Nutrient media is circulated through the fibers to sustain the cells.
- During use, plasma is removed from the patients blood. The patient's plasma is fed into the space surrounding the fibers.
- The fibers, which are composed of a semi-permeable membrane, facilitate transfer of toxins, nutrients and other chemicals between the blood and the suspended cells. The membrane also keeps immune bodies, such as immunoglobulins, from passing to the cells to prevent an immune system rejection.

LIVERAID & ELAD

- **LIVERAID**
- In LIVERAID pig hepatocytes are used.
- Phase I trial is conducted in which 32 patients were successfully bridged to liver transplantation.
- 30 day patient trial was 90%.
- **ELAD**
- Human hepatocytes (440g) are used
- Phase I trial completed and pilot studies have been reported.

Bioartificial Organs



Artificial Organ

- An Artificial organ may be defined as a human made device designed to replace, duplicate or augment, functionally or cosmetically a missing, diseased or otherwise incompetent part of the body, either temporarily or permanently and which requires a non-biologic material interface with the living tissue.

Introduction

- A biological device that contains a biological component and a device component that function in a unique association
- **Biologic component**: Isolated cells, cell lines or living tissues.
- **Synthetic component**: biopolymers, plastic or permeability selective membranes among other parts.

Different Aspects of Artificial Organs

- Artificial organs can conveniently be classed into four groups:

I. Bone/Joint Replacements (e.g. hip, knee, finger, total limb),

II. Skin/Soft Tissue Replacements (e.g. skin, breast, muscle),

III. Internal Organs (e.g. heart, kidney, blood vessels, liver, pancreas)

IV. Sensory Organs (e.g. eye, ear).

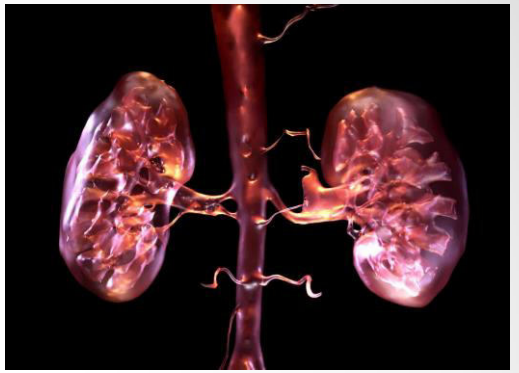
Manufacturing Processes of Artificial Organs

- The various processes involved in the manufacturing of artificial organs involve the following:
- **Tissue Engineering**: This is the use of a combination of cells, engineering and materials methods, and suitable biochemical and physico-chemical factors to improve or replace biological functions.
- **Bioprinting or 3D printing of tissues**: Bioprinting, where living cells are precisely printed in a certain pattern, has great potential and promise for fabricating engineered living organs.

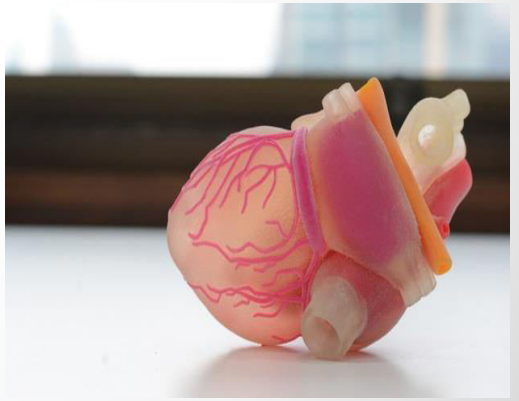
- **Industrially** by using biomaterials which includes metals, ceramics, polymers and composites.
- In the laboratory by using stem cells under **Regenerative Medicine**

Areas of Applications

- Artificial Bone
- Artificial Skin
- Bionics
- Biomedical Engineering
- Cochlear Implant
- Ocular prosthetic
- Organ Anatomy
- Organ Transplant
- Plastic Surgery
- Prosthesis
- Tissue Scaffold







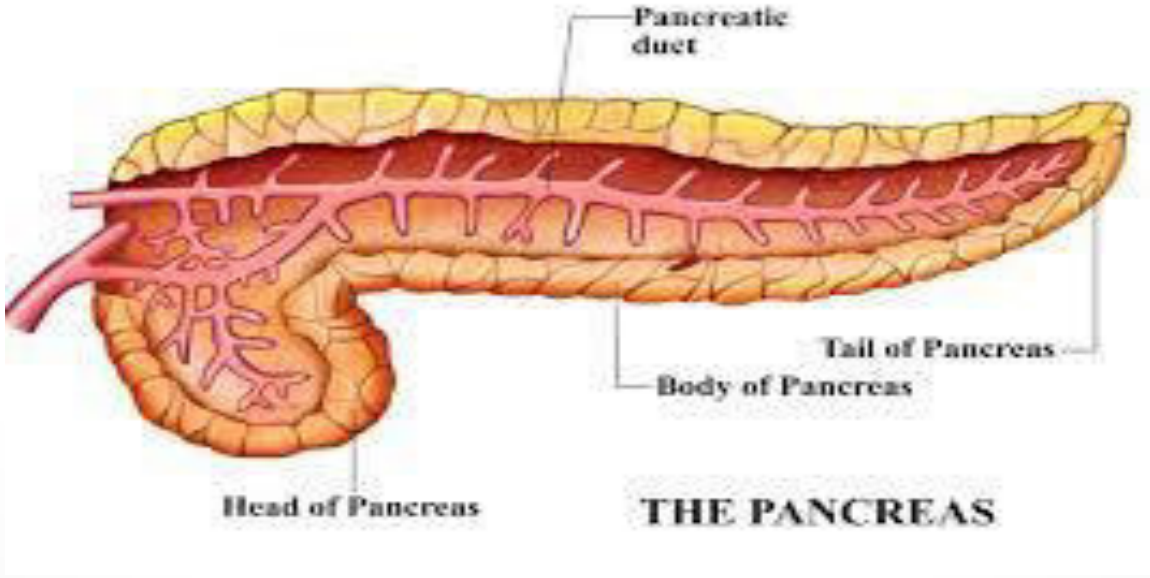
Advantages

- Artificial organs can replace diseased or damaged organs, thereby, providing the ailing patient with an opportunity to lead a healthy and normal life.
- Artificial organs can meet the huge demand of healthy donor organs. There is a huge list of patients who are in urgent need of healthy organs but are unable to find a suitable willing donor.
- A major stumbling block in the form of organ rejection can be solved due to artificial organs. As artificial organs are created by taking the stem cells of the same person and of the same organ, the possibility of rejection has been reduced significantly.
- With the help of regenerative medicine or artificial organ therapy, burn victims can even have a new skin.

Disadvantages

- A major concern is the possible presence of the disease in the base tissue which is used to create the organ. Sometimes, even a foreign body tissue is used to regenerate or reconstruct the organ. In such cases, there is a possibility that the tissue is already infected by other diseases.
- The entire cost of growing and transplanting an artificial organ is prohibitive, and thus, limit the scope of its application to the general public.
- There are high chances of organ failure, and the body may even take some time to adapt to the new organ. How the body reacts to the new organ may vary from person to person.
- If there is a problem with the functioning of the organ, you might need to go for another transplant

Bioartificial Pancreas



Normal Functioning Pancreas

- The pancreas is an organ located in the abdomen.
- It plays an essential role in converting the food we eat into fuel for the body's cells.
- The pancreas has two main functions: an exocrine function that helps in digestion and an endocrine function that regulates blood sugar.

Why the need for an Artificial pancreas?

- Pancreas fails to create proper amounts of insulin to counter blood glucose levels
- Patients must frequently screen blood glucose by finger sticks
- Self-injections or machine injections of insulin
- Difficult to determine correct amount of insulin based on meals, exercise, stress and changing schedules
- Diabetes maintenance is tedious and expensive
- The FDA has been cooperating with diabetes patients, care providers, medical device manufactures, and researchers to progress the advancement of an artificial pancreas

Bioartificial Pancreas

- Bioartificial pancreas has the ability in preventing or reversing complications associated with the disease
- Construct is based on encapsulation of islet cells with a semipermeable membrane so that cells can be protected from host immune system
- Encapsulation of islets eliminates the requirement of immunosuppressive drugs
- **Requirements for Bioartificial Pancreas**
- A steady supply of islets cells
- Islets cell viability must be maintained with appropriate oxygen supply and nutrients
- Islets must be protected by immune system
- Practical means of implantation must exist

- Hydrogel hybrid-type organs or pancreas.
- In this system islets are encapsulated inside a semipermeable membrane, which secrete specific compounds in response to body fluids.
- **Two methods used to obtain such systems are**
- Microencapsulation
- Special large containers.

Encapsulation

- First attempt is based on additional covering of alginate matrix by poly vinyl layer crosslinked with glutaraldehyde.
- Second approach is additional covering of alginate matrix by poly (methyl methacrylate) membrane.
- By selecting appropriate condition of encapsulation low molecular weight compounds and low molecular weight proteins can be allowed to pass through the semipermeable membrane.

Design objectives

1. **Keep the cells alive**
- Material containing islets is biocompatible
- Process for fabrication must not damage islets
- Dimensions permit rapid diffusion of nutrients

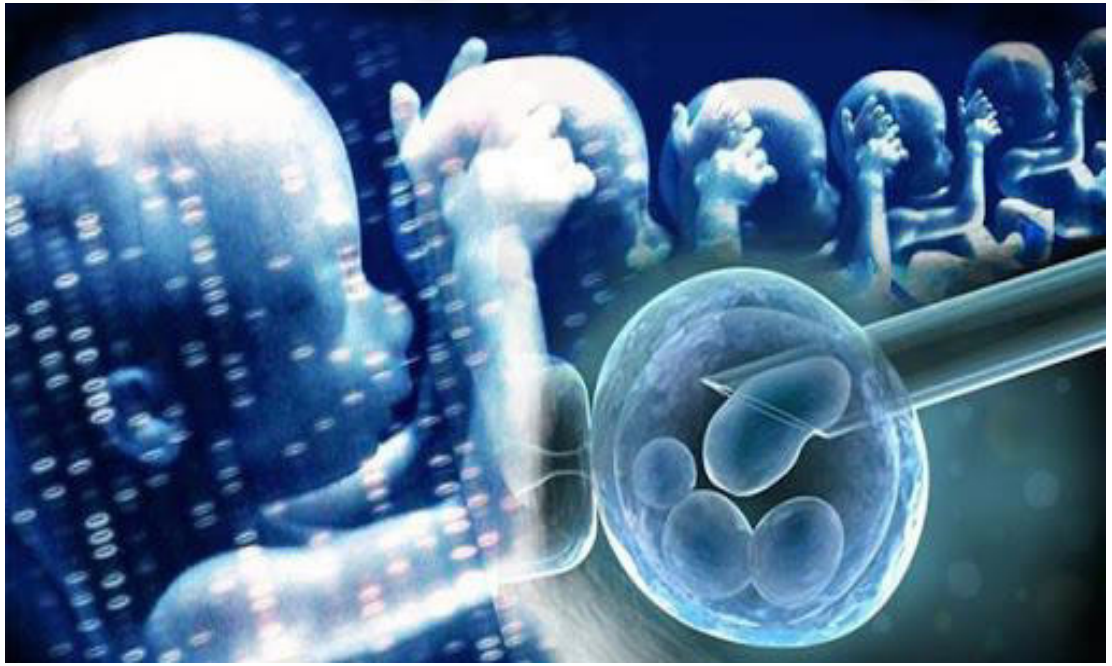
2. **Prevent destructive Host response**
- Outer surface should be totally bio invisible
- All islets must be completely covered to prevent immune response
- Permeability is controllable

3. **Assure practical surgical implantation**
- Islets density must be high
- Device retrievable
- Bio-stable long –term
- Chemically and physically durable

Advantages of Bioartificial Pancreas.

- No immune suppression required
- Xenografts (use of animal tissue)
- Retrievable
- Replaceable.
- **Disadvantages**
- To date no product in in market.
- Cells inside the encapsulation die because of no oxygen availability because of induction of immune response.
- Dimensions does not allow penetration of oxygen towards the core and cells die.

Cloning Strategies



Cloning

- 'Cloning' describes many different processes used to produce genetically identical copies of biological matter.
- The copied material is called the 'clone’.
- The clone has the same genetic makeup as the original.

Clones

1938 – the first idea of cloning

1952 – an attempt to clone a Rana pipiens frog

1996 – the first mammal cloned from a cell taken from an adult animal – Dolly the sheep

1998 – the first cloned mouse (it was called Cumulina)

March–April 2003 – a rabbit is cloned in France and Southern Korea

2009 – the first animal from an extinct species cloned: Pyrenean ibex. The animal lived seven minutes. It died of lung malformations

Types of Cloning

- Recombinant cloning
- Therapeutics cloning
- Reproductive cloning

Recombinant Cloning

- Recombinant DNA (rDNA) molecules are DNA molecules formed by laboratory methods of genetic recombination to bring together genetic material from multiple sources, creating sequences that would not otherwise be found in the genome.

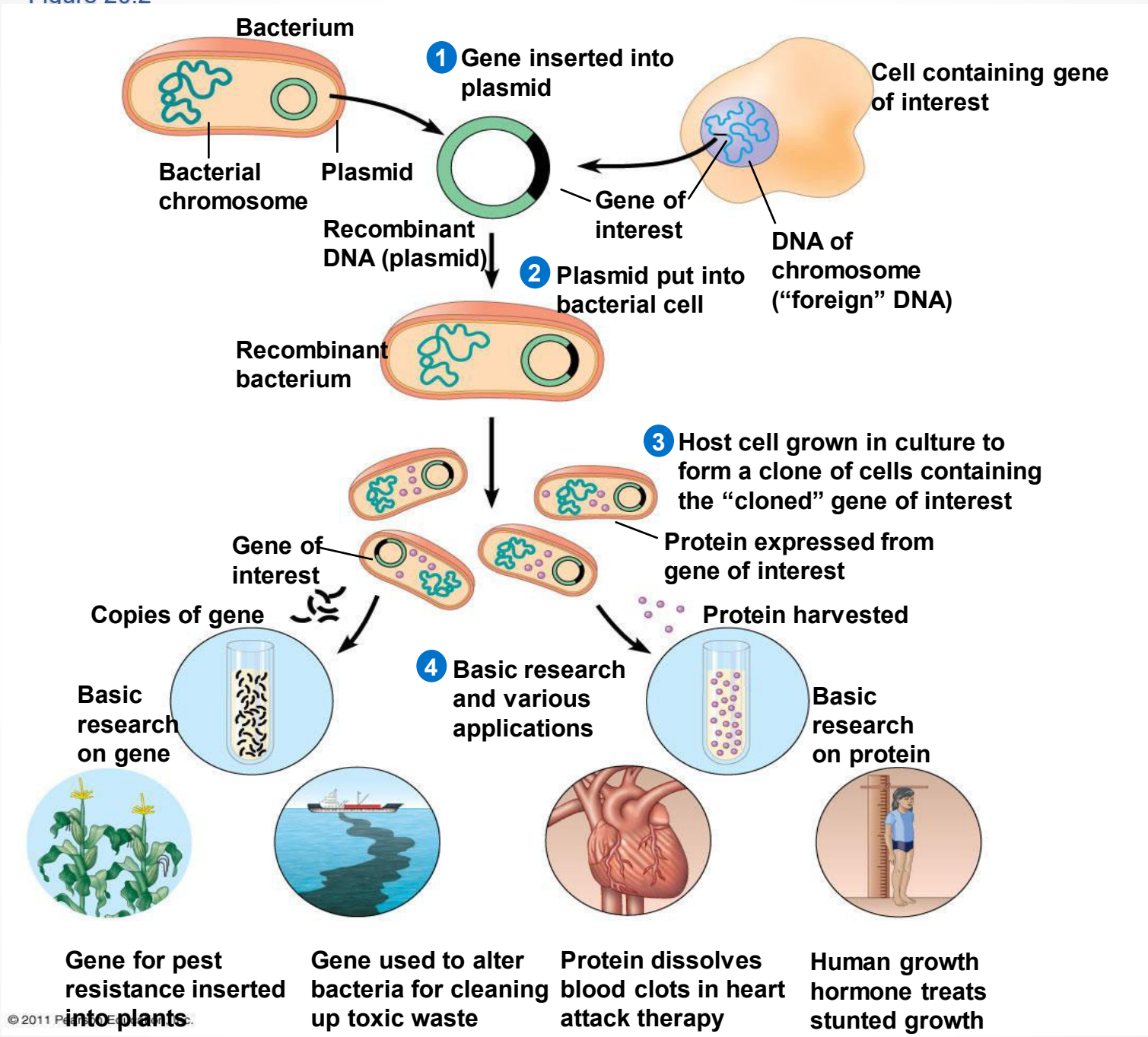
Cloning Strategies

- The cloning strategy is divided into four stages :
  - Obtaining DNA fragments
  - Joining to vectors
  - Introduction into host cell
  - Screening or Selection

Cloning Tools

- Endonucleases
- Ligases
- Vector
- Host
- Transformation

Figure 20.2



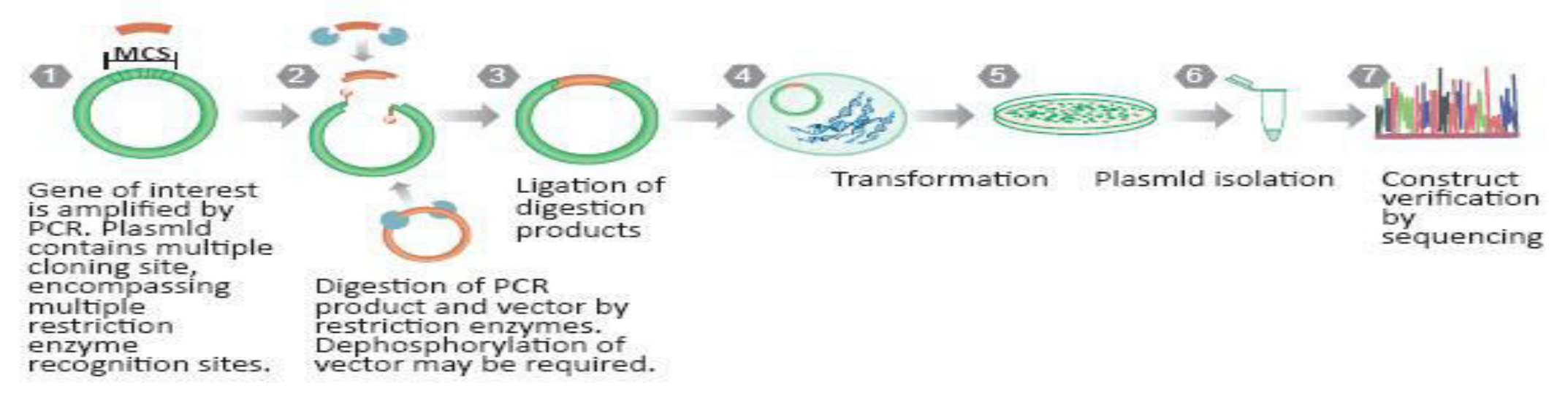
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Types of Recombinant Cloning

- Traditional cloning
- TA cloning
- Seamless cloning

Traditional cloning

- *Traditional cloning*, requires the use of the polymerase chain reaction (PCR) to amplify the template sequence of interest and add restriction sites to the ends of the sequence.



TA cloning

- TA cloning is one of the simplest forms of cloning.
- In this method, vectors containing 5' thymine overhangs are used to accept PCR products in which additional 3' adenosine overhangs have been added on by the nature of TAQ polymerase amplification.

Seamless cloning

- Seamless cloning technologies eliminate the requirement for restriction enzymes.
- This can be advantageous when an insert contains a number of restriction sites within its sequence, making it difficult to identify restriction enzymes that will not cut the gene of interest during the cloning procedure.

Benefits

- Many copies of genetic material available in the laboratory for the purpose of experimentation
- Production of pharmaceutical drugs, insulin, clotting factors, human growth hormone, and industrial enzymes
- Important for human research
- For production of GMOs

Reproductive Cloning

- The production of a genetic duplicate of an existing organism

- Somatic cell nuclear transfer

Therapeutic Cloning

- Cloning designed as therapy for a disease.
- In therapeutic cloning, the nucleus of a cell, typically a skin cell, is inserted into a fertilized egg whose nucleus has been removed.
- The nucleated egg begins to divide repeatedly to form a blastocyst.

Grafting Techniques



Grafting

- Grafting refers to a surgical procedure to move tissue from one site to another on the body, or from another person, without bringing its own blood supply with it.
- Instead, a new blood supply grows in after it is placed. In some instances a graft can be an artificially manufactured device.
- Examples of this are a tube to carry blood flow across a defect or from an artery to a vein for use in hemodialysis.

Classification

- **AUTOGRAFT:** graft taken from one part of the body of an individual and transplanted onto another site in the same individual, e.g., skin graft.
- **ISOGRAFT:** graft taken from one individual and placed on another individual of the same genetic constitution, e.g., grafts between identical twins.
- **ALLOGRAFT:** graft taken from one individual placed on genetically non-identical member of the same species.
- **XENOGRAFT:** graft taken from one individual placed on an individual belonging to another species, e.g., animal to man.

Types of Grafting

- The term grafting is most commonly applied to skin grafting.
- Specific types include:
  - Skin grafting
  - Bone grafting
  - Tendon grafting
  - Cartilage grafting

Skin Grafting

- A skin graft is a procedure performed where healthy skin is removed from one area of the body, the donor site, and transplanted to another, the recipient site.
- The areas of the body that are most commonly used as donor sites for skin graft are the legs, inner thigh, upper arm and fore arm.

Types of Skin Grafting

- 1.Split thickness graft
- 2.Full thickness graft
- 3.Composite skin graft

Split or Partial Thickness Graft

- This is the most common type of graft.
- The epidermis and part of dermis are removed from donor site and transplanted on the damaged area.

Full Thickness Graft

- The entire epidermis and dermis are transplanted to the recipient site.
- The donor site needs to be either closed with stitches or have a partial thickness graft transplanted.

Composite Skin Graft

- A small graft containing skin and underlying cartilage or other tissue.
- Usually skin and fat or skin cartilage often taken from ear margin and useful for rebuilding missing elements of nose, eyelids and fingertips.

Bone Grafting

- Used for the replacement or augmentation of the bone
- Used in dental implants, as well as other instances.
- The bone may be autologous, typically harvested from the iliac crest of the pelvis, or banked bone.

Types of Bone Grafting

1.Bone blend

2.Bone swagging

Bone Blend

- Uses an autoclave capsule and pestle
- Bones removed from predetermined site, triturated in capsule in a workable plastic like mass and packed in a bony defects.

Bone Swagging

- Techniques required existence of an edentulous area adjacent to the defect from which bone is pushed in contact with root surface without fracturing the bone at its base

Tendon Graft

- Contraindicated if there is stiff joints, adherent extensor tendons, and inadequate skin cover
- Only autograft
- Unacceptable amount of host reaction and adhesion after allografts and xenografts

Cartilage Graft

- Cartilage has no intrinsic blood supply
- The use of cartilage autografts is widespread and includes nasal, auricular, craniofacial skeleton, and joint reconstruction
- Both chondrocytes and matrix are subject to xenogeneic mechanisms of rejection with a generally poorer outcome in comparison.
- There is only small number of usage

Genome Wide Association Study (GWAS)

Genome-Wide Association Study (GWAS)

- A genome-wide association study (**GWAS**) is an approach used in genetics research to associate specific genetic variations with particular diseases.
- The method involves scanning the genomes from many different people and looking for genetic markers that can be used to predict the presence of a disease.

Central Goal of Human Genetics

- To identify genetic risk factors for common, complex diseases

Goal of GWAS to use genetic risk factors to predict who is at risk

- Identify the biological underpinnings of disease susceptibility for developing new prevention and treatment strategies

Concepts underlying the study design

- SNP---single nucleotide polymorphism
- Single base pair changes in the DNA sequence that occur with high frequency in the human genome
- SNP (common) vs. Mutation (rare)
- Cystic fibrosis---mutations in the CFTR gene
- Linkage analysis---genotyping families affected by cystic fibrosis using a collection of genetic markers across the genome and examining how these genetic markers segregate with the disease across multiple families

Common Disease Common Variant Hypothesis

- Common disorders are likely influenced by genetic variation that is also common in the population

I. If common genetic variants influence disease, the effect size (or penetrance) for any one variant must be small relative to that found for rare disorders.

II. If common alleles have small genetic effects (low penetrance), but common disorders show heritability (inheritance in families), then multiple common alleles must influence disease susceptibility.

Capturing Common Variation

- Location and density of commonly occurring SNPs is needed to identify the genomic regions and individual sites that must be examined by genetic studies
- Population-specific differences in genetic variation must be cataloged so that studies of phenotypes in different populations can be conducted with the proper design
- Correlations among common genetic variants must be determined so that genetic studies do not collect redundant information

International Hap Map Project

- Used a variety of sequencing techniques to discover and catalog SNPs in European descent populations, the Yoruba populations of African origin, Han Chinese individuals from Beijing, and Japanese individuals from Tokyo
- Has since been expanded to include 11 human populations

Linkage Disequilibrium

- A property of SNPs on a contiguous stretch of genomic sequence that describes the degree to which an allele of a SNP is inherited or correlated with an allele of another SNP within a population
- Linkage between markers on a population scale

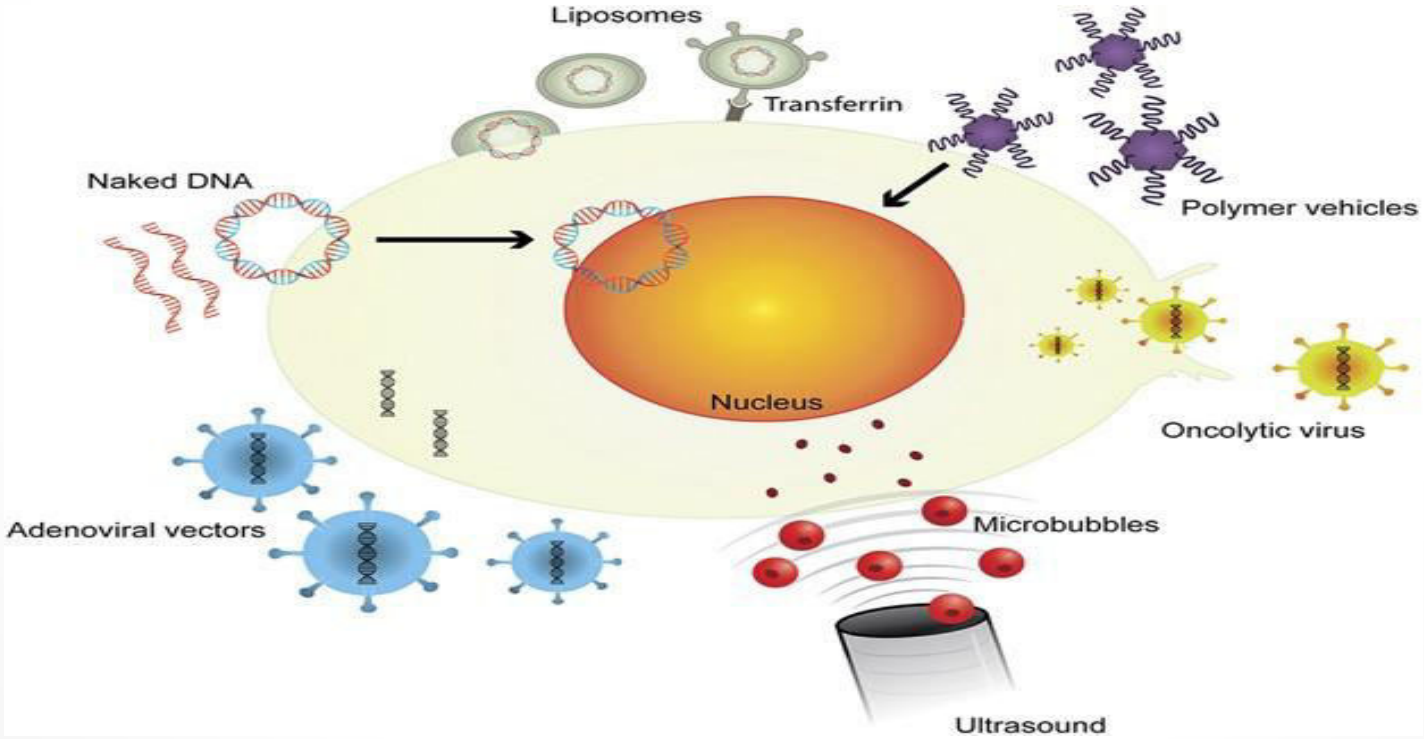
Direct vs. Indirect Association

- LD creates two possible positive outcomes from a genetic association study
- Direct association----the SNP influencing a biological system that leads to the phenotype is directly genotyped in the study
- Indirect association----the influential SNP is not directly typed, but instead a tag SNP in high LD with the influential SNP is typed
- Therefore, a significant SNP association from a GWAS should not be assumed as the causal variant

Application in Pharmacology

- Identifying DNA sequence variations associated with drug metabolism and efficacy as well as adverse effects
- Example, warfarin---determining the appropriate dose
- Personalized medicine

Nucleic Acid And Cell Based Therapies



Nucleic Acid And Cell Based Therapies

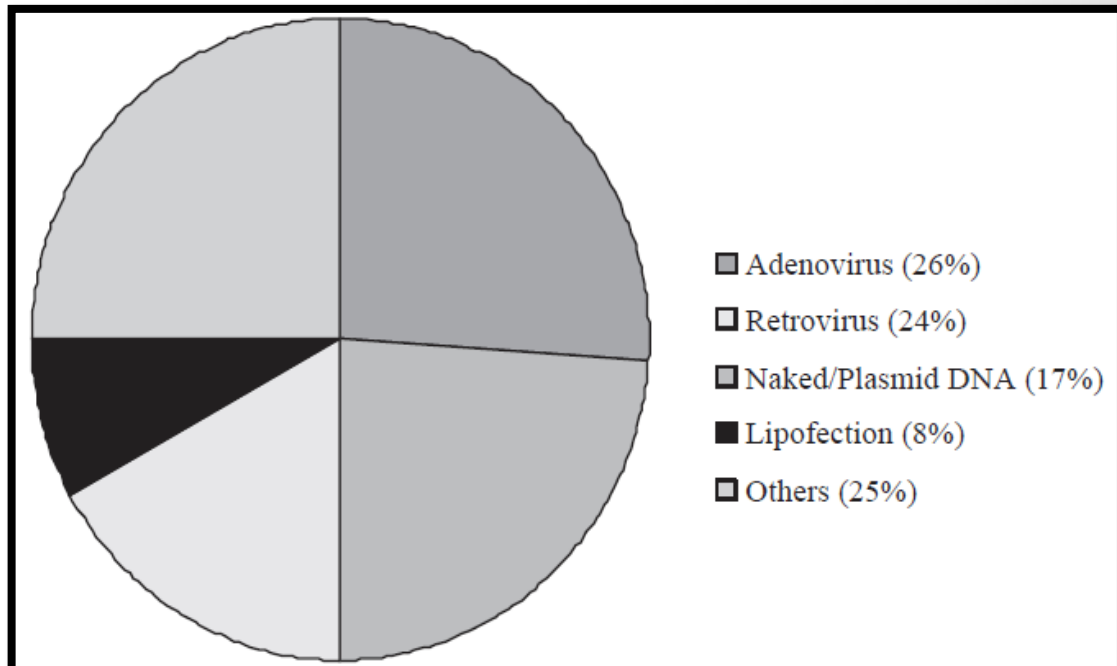
"Therapeutic techniques and methods that have a curative approach while using nucleic acids(DNA/RNA) and certain cells of the body."

- **Nucleic acid based:**
  - Gene therapy
  - Antisense technology
  - Aptamers
  - RNAi

Cell based therapies: Stem cell

Gene Therapy

- The stable introduction of a gene into the genetic complement of a cell, such that subsequent expression of the gene achieves a therapeutic goal. The potential of gene therapy as a curative approach for inborn errors of metabolism and other conditions induced by the presence of a defective copy of a specific gene (or genes) is obvious.
- Cancer, AIDS, cystic fibrosis, SCID, familial hypercholesterolemia etc. are under clinical trials.
- Requirements for the introduction of gene of interest
- Vector (viral or non-viral)
- Cell line (opted according to the need)



Antisense Technology

- Antisense therapy is a form of treatment for genetic disorders or infections.
- When the genetic sequence of a particular gene is known to be causative of a particular disease, it is possible to synthesize a strand of nucleic acid (DNA, RNA or a chemical analogue) that will bind to the messenger RNA (mRNA) produced by that gene and inactivate it, effectively turning that gene "off".

Aptamers

- Aptamers are single-stranded DNA or RNA-based sequences that fold up to adopt a unique three dimensional structure, allowing them to bind a specific target molecule.
- Highly specific , high binding affinity
- DNA and RNA based aptamers libraries synthesized by chemical methods.
- Only one product available in market by the name of “Macugen”.

RNA Interference (RNAi)

- Biological process in which RNA molecules inhibit gene expression, typically by causing the destruction of specific mRNA molecules via RISC(RNA Induced splicing complex).
- A method of gene silencing
- Viral infections
- Cancer therapy

Cell Based Therapies (Stem Cells)

- On the basis of source:
- Adult (adipose tissue, blood, bone marrow, umbilical cord)
- Embryonic (pre implant stage embryos)

- This technique is used in:

- Transplants
- Metabolic diseases (diabetes)
- Cancer therapy
- Induced pluripotent cells

Advantages

- Less chances of adverse effects
- Low mortality rate
- Cheaper and reliable
- AIDS and various cancer treatment trials are in process.

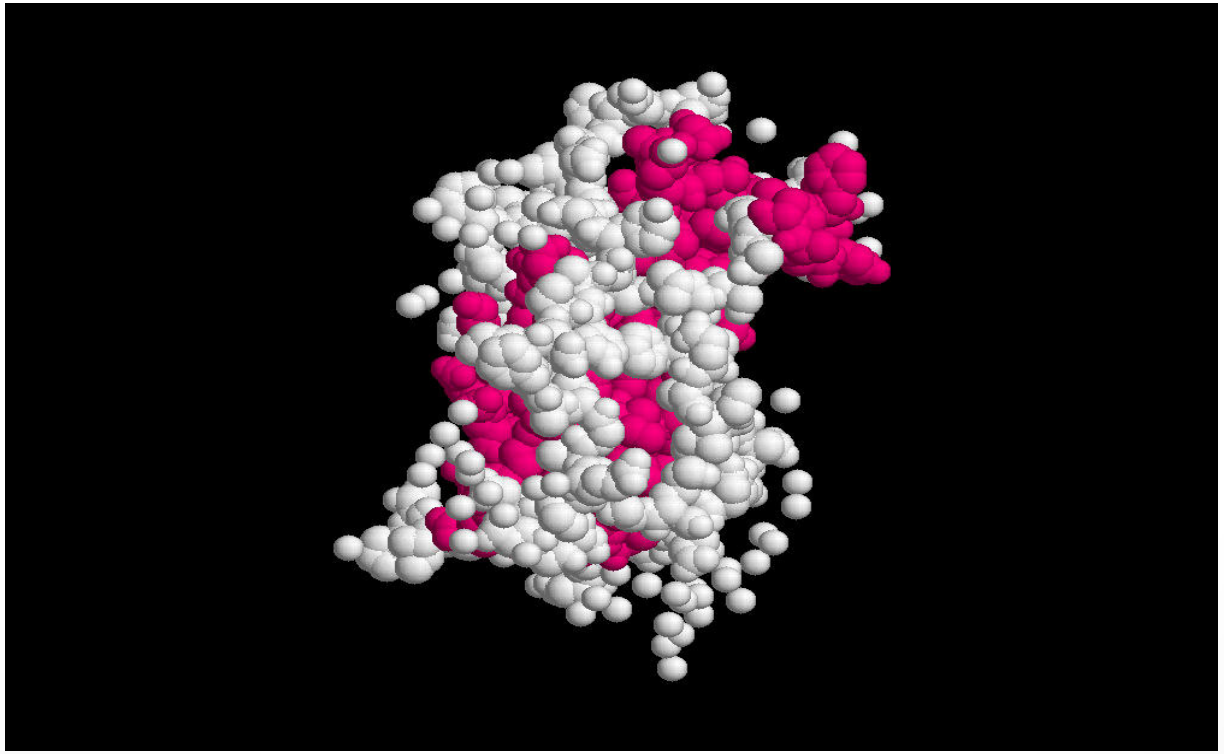
Disadvantages

- Sometimes immunosuppression is required
- Tumor formation is common is transplantation
- In case of gene therapies, multigene disorders can’t be addressed through this technique.

Current Status

- Only three nucleic acid based medicines in market. one antisense-based product (trade name Vitravene), one aptamer (trade name Macugen) and one gene therapy product (trade name Gendicine, approved only in China).
- Till now, 7 deaths are reported during the clinical trials.
- 1200 clinical trials of gene therapy carried out worldwide.

Recombinant Hormones



Introduction

- **Hormone**

Hormones are type of growth factors, the act as a messenger between cells and are vital for the function of multi-cellular organisms

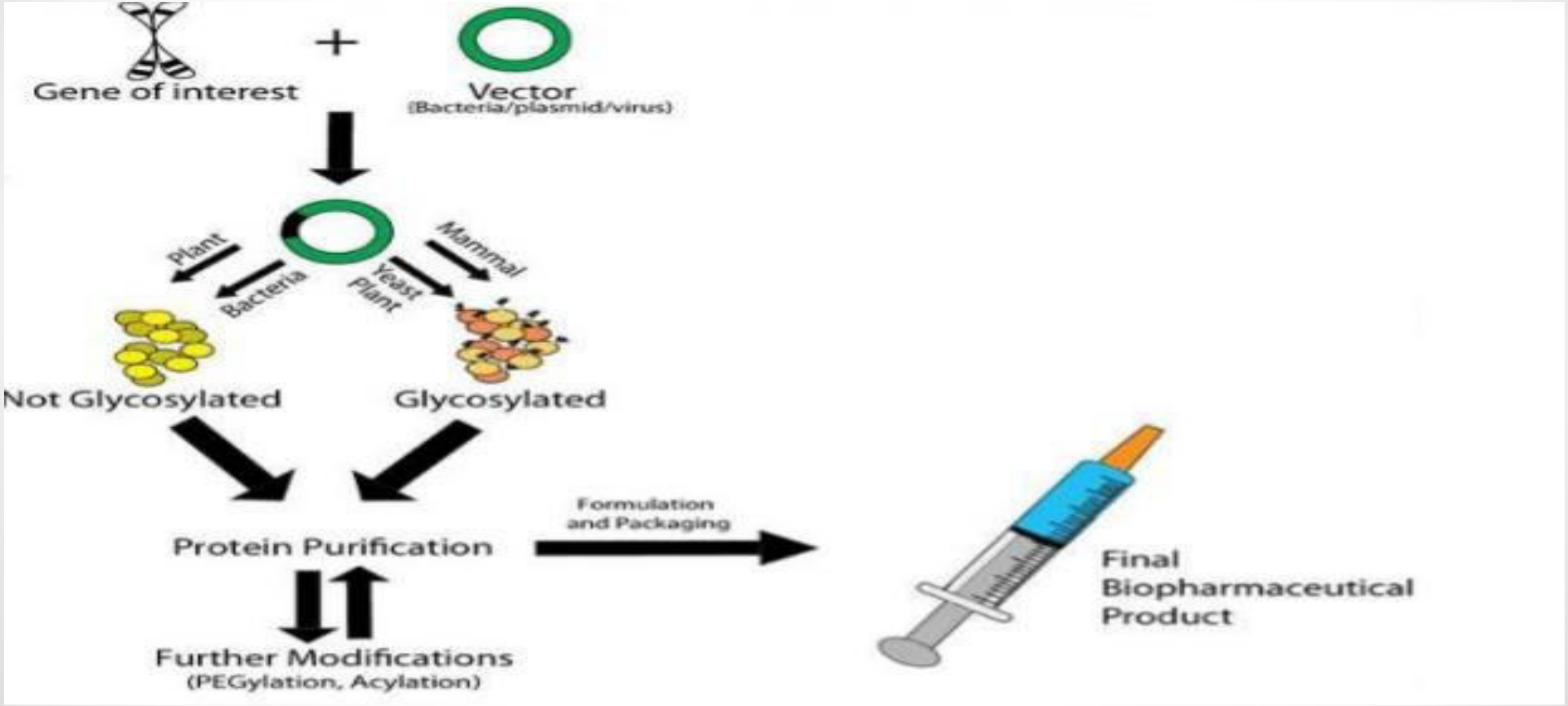
- **Recombinant Hormones:**

Synthetic hormones which are being produced by genetic engineering rather than extracting them from animals and humans. e.g. HGH, Insulin, FSH

Endocrine System

- The endocrine system is made up of glands that produce and secrete hormones.
- Hormones regulate:
  - The body's growth
  - Metabolism
  - Sexual development and function

Recombinant Hormone Production



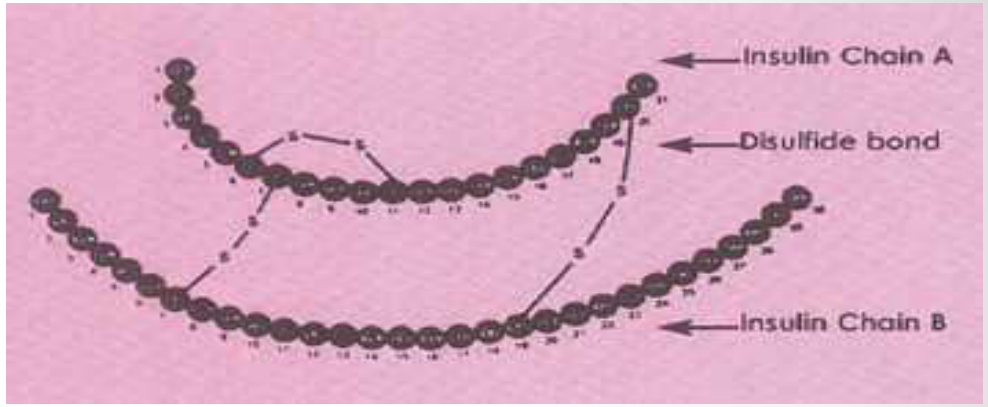
List of Recombinant Hormones

- Human growth hormone (rHGH)
- Biosynthetic human insulin (BHI)
- Factor VIII
- Erythropoietin (EPO)
- β-ENDORPHIN
- Follicle-stimulating hormone (FSH)
- Bovine somatotropin (bST)
- Interferon (IF)
- Somatostatin

Recombinant Hormones	Strain / Cell line used	Function / Role
Insulin	E. coli	Hormone which lowers blood sugar used by diabetics
Human Growth hormone	E. Coli CHO cells	For normal body growth & development
Erythropoietin	CHO cells	production of red and white blood cells
Factor VIII	CHO or BHK cells	Blood protein necessary for clotting
β-ENDORPHIN	Mouse pituitary cell line	Pain suppressors
Bovine somatotropin (bST)	E. coli	Regulate metabolic processes. Enhance milk production
Somatostatin	CHO	Regulate endocrine system

Structure of Insulin

- Chemically, insulin is a small, simple protein
- Preproinsulin=108 amino acid
- 23 a.a signal peptide.



- 34 a.a long pro-part of the insulin is removed in secretory granules.
- Mature insulin consists of 51 amino acid, 30 of which constitute B-chain
- 21 of which comprise A-chain.
- The two chains are linked by a disulfide bond

Protein Extraction and Purification

- Cell disruption
- Centrifugation
- Ultra filtration
- Diafiltration
- Chromatography
- Addition of excipients
- Sterile filtration
- Powder or liquid preparation
- Sealing, labelling and packing of final product

Human Growth Hormone/ Somatotropin

- Secreted by the pituitary gland
- Responsible for normal body growth and development
- Stimulate protein production in muscle cells
- Energy release from the breakdown of fats
- Stimulates the development of bones
- **Growth Hormone:** 191 amino acids, single chain

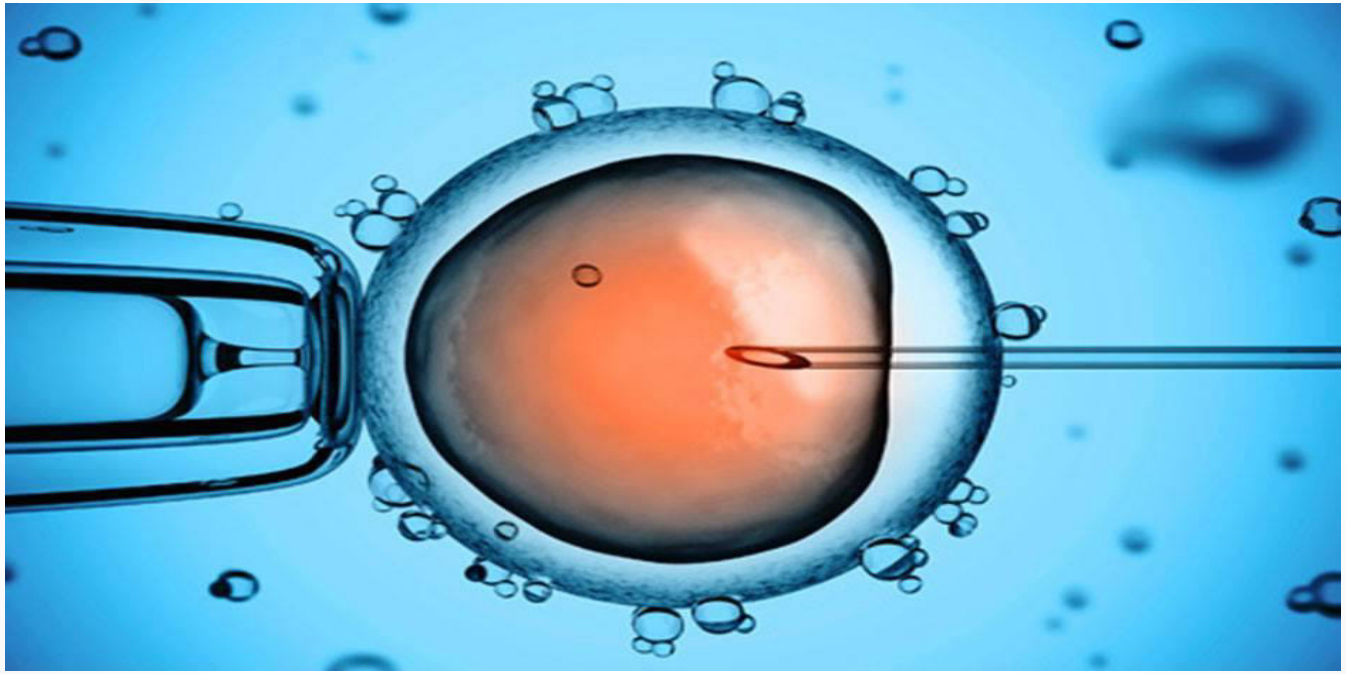
Erythropoietin

- A glycoprotein, acts on the bone marrow to increase the production of red and white blood cells.
- Stimuli such as bleeding or moving to high altitudes (where oxygen is scarce) trigger the release of erythropoietin
- Kidney is the principal production site of Erythropoietin
- Kidney cells specifically sense the oxygen deficit in the blood and start producing Erythropoietin

Recombinant hormones now approved

- Glucagon
- Thyroid stimulating hormone
- Parathyroid hormone
- calcitonin

Reproductive Technologies



Infertility

- Inability to conceive after 1 year of properly timed unprotected intercourse.
- It is world wide problem effecting 13-14% of reproductive –aged couples.
- **Infertility may be**
  - Primary, where no history of pregnancy ever having occurred
  - Secondary, inability to conceive after one successful pregnancy or after an abortion

Causes of Infertility

Female

- Dietary disturbances, anxiety, fear etc.
- Uterine absence, hypoplasia etc.
- Pituitary failure, thyroid disturbances, polycystic disease.
- Pelvic inflammation, T.B. etc.

Male

- Fatigue, excess smoking, alcohol etc.
- Undescended testis, other abnormalities
- Pituitary failure, thyroid deficiency etc.
- Mumps, arthritis, venereal disease etc.

Assisted Reproductive Technologies (ART)

- ART include those infertility treatments in which both eggs and sperms are handled in the lab for the purpose of establishing a pregnancy
- Birth of Louise Brown in 1978, world's first test tube baby.
- Various Assisted Reproductive Techniques
  - **Intra Uterine Insemination (IUI)**
  - **Intracytoplasmic sperm injection (ICSI)**
  - **In Vitro Fertilization (IVF)**
  - **Gamete Intra Fallopian Transfer (GIFT)**



Intra Uterine Insemination (IUI)

- Suggested for subtle male factor infertility and an abnormal sperm mucus interaction
- **Procedure**
  - One million motile sperms are usually placed at the top of uterine cavity near the opening of the fallopian tubes.
  - The whole procedure takes place during the fertile days i.e.13-15 of her menstrual cycle.
- **Complications**
  - Uterine perforation vary rare.
  - Infection during insemination.
  - Localized trauma of the endometrium can cause bleeding but has no risk
  - Discomfort can cause uterine cramp.

Intracytoplasmic sperm injection (ICSI)

- Suggested for men incapable of reproduction
- **Procedure**
  - The ovum is injected with single sperm in vitro with the help of micromanipulation devices.
  - After the procedure the oocyte is placed into cell culture and checked for signs of fertilization.
  - Once fertilization is accomplished, cell division occur and embryo is transferred into women's uterus.
- **Complications**
  - Naturally competent cells initiate fertilization which does not occur in in-vitro.
  - Hyaluronan is used as a drop in the plate and sperms compete for it. Those sperms are selected and used for fertilization

In Vitro Fertilization (IVF)

- This technology is developed to overcome the infertility due to the problems of:
  - Couples with endometriosis, drug resistant polycystic ovarian syndrome etc.
  - Age related ovarian dysfunction, ovarian failure or surgically removed ovaries
  - Blocked fallopian tubules, ovulation problem
  - Seminal antibodies in women
  - Combine ICSI and IVF can be used to treat sperm disorders.

Procedure

- **Four stage procedure**
  - Stage 1- Ovarian stimulation and monitoring
  - Stage 2- Egg retrieval
  - Stage 3- Fertilization
  - Stage 4- Embryo transfer

Procedure

- Women is given drugs to produce super-ovulation i.e. produces many eggs
- Fluid collected from fallopian tubes and eggs separated
- Semen sample taken at the same time.
- Eggs and semen are placed in a culture dish to fertilize under special conditions.
- Fertilization observed in the laboratory.
- After three days of fertilization each successful embryo has divided to 8 cell level.
- At this time three healthy embryos are transferred to the uterus.
- Patient is given drugs to help support implantation
- Generally after 12 to 14 days after embryo transplantation blood test is done to confirm pregnancy.

Complications and Cure

- **Complications**
- Hyperstimulation syndrome
- Estrogen and kallikrein-kinin and vascular endothelial growth factor (VEGF) is poured from ovaries into abdominal cavity.
- These coat the lining of abdominal cavity and make it permeable i.e. leaky.
- Serum from blood pours into the cavity resulting balloon ovaries, abdomen swells, low blood pressure and may get dizzy due to it.
- **Cure**
- Drainage of extra fluid
- Intravenous fluid intake.

Gamete Intra Fallopian Transfer (GIFT)

- Developed in 1983, by Dr. Ricardo Asch.
- Suggested for woman with unexplained infertility or infertility due to cervical or immunological factors, mild endometriosis.
- In GIFT mixture of sperm and egg is injected into mothers fallopian tube where fertilization occurs.
- Rate of pregnancy is higher though as compared to other methods i.e. IVF and ICSI

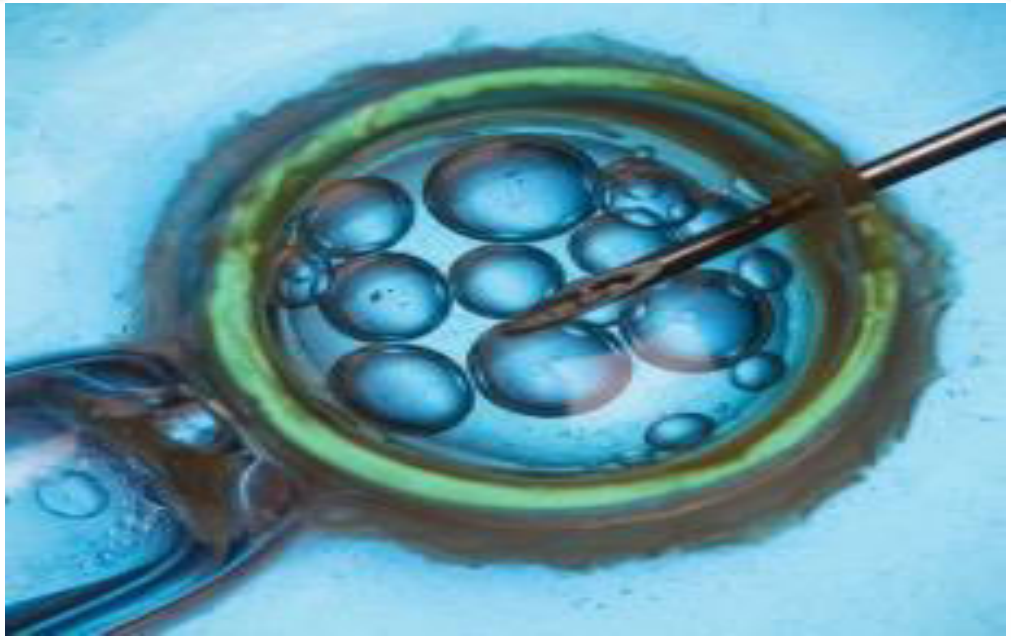
Procedure

- **Step 1.** Over stimulation and monitoring
- **Step 2**. Egg retrieval
- Oocyte is aspirated either by transvaginal with ultra sound guidance or transabdominally by laparoscopic guidance.
- **Step 3**. Gamete transfer
- Sperm and egg are combined and incubated for 10 min. to allow sperm binding.
- 25,000 sperms and 2 oocyte are placed in fallopian tube with extraneous progesterone.

Limitations

- Abnormal fallopian tubes not eligible.
- GIFT requires surgery for gamete transfer.

Reproductive Technologies



Zygote Intrafallopian Transfer (ZIFT)

- Oocyte is retrieved from the ovary as in GIFT.
- Fertilization occurs in laboratory.
- After 24 hrs. 2-3 embryos are transferred via laparoscopy into one of the fallopian tubes.
- If embryos are allowed to grow greater than 2 cell stage then its called tubal embryo transfer (TET).
- **Only benefit ZIFT over IVF is for women who are thought to have compromised embryo quality due to embryo in vitro culture.**
- Placing zygotes back into their own natural incubators is thought to enhance development and improve pregnancy.

Blastocyst culture and Transfer

For this

1. Have 3 or more IVF cycles and no pregnancy
2. Over 40 years of age
3. Want single pregnancy
4. Undergoing preimplantation genetic disorders.

Procedure

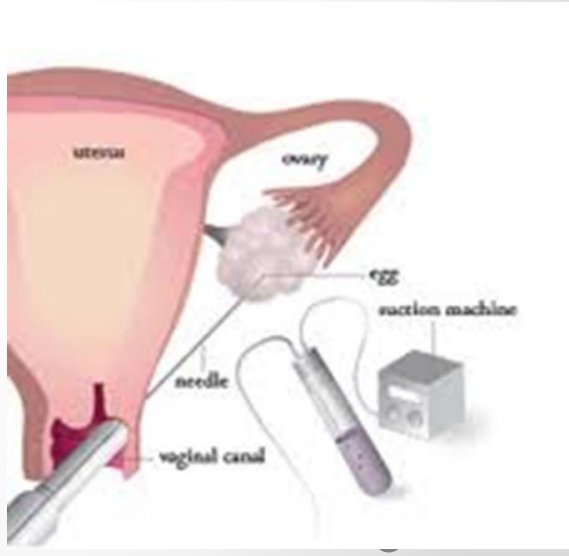
- Embryos are grown for 5 days until blastocyst stage and then transplanted. This ensures
- Max. pregnancy rates
- Minimizing the risk of multiple pregnancies.
- Good healthy blastocyst are transplanted via catheter. Remaining are frozen for further use.
- Occasionally cover of blastocyst is dissolved by an enzyme this increases the chance of implantation.

Preimplantation Genetic Diagnosis PGD

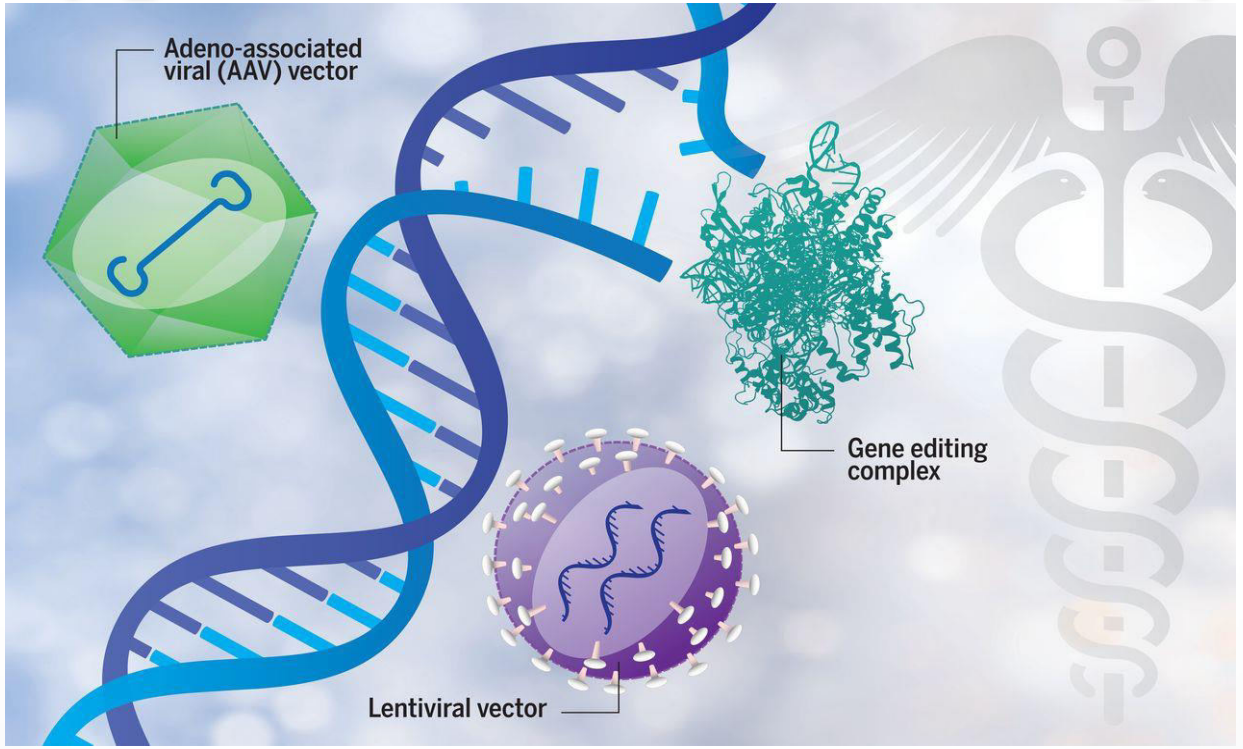
- PGD improves the likelihood of successful pregnancy and is requested by prospective parents who are concerned about passing an incurable genetically based disease or disorder to their child.
- First PGD baby was born in 1989 and by 1997 over 30 babies were born world wide.
- PGD can monitor 2 dozen disorders.

In Vitro Maturation (IVM)

- In vitro maturation (IVM) is the technique of letting the contents of ovarian follicles and the oocytes inside mature in vitro.
- It can be offered to women with infertility problems, combined with IVF, offering women pregnancy without ovarian stimulation.
- In vitro maturation of ova and less risk of over stimulation of mother.
- Gn-Rha (prevent ovulation till maturation)
- FSH (stimulate multiple follicle production)
- HCG (final maturation of eggs)



Types of Gene Therapy



Introduction

- **Somatic gene therapy:** Transfer of a section of DNA to any cell of the body that doesn't produce sperm or eggs. Effects of gene therapy will not be passed onto the patient's children.
- **Germ line gene therapy:** Transfer of a section of DNA to cells that produce eggs or sperm. Effects of gene therapy will be passed onto the patient's children and subsequent generations

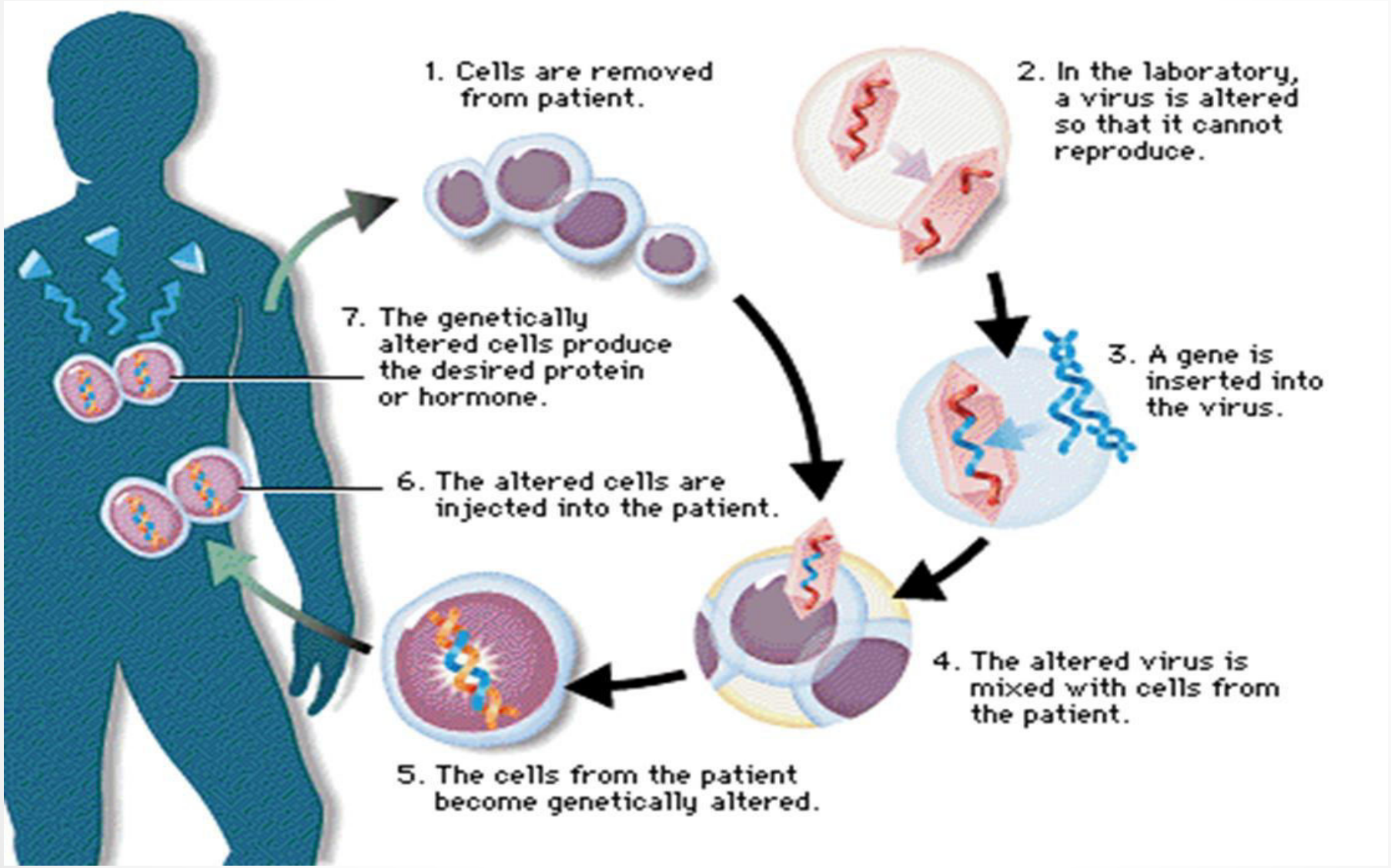
Somatic Gene Therapy Types

- **In vivo Gene Therapy**
  - Delivery of new Genetic material directly to target cells in the body
  - Transient expression
  - Treatment of cystic fibrosis
  - Lungs ,Brain

Ex vivo Gene Therapy

- Cells are modified outside the body and then transplanted back it again
- Long-term expression
- Fibroblast cells, Hematopoietic cells

Ex Vivo Gene Therapy



GENE AUGMENTATION THERAPY

- This is used to treat diseases caused by a mutation that stops a gene from producing a functioning product, such as a protein.
- This therapy adds DNA containing a functional version of the lost gene back into the cell.
- The new gene produces a functioning product at sufficient levels to replace the protein that was originally missing.
- This is only successful if the effects of the disease are reversible or have not resulted in lasting damage to the body.
- For example, this can be used to treat loss of function disorders such as cystic fibrosis by introducing a functional copy of the gene to correct the disease

GENE INHIBITION THERAPY

- Suitable for the treatment of infectious diseases, cancer and inherited disease caused by inappropriate gene activity.
- The aim is to introduce a gene whose product either:
- inhibits the expression of another gene
- interferes with the activity of the product of another gene.
- The basis of this therapy is to eliminate the activity of a gene that encourages the growth of disease-related cells.
- For example, cancer is sometimes the result of the over-activation of an oncogene (gene which stimulates cell growth). So, by eliminating the activity of that oncogene through gene inhibition therapy, it is possible to prevent further cell growth and stop the cancer in its tracks.

Risk of Gene Therapy

- This technique presents the following risks:
- **Unwanted immune system reaction**. Your body's immune system may see the newly introduced viruses as intruders and attack them. This may cause inflammation and, in severe cases, organ failure.
- **Targeting the wrong cells**. Because viruses can affect more than one type of cells, it's possible that the altered viruses may infect additional cells — not just the targeted cells containing mutated genes. If this happens, healthy cells may be damaged, causing other illness or diseases, such as cancer.
- **Infection caused by the virus**. It's possible that once introduced into the body, the viruses may recover their original ability to cause disease.
- **Possibility of causing a tumor**. If the new genes get inserted in the wrong spot in your DNA, there is a chance that the insertion might lead to tumor formation.

DNA Fingerprinting



DNA Fingerprinting

- DNA Fingerprinting is a way to identify a certain individual, rather than simply identifying a species or a particular trait
- A technique used by scientists to distinguish between individuals of the same species using only samples of their DNA
- The process of DNA fingerprinting was invented by Alex Jeffreys in 1985

Biological Sample used for DNA Fingerprinting

- Blood
- Hair
- Saliva
- Semen
- Body tissue cells

Steps to DNA Fingerprinting

- There are 8 steps for DNA Fingerprinting
- **Step 1: Isolation of DNA**
  - DNA must be recovered from cells or tissue.
  - Only a small amount of blood, hair, or skin is needed to isolate DNA

Steps to DNA Fingerprinting

- **Step 2:Digestion of DNA by restriction endonucleases**
  - The DNA is cut into fragments using restriction enzymes
  - Each restriction enzyme cuts DNA at a specific base sequence
  - The sections of DNA that are cut out are called restriction fragments
  - This yields thousands of restriction fragments of all different sizes because the base sequences being cut may be far apart (long fragment) or close together (short fragment)
  - Commonly used RE are (hae III, Hinf I, Alu I etc.)
  - Reaction mixture is incubated overnight at 37C

Steps to DNA Fingerprinting

- **Step 3: Electrophoretic separation of different fragments**
- Fragments are separated on the basis of size using a process called gel electrophoresis
- DNA fragments are injected into wells and an electric current is applied along the gel.
- DNA is negatively charged so it is attracted to the positive end of the gel.
- The shorter DNA fragments move faster than the longer fragments
- DNA is separated on basis of size.

Steps to DNA Fingerprinting

- **Step 4: Transfer DNA on Nylon/Nitrocellulose membrane**
  - The DNA fragments are transferred to a nylon sheet by placing the sheet on the gel and soaking them overnight by the process southern blot.

- **Step5: Probing/probe labeling**
  - Adding radioactive or colored probes to the nylon sheet which is complementary to target sequences
  - Each probe only sticks to one or two specific places on the sheet

Steps to DNA Fingerprinting

- **Step 6: Hybridization**
  - Labeled probe DNA should be hybridized with the complementary sequences located on nylon membrane for the detection of position of later
  - Membrane is washed to remove non specific binding and clearing of the background
- **Step 7: Autoradiography**
  - To detect the sequences in genome bound with the hybridized radioactive probe on membrane
  - Technique involves alignment of hybridized membrane with X-ray film
  - The X-ray film alter its development shows multiple no. of bands that looks like bar codes and known as DNA fingerprints.

Steps to DNA Fingerprinting

- **Step 8: Interpretation of band patterns**

  - Analysis of band patterns of different individuals
  - Comparison of position of bands
  - Computer software are also available for the analysis of DNA fingerprints.

Uses

- **Diagnosis of Inherited Disorders**

  - Helps diagnose disorders in both prenatal and newborn babies

  - Disorders may include cystic fibrosis, hemophilia, Huntington's disease, familial Alzheimer's, sickle cell anemia, thalassemia, and much more.

Uses

- **Crime**
  - Forensic science is the use of scientific knowledge in legal situations.
  - The DNA profile of each individual is highly specific.
  - The chances of two people having exactly the same DNA profile is 30,000 million to 1 (except for identical twins).
- **Personal Identification**
  - The U.S. armed services are just beginning a program where they collect DNA fingerprints from all personnel for later use, incase they need to identify casualties or missing people

Some Other Uses of DNA fingerprinting

- Identification of carcass of tissues
- Detection of somatic mutations or cancer
- Pathogen identification
- Detection of loci controlling quantitative traits or disease resistance
- Sex determination
- Individual identification